```
# ==== run.py ====
# -*- coding: utf-8 -*-
"""系统启动入口。
运行：
    python run.py                启动并自动打开浏览器
    python run.py --no-browser   只启动服务，不打开浏览器
    python run.py --dev          开发模式（开启代码热重载与调试器）
    python run.py --port 5001    指定端口
默认以 127.0.0.1:5000 启动，就绪后自动打开浏览器。
"""
import os
import sys
import threading
import time
import webbrowser
if hasattr(sys.stdout, "reconfigure"):
    sys.stdout.reconfigure(encoding="utf-8")   # 控制台中文正常显示
from app import create_app
from app.models import db
PORT = 5000
for i, a in enumerate(sys.argv):
    if a == "--port" and i + 1 < len(sys.argv):
        PORT = int(sys.argv[i + 1])
URL = "http://127.0.0.1:%d" % PORT
NO_BROWSER = "--no-browser" in sys.argv
DEV = "--dev" in sys.argv
app = create_app()
def open_browser_later(delay=1.2):
    """等服务工作起来再打开浏览器，避免打开时页面还没就绪。
    放在后台线程里，不用 sleep 阻塞服务启动；线程设为守护线程，
    用户 Ctrl+C 停止服务时不会残留。
    """
    def run():
        time.sleep(delay)
        try:
            webbrowser.open(URL)
        except Exception:
            pass            # 打开失败不影响服务本身
    t = threading.Thread(target=run, daemon=True)
    t.start()
if __name__ == "__main__":
    with app.app_context():
        db.create_all()     # 首次运行时自动建表
        from app.utils.schema import ensure_schema
        added = ensure_schema(db)     # 旧库补齐新增字段
        if added:
            print("已补充字段：%s" % "、".join(added))
    print("=" * 58)
    print("   历史遗留废弃矿山自然恢复适宜性评价系统  Ｖ１．０")
    print("=" * 58)
```

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    print("   访问地址：%s" % URL)
    print("   演示账号：admin      密码：admin123")
    print("   停止运行：按 Ctrl+C，或直接关闭本窗口")
    print("=" * 58)
    if not NO_BROWSER and not DEV:
        print("   正在打开浏览器……")
    print()
    # 开发模式才启用热重载（重载会二次启动进程，故此时不自动开浏览器）
    if not NO_BROWSER and not DEV:
        open_browser_later()
    app.run(host="127.0.0.1", port=PORT, debug=DEV,
    use_reloader=DEV)
# ==== app/__init__.py ====
# -*- coding: utf-8 -*-
"""Flask 应用工厂。"""
from datetime import date
from flask import Flask
from flask_login import LoginManager
from app.config import Config
from app.models import db, User
from app.utils.constants import ROLE_LABELS
login_manager = LoginManager()
login_manager.login_view = "auth.login"
login_manager.login_message = "请先登录"
@login_manager.user_loader
def load_user(user_id):
    return db.session.get(User, int(user_id))
def create_app(config_class=Config):
    app = Flask(__name__)
    app.config.from_object(config_class)
    db.init_app(app)
    login_manager.init_app(app)
    # 注册视图（各模块路由）
    from app.views import auth, project, unit, standard,
    constraint, \
        ahp, indicator, evaluate, report, search
    app.register_blueprint(auth.bp)
    app.register_blueprint(project.bp)
    app.register_blueprint(unit.bp)
    app.register_blueprint(standard.bp)
    app.register_blueprint(constraint.bp)
    app.register_blueprint(ahp.bp)
    app.register_blueprint(indicator.bp)
    app.register_blueprint(evaluate.bp)
    app.register_blueprint(report.bp)
    app.register_blueprint(search.bp)
    # 全局登录保护：除登录页与静态资源外，均需登录
    from flask import request, redirect, url_for
    from flask_login import current_user, login_user
    PUBLIC_ENDPOINTS = {"auth.login", "static"}
```

```
@app.before_request
def require_login():
    #  截图模式（仅用于生成说明书配图）：自动以管理员身份登录
    if app.config.get("SCREENSHOT_MODE"):
        if (request.endpoint not in ("auth.login",
"static")
                and not current_user.is_authenticated):
            admin = User.query.filter_by(role="admin").f
irst()
            if admin:
                login_user(admin)
        return None
    if request.endpoint in PUBLIC_ENDPOINTS or
request.endpoint is None:
        return None
    if not current_user.is_authenticated:
        return redirect(url_for("auth.login"))
    return None
#  侧边栏导航：按论文评价流程的顺序编排，分为「评价流程」与「系统管理」两组。
#  不使用模块编号，避免出现与实际流程不一致的代号。
#  每项带图标名（内联 SVG，见 templates/_macros.html 的 ICONS）
#  与可选的待办角标（该步骤还没做完的图斑数）。
FLOW_LINKS = [
    ("项目与矿山档案", "/project/", "folder"),
    ("政策与标准知识库", "/standard/", "book"),
    ("限制性指标判定", "/constraint/", "shield"),
    ("微地质单元", "/unit/", "layers"),
    ("指标体系与权重", "/ahp/", "sliders"),
    ("指标数据采集", "/indicator/", "database"),
    ("适宜性定量评价", "/evaluate/", "target"),
    ("报告与结果输出", "/report/", "file"),
]
ADMIN_LINKS = [
    ("用户与操作日志", "/users", "users"),
]
@app.context_processor
def inject_nav():
    path = request.path or "/"
    def hit(prefix):
        """当前路径是否属于某个入口（/unit/ 也匹配 /unit/3）。"""
        return path.startswith(prefix.rstrip("/"))
    badges = {}
    if current_user.is_authenticated:
        from app.services import dashboard as dash
        badges = dash.nav_badges()
    def build(links):
        return [{"label": label, "href": href, "icon":
icon,
                 "active": hit(href), "badge":
badges.get(href, 0)}
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                    for label, href, icon in links]
    groups = [
        {"title": "评价流程", "links": build(FLOW_LINKS)},
        {"title": "系统管理", "links": build(ADMIN_LINKS)},
    ]
    # 面包屑用的当前页名：按路径命中的入口取，未命中时给个通用名
    if path == "/":
        page_title = "工作台"
    elif path.startswith(("/users", "/logs")):
        page_title = "用户与操作日志"
    elif path.startswith("/search"):
        page_title = "搜索结果"
    else:
        page_title = "工作台"
        for label, href, _icon in FLOW_LINKS:
            if hit(href):
                page_title = label
                break
    # 层级引导用：项目是归档单位，图斑才是评价对象。
    # 下游各页面按图斑组织数据，因此需要提示「哪些项目下还没有图斑」。
    from app.models import Project, MinePatch
    projects = Project.query.order_by(Project.id).all()
    return {
        "nav_groups": groups,
        "home_active": path == "/",
        "page_title": page_title,
        "today": date.today().strftime("%Y-%m-%d"),
        "search_q": request.args.get("q", ""),
        "role_labels": ROLE_LABELS,
        "project_count": len(projects),
        "patch_count": MinePatch.query.count(),
        "projects_without_patch": [p for p in projects
                                   if p.patches.count()
== 0],
    }
# 操作审计：统一在响应阶段记录成功的写操作（操作日志）
from app.services import audit
@app.after_request
def audit_writes(response):
    if request.method in ("POST", "PUT", "PATCH",
"DELETE"):
        audit.record(request.endpoint,
response.status_code)
    return response
# 错误页：给出可读提示，而不是默认的英文错误页
from flask import render_template
@app.errorhandler(403)
def forbidden(e):
    return render_template("error.html", code=403,
                           message="没有权限执行该操作"), 403
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    @app.errorhandler(404)
    def not_found(e):
        return render_template("error.html", code=404,
                               message="请求的页面不存在"), 404
    return app
# ==== app/config.py ====
# -*- coding: utf-8 -*-
"""应用配置。"""
import os
from datetime import timedelta
BASE_DIR = os.path.abspath(os.path.dirname(__file__))
class Config:
    SECRET_KEY = os.environ.get("SECRET_KEY",
    "mine-restoration-dev-key")
    # 单机部署，SQLite 数据库文件放在应用目录下
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE_DIR, "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    # 禁止跨站请求伪造（CSRF），登录与表单提交依赖此项
    WTF_CSRF_ENABLED = True
    # 「记住我」：登录时勾选则下发持久 cookie，关闭浏览器后仍可自动登录
    REMEMBER_COOKIE_DURATION = timedelta(days=30)
    REMEMBER_COOKIE_HTTPONLY = True
    REMEMBER_COOKIE_SAMESITE = "Lax"
    # 影像与数据文件上传（遥感影像挂接、批量导入）
    UPLOAD_FOLDER = os.path.join(BASE_DIR, "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024          # 单次上传上限
    32 MB
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
    ALLOWED_DATA_EXT = {"csv", "xlsx", "xls"}
# ==== app/models.py ====
# -*- coding: utf-8 -*-
"""数据模型：系统全部数据表。
本文件由甲在阶段 0 一次性写完，作为两人独立开发的地基。
约定：两人只读此文件；确需改表时，由乙提需求、甲统一修改。
"""
import json
from datetime import datetime
from flask_login import UserMixin
from flask_sqlalchemy import SQLAlchemy
db = SQLAlchemy()
class User(UserMixin, db.Model):
    """用户（用户与权限管理）。"""
    __tablename__ = "users"
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(64), unique=True,
    nullable=False, index=True)
    password_hash = db.Column(db.String(256),
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    nullable=False)                                                    251
    # 角色：管理员 admin / 评价员 evaluator / 查看者 viewer              252
    role = db.Column(db.String(20), nullable=False,                    253
    default="viewer")                                                  254
    active = db.Column(db.Boolean, nullable=False,                     255
    default=True)                                                      256
    created_at = db.Column(db.DateTime,                                257
    default=datetime.utcnow)                                           258
    def to_dict(self):                                                 259
        return {"id": self.id, "username": self.username,              260
                "role": self.role, "active": self.active}              261
class OperationLog(db.Model):                                          262
    """操作日志。"""                                                   263
    __tablename__ = "operation_logs"                                   264
    id = db.Column(db.Integer, primary_key=True)                       265
    user_id = db.Column(db.Integer, db.ForeignKey("users.id"           266
    ), nullable=True)                                                  267
    action = db.Column(db.String(128), nullable=False)                 268
    # 操作类型                                                         269
    target_type = db.Column(db.String(64), nullable=True)              270
    # 对象类型                                                         271
    target_id = db.Column(db.Integer, nullable=True)                   272
    # 对象主键                                                         273
    detail = db.Column(db.Text, nullable=True)                         274
    # 操作详情                                                         275
    created_at = db.Column(db.DateTime,                                276
    default=datetime.utcnow, index=True)                               277
class Project(db.Model):                                               278
    """评价项目（项目与矿山档案）。"""                                 279
    __tablename__ = "projects"                                         280
    id = db.Column(db.Integer, primary_key=True)                       281
    name = db.Column(db.String(128), nullable=False)                   282
    code = db.Column(db.String(64), nullable=True)                     283
    # 项目编号                                                         284
    description = db.Column(db.Text, nullable=True)                    285
    status = db.Column(db.String(20), default="进行中")                286
    # 进行中 / 已完成 / 已归档                                         287
    created_by = db.Column(db.Integer,                                 288
    db.ForeignKey("users.id"), nullable=True)                          289
    created_at = db.Column(db.DateTime,                                290
    default=datetime.utcnow)                                           291
    patches = db.relationship("MinePatch",                             292
    backref="project", lazy="dynamic",                                 293
                                  cascade="all, delete-orphan")        294
class MinePatch(db.Model):                                             295
    """矿山图斑。"""                                                   296
    __tablename__ = "mine_patches"                                     297
    id = db.Column(db.Integer, primary_key=True)                       298
    project_id = db.Column(db.Integer,                                 299
    db.ForeignKey("projects.id"), nullable=False)                      300
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    name = db.Column(db.String(128), nullable=False)
    admin_region = db.Column(db.String(128), nullable=True)
    # 所在行政区
    center_x = db.Column(db.Float, nullable=True)
    # 中心经度 / 横坐标
    center_y = db.Column(db.Float, nullable=True)
    # 中心纬度 / 纵坐标
    area_ha = db.Column(db.Float, nullable=True)
    # 图斑面积（公顷）
    mineral_type = db.Column(db.String(64), nullable=True)
    # 开采矿种
    mining_method = db.Column(db.String(64), nullable=True)
    # 开采方式
    stop_years = db.Column(db.Float, nullable=True)
    # 停产年限
    responsible_entity_missing = db.Column(db.Boolean,
    default=True)  # 责任主体是否缺失
    # 遥感底图 / 无人机正射影像挂接
    image_path = db.Column(db.String(256), nullable=True)
    # 相对 static 的路径
    image_note = db.Column(db.String(256), nullable=True)
    # 影像说明（如来源、时相）
    image_uploaded_at = db.Column(db.DateTime,
    nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    units = db.relationship("MicroUnit", backref="patch",
    lazy="dynamic",
                            cascade="all, delete-orphan")
class ReferenceEcosystem(db.Model):
    """参照生态系统：标准化基准值，与图斑绑定。"""
    __tablename__ = "reference_ecosystems"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    coverage_benchmark = db.Column(db.Float, default=75.0)
    # 植被覆盖度基准 %
    shannon_benchmark = db.Column(db.Float, default=0.7)
    # Shannon-Wiener 基准
    dominant_species = db.Column(db.String(256),
    nullable=True)  # 建群种
    soil_benchmark = db.Column(db.String(256),
    nullable=True)    # 土壤基准
class Standard(db.Model):
    """地方标准 / 政策文件（政策与标准知识库）。"""
    __tablename__ = "standards"
    id = db.Column(db.Integer, primary_key=True)
    province = db.Column(db.String(32), nullable=False)
    # 省份
    name = db.Column(db.String(256), nullable=False)
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    # 名称                                                                    351
    doc_number = db.Column(db.String(128), nullable=True)                     352
    # 标准编号 / 文号                                                         353
    publish_date = db.Column(db.String(32), nullable=True)                    354
    implement_date = db.Column(db.String(32), nullable=True)                  355
    # 类别：地方标准 / 政策文件                                              356
    category = db.Column(db.String(32), nullable=False,                       357
    default="地方标准")                                                       358
    summary = db.Column(db.Text, nullable=True)                               359
    # 特点概述                                                                360
    rules = db.relationship("StandardRule",                                   361
    backref="standard",                                                       362
                                cascade="all, delete-orphan")                 363
class StandardRule(db.Model):                                                 364
    """结构化规则：将标准条文拆解为可执行规则。"""                            365
    __tablename__ = "standard_rules"                                          366
    id = db.Column(db.Integer, primary_key=True)                              367
    standard_id = db.Column(db.Integer,                                       368
    db.ForeignKey("standards.id"), nullable=False)                            369
    item = db.Column(db.String(128), nullable=False)                          370
    # 判定项                                                                  371
    operator = db.Column(db.String(8), nullable=False)                        372
    # 比较符：> >= < <= ==                                                    373
    threshold = db.Column(db.String(64), nullable=True)                       374
    # 阈值                                                                    375
    condition_type = db.Column(db.String(32),                                 376
    default="必要条件")   # 必要条件 / 充分条件                              377
    region = db.Column(db.String(64), nullable=True)                          378
    # 适用分区（宁夏分区差异化）                                             379
    note = db.Column(db.Text, nullable=True)                                  380
class ConstraintCheck(db.Model):                                              381
    """限制性判定记录。"""                                                    382
    __tablename__ = "constraint_checks"                                       383
    id = db.Column(db.Integer, primary_key=True)                              384
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat                  385
    ches.id"), nullable=False)                                                386
    policy_pass = db.Column(db.Boolean, nullable=True)                        387
    # 政策约束                                                                388
    location_pass = db.Column(db.Boolean, nullable=True)                      389
    # 区位约束                                                                390
    pollution_pass = db.Column(db.Boolean, nullable=True)                     391
    # 土壤污染                                                                392
    geo_pass = db.Column(db.Boolean, nullable=True)                           393
    # 地质灾害                                                                394
    farmland_pass = db.Column(db.Boolean, nullable=True)                      395
    # 耕地属性                                                                396
    passed = db.Column(db.Boolean, nullable=True)                             397
    # 总体是否通过                                                            398
    reason = db.Column(db.Text, nullable=True)                                399
    # 不通过原因                                                              400
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    checked_at = db.Column(db.DateTime,                                  401
    default=datetime.utcnow)                                             402
class MicroUnit(db.Model):                                               403
    """微地质单元：岩质边坡 / 碎石边坡 / 平台。"""                                  404
    __tablename__ = "micro_units"                                        405
    id = db.Column(db.Integer, primary_key=True)                         406
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat             407
    ches.id"), nullable=False)                                           408
    unit_type = db.Column(db.String(32), nullable=False)                 409
    # 单元类型                                                           410
    area = db.Column(db.Float, nullable=True)                            411
    # 面积                                                               412
    ratio = db.Column(db.Float, nullable=True)                           413
    # 面积占比（自动计算）                                                 414
    description = db.Column(db.Text, nullable=True)                      415
    # 现场描述                                                           416
class Indicator(db.Model):                                               417
    """指标定义：指标树，含角色与启用单元类型。"""                                   418
    __tablename__ = "indicators"                                         419
    id = db.Column(db.Integer, primary_key=True)                         420
    code = db.Column(db.String(32), unique=True,                         421
    nullable=False)  # 指标编码                                          422
    name = db.Column(db.String(64), nullable=False)                      423
    # 指标名称                                                           424
    level = db.Column(db.Integer, nullable=False)                        425
    # 1/2/3 级                                                           426
    parent_code = db.Column(db.String(32), nullable=True)                427
    # 父指标编码                                                         428
    # 角色：土壤污染指标同时作为限制性指标与评价指标                                  429
    is_constraint = db.Column(db.Boolean, default=False)                 430
    is_evaluative = db.Column(db.Boolean, default=True)                  431
    unit = db.Column(db.String(32), nullable=True)                       432
    # 单位                                                               433
    ideal_value = db.Column(db.String(64), nullable=True)                434
    # 理想值（可配置）                                                     435
    ideal_type = db.Column(db.String(32), nullable=True)                 436
    # benefit/cost/mid/discrete                                          437
    # 启用该指标的微地质单元（逗号分隔）。岩质边坡不启用土壤指标、平台不启用地形指标                    438
    enabled_units = db.Column(db.String(128), nullable=True)             439
    description = db.Column(db.Text, nullable=True)                      440
    # 取值依据说明                                                       441
    def units_list(self):                                                442
        return [s for s in (self.enabled_units or                        443
    "").split(",") if s]                                                 444
class WeightScheme(db.Model):                                            445
    """权重方案：按微地质单元类型分别保存。"""                                      446
    __tablename__ = "weight_schemes"                                     447
    id = db.Column(db.Integer, primary_key=True)                         448
    unit_type = db.Column(db.String(32), nullable=False)                 449
    # 所属单元类型                                                       450
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    name = db.Column(db.String(64), nullable=True)
    # 方案名称
    level = db.Column(db.Integer, nullable=False,
    default=1)          # 1=一级指标 2=二级指标
    is_default = db.Column(db.Boolean, default=False)
    # 是否默认（论文表 3）
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class WeightItem(db.Model):
    """权重明细：方案内的各项权重。"""
    __tablename__ = "weight_items"
    id = db.Column(db.Integer, primary_key=True)
    scheme_id = db.Column(db.Integer,
    db.ForeignKey("weight_schemes.id"), nullable=False)
    indicator_code = db.Column(db.String(32),
    nullable=False)        # 指标编码
    weight = db.Column(db.Float, nullable=False)
    # 权重值
class ExpertMatrix(db.Model):
    """专家判断矩阵：多专家打分的原始矩阵。"""
    __tablename__ = "expert_matrices"
    id = db.Column(db.Integer, primary_key=True)
    scheme_id = db.Column(db.Integer,
    db.ForeignKey("weight_schemes.id"), nullable=True)
    # 直接标注适用对象，便于无需先建权重方案即可录入专家矩阵
    unit_type = db.Column(db.String(32), nullable=True)
    # 微地质单元类型
    l1_name = db.Column(db.String(64), nullable=True)
    # 所属一级指标（二级矩阵用）
    expert_name = db.Column(db.String(64), nullable=True)
    # 专家姓名
    level = db.Column(db.Integer, nullable=False,
    default=1)          # 矩阵对应指标层级
    matrix_json = db.Column(db.Text, nullable=False)
    # 判断矩阵（JSON 二维数组）
    cr = db.Column(db.Float, nullable=True)
    # 该专家矩阵的一致性比例
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class IndicatorData(db.Model):
    """指标实测数据：按微地质单元组织。"""
    __tablename__ = "indicator_data"
    id = db.Column(db.Integer, primary_key=True)
    unit_id = db.Column(db.Integer, db.ForeignKey("micro_uni
    ts.id"), nullable=False, unique=True)
    soil_ratio = db.Column(db.Float, nullable=True)
    # 土石比
    thickness = db.Column(db.Float, nullable=True)
    # 土层厚度 cm
    ph = db.Column(db.Float, nullable=True)
```

```
    organic_matter = db.Column(db.Float, nullable=True)
    # 有机质 g/kg
    # 重金属含量，JSON 字符串：{"Hg": 0.05, ...}，未检测项缺失或为 null
    heavy_metals = db.Column(db.Text, nullable=True)
    coverage = db.Column(db.Float, nullable=True)
    # 植被覆盖度 %
    shannon = db.Column(db.Float, nullable=True)
    # Shannon-Wiener 多样性指数
    slope = db.Column(db.Float, nullable=True)
    # 坡度 °
    aspect = db.Column(db.String(16), nullable=True)
    # 坡向：阳坡/阴阳坡/阴坡
    def heavy_metals_dict(self):
        return json.loads(self.heavy_metals) if
    self.heavy_metals else {}
class EvaluationResult(db.Model):
    """评价结果。"""
    __tablename__ = "evaluation_results"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    final_score = db.Column(db.Float, nullable=True)
    # 最终评价因子 F
    grade = db.Column(db.String(32), nullable=True)
    # 适宜性等级
    measures = db.Column(db.Text, nullable=True)
    # 推荐措施
    unit_scores = db.Column(db.Text, nullable=True)
    # 各单元得分 JSON
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class ScoreDetail(db.Model):
    """得分明细：每一层指标的权重、标准化值、加权贡献，供追溯与复核。"""
    __tablename__ = "score_details"
    id = db.Column(db.Integer, primary_key=True)
    result_id = db.Column(db.Integer,
    db.ForeignKey("evaluation_results.id"), nullable=False)
    unit_id = db.Column(db.Integer, nullable=True)
    l1_name = db.Column(db.String(64), nullable=True)
    # 一级指标
    l2_name = db.Column(db.String(64), nullable=True)
    # 二级指标
    raw_value = db.Column(db.String(64), nullable=True)
    # 实测值
    std_value = db.Column(db.Float, nullable=True)
    # 标准化值
    weight = db.Column(db.Float, nullable=True)
    # 权重
    contribution = db.Column(db.Float, nullable=True)
    # 加权贡献
```

```
# ==== app/forms.py ====
# -*- coding: utf-8 -*-
"""表单定义（共享，由甲统一维护）。
各模块的 Flask-WTF 表单集中于此。
"""
from flask_wtf import FlaskForm
from wtforms import SelectField, SubmitField, StringField,
    PasswordField, BooleanField
from wtforms.validators import DataRequired, Length
from app.utils.constants import UNIT_TYPES, ASPECT_TYPES
class AHPSelectForm(FlaskForm):
    """选择微地质单元类型与指标层级，以确定判断矩阵规模。"""
    unit_type = SelectField("微地质单元类型",
    validators=[DataRequired()])
    level = SelectField("指标层级",
                        choices=[("1", "一级指标"), ("2",
    "二级指标")],
                        validators=[DataRequired()])
    l1 = SelectField("一级指标", validators=[],
    validate_choice=False)  # 动态填充，层级=2 时使用
    submit = SubmitField("生成判断矩阵")
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        self.unit_type.choices = [(t, t) for t in
    UNIT_TYPES]
class EmptyForm(FlaskForm):
    """仅用于承载 CSRF 令牌的空表单（矩阵录入等自定义表单使用）。"""
    pass
class LoginForm(FlaskForm):
    """登录表单。支持「记住我」以在关闭浏览器后保持登录状态。"""
    username = StringField("用户名", validators=[DataRequired()
    , Length(max=64)])
    password = PasswordField("密码", validators=[DataRequired(
    )])
    remember = BooleanField("记住我")
    submit = SubmitField("登录")
# ==== app/utils/constants.py ====
# -*- coding: utf-8 -*-
"""评价指标常量表。
全部数值取自论文，作为系统计算的唯一基准：
- 论文 4.2.2 节：指标标准化理想极值与取值依据
- 论文 4.2.3 节 / 表 4：适宜性等级划分
- 论文表 3：按微地质单元分别确定的一级、二级指标权重
- 论文 4.2.1 节：平均随机一致性指标 RI
"""
# ------------------------------------------------------------------
    ----------------
# 一、微地质单元类型
# ------------------------------------------------------------------
    ----------------
```

```
#  论文  4．1  节："按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡
#  三种地质微单元类型"
UNIT_TYPES  =  ("岩质边坡",  "碎石边坡",  "平台")
#  每个单元类型启用的指标类别（论文表  3  隐含：岩质边坡无土壤指标，平台无地形指标）
#  供数据录入界面与评价引擎判断"该单元是否需要某类指标"
UNIT_ENABLED_INDICATORS  =  {
      "岩质边坡":  ("植被指标",  "地形指标"),                    #  不设土壤指标
      "碎石边坡":  ("土壤指标",  "植被指标",  "地形指标"),
      "平台":  ("土壤指标",  "植被指标"),                          #  不设地形指标
}
#  -----------------------------------------------------------------------
      ------------------
#  二、指标标准化理想极值（论文  4．2．2  节）
#  -----------------------------------------------------------------------
      ------------------
IDEAL_SOIL_RATIO  =  0．8            #  土石比理想值
IDEAL_THICKNESS  =  20．0            #  土层厚度理想值  （cm）
IDEAL_PH_RANGE  =  (5．0,  9．0)     #  pH  理想区间（适中型指标）
IDEAL_ORGANIC  =  30．0              #  有机质理想值
      (g/kg)，全国第二次土壤普查养分等级划分标准二等最低值
IDEAL_COVERAGE  =  75．0             #  植被覆盖度参照生态系统  (%)
IDEAL_SHANNON  =  0．7               #  Shannon-Wiener  多样性指数参照生态系统
#  重金属极值  (mg/kg)，《绿化种植土壤》CJ/T  340—2016  土壤重金属含量  IV  级最高值
#  论文原文：镉、汞、铅、铬、砷、镍、铜、锌分别为  2、2、530、400、45、220、600、800
HEAVY_METALS  =  ("Cd",  "Hg",  "Pb",  "Cr",  "As",  "Ni",  "Cu",
      "Zn")
HEAVY_METAL_LIMIT  =  {
      "Cd":  2．0,  "Hg":  2．0,  "Pb":  530．0,  "Cr":  400．0,
      "As":  45．0,  "Ni":  220．0,  "Cu":  600．0,  "Zn":  800．0,
}
#  污染指标下  8  种重金属"重要性相当，权重平均分配"（论文  4．2．1  节）
HEAVY_METAL_EQUAL_WEIGHT  =  1．0  /  8．0
#  坡度分界，论文  4．2．2  节引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772  号
SLOPE_BREAK  =  25．0
#  坡向离散赋值，论文  4．2．2  节："阳坡、阴阳坡、阴坡分别赋值  0．2、0．6、1"
ASPECT_SCORES  =  {"阳坡":  0．2,  "阴阳坡":  0．6,  "阴坡":  1．0}
ASPECT_TYPES  =  ("阳坡",  "阴阳坡",  "阴坡")
#  pH  适中型标准化的容差边界（区间外线性衰减，避免得分突变）
PH_TOL_LOW  =  4．0
PH_TOL_HIGH  =  10．0
#  -----------------------------------------------------------------------
      ------------------
#  三、适宜性等级划分（论文表  4）
#  -----------------------------------------------------------------------
      ------------------
GRADE_LEVELS  =  (
      {"grade":  "不适宜",  "min":  None,  "max":  0．3,
        "meaning":  "存在限制性条件或区域生态本底较差，自然恢复影响因子严重制约自然恢复",
        "measure":  "生态重建、人工辅助"},
      {"grade":  "基本适宜",  "min":  0．3,  "max":  0．7,
```

```
        "meaning": "生态本底良好，核心自然恢复影响因子基本满足自然恢复需求",
        "measure": "自然恢复为主，可辅以封闭场地和少量补种、补植、飞播等"},
    {"grade": "适宜", "min": 0.7, "max": None,
        "meaning": "各自然恢复影响因子均能满足自然恢复需求，自然演替速度快",
        "measure": "自然恢复"},
)
def grade_of(score):
    """按论文表 4 判定适宜性等级。"""
    for g in GRADE_LEVELS:
        lo, hi = g["min"], g["max"]
        if lo is None and score < hi:
            return g
        if hi is None and score >= lo:
            return g
        if lo is not None and hi is not None and lo <=
    score < hi:
            return g
    # score 恰好等于 0.7 时落入"适宜"
    return GRADE_LEVELS[-1]
# ---------------------------------------------------------------------
    ------------------
# 四、平均随机一致性指标 RI（论文 4.2.1 节给出 n=2,3,4,5，其余为常用扩展值）
# ---------------------------------------------------------------------
    ------------------
RI_TABLE = {1: 0.0, 2: 0.0, 3: 0.58, 4: 0.90, 5: 1.12, 6:
    1.24,
            7: 1.32, 8: 1.41, 9: 1.45, 10: 1.49}
# 一致性比例阈值（论文：当 CR<0.1 时，判断矩阵满足一致性要求）
CR_THRESHOLD = 0.1
# ---------------------------------------------------------------------
    ------------------
# 五、默认权重方案（论文表 3）
# ---------------------------------------------------------------------
    ------------------
# 论文 4.2.2 节："划分岩质边坡、碎石土边坡、平台三类微地质单元开展独立评价，
# 分别计算评价指标权重"（论文摘要、图 2 流程与表 5 均作"碎石边坡"，
# 系统统一采用"碎石边坡"）。权重由 10 位专家打分后经 AHP 方法确定。
# 下表为论文表 3 所载的归一化权重，作为系统默认方案。
DEFAULT_WEIGHTS = {
    "岩质边坡": {
        "l1": {"植被指标": 0.45, "地形指标": 0.55},
        "l2": {
            "植被指标": {"植被覆盖度": 0.67, "生物多样性": 0.33},
            "地形指标": {"坡度": 0.64, "坡向": 0.36},
        },
    },
    "碎石边坡": {
        "l1": {"土壤指标": 0.44, "植被指标": 0.31, "地形指标": 0.25},
        "l2": {
            "土壤指标": {"土石比": 0.25, "土层厚度": 0.25, "pH": 0.17,
```

```
                                        "有机质": 0.17, "污染指标": 0.17},
                    "植被指标": {"植被覆盖度": 0.64, "生物多样性": 0.36},
                    "地形指标": {"坡度": 0.67, "坡向": 0.33},
                },
            },
            "平台": {
                "l1": {"土壤指标": 0.54, "植被指标": 0.46},
                "l2": {
                    "土壤指标": {"土石比": 0.25, "土层厚度": 0.25, "pH": 0.17,
                                 "有机质": 0.17, "污染指标": 0.17},
                    "植被指标": {"植被覆盖度": 0.61, "生物多样性": 0.39},
                },
            },
        }
        # ---------------------------------------------------------------------
        # 六、限制性指标（论文表 2 + 4.1 节流程）
        # ---------------------------------------------------------------------
        # 论文摘要与 4.1 节：四项限制性指标；流程段另明确"是否为耕地"为约束条件
        CONSTRAINT_ITEMS = ("政策", "区位约束", "土壤污染", "地质灾害", "耕地属性")
        # ---------------------------------------------------------------------
        # 七、用户角色
        # ---------------------------------------------------------------------
        ROLE_LABELS = {"admin": "管理员", "evaluator": "评价员",
            "viewer": "查看者"}
        # ==== app/utils/schema.py ====
        # -*- coding: utf-8 -*-
        """轻量结构迁移。
        SQLite 下 `db.create_all()` 只建缺失的表，不会给已有表补列。
        本模块在启动时检查目标列是否存在，缺失则 ALTER TABLE 补上，
        使旧数据库无需重建即可用上新字段。
        """
        from sqlalchemy import text
        # (表名, 列名, 列定义)
        REQUIRED_COLUMNS = [
            # 遥感影像挂接
            ("mine_patches", "image_path", "VARCHAR(256)"),
            ("mine_patches", "image_note", "VARCHAR(256)"),
            ("mine_patches", "image_uploaded_at", "DATETIME"),
            # 指标维护
            ("indicators", "enabled_units", "VARCHAR(128)"),
            ("indicators", "description", "TEXT"),
            # 多专家矩阵
            ("expert_matrices", "unit_type", "VARCHAR(32)"),
            ("expert_matrices", "l1_name", "VARCHAR(64)"),
            ("expert_matrices", "cr", "FLOAT"),
        ]
```

```
def ensure_schema(db):
    """为已有表补齐缺失列。返回补列清单。"""
    added = []
    try:
        with db.engine.connect() as conn:
            for table, column, ddl in REQUIRED_COLUMNS:
                rows = conn.execute(text("PRAGMA
    table_info(%s)" % table)).fetchall()
                if not rows:
                    continue                              #
    表还不存在，create_all 会建
                existing = {r[1] for r in rows}
                if column in existing:
                    continue
                conn.execute(text(
                    "ALTER TABLE %s ADD COLUMN %s %s" %
    (table, column, ddl)))
                conn.commit()
                added.append("%s.%s" % (table, column))
    except Exception as e:                                #
    迁移失败不应阻断启动
        print("[schema] 结构检查跳过：%s" % e)
    return added
# ==== app/utils/uploads.py ====
# -*- coding: utf-8 -*-
"""上传文件处理（遥感影像、批量导入）。"""
import os
import re
import uuid
from datetime import datetime
from werkzeug.utils import secure_filename
def _safe_ext(filename):
    return os.path.splitext(filename)[1].lower().lstrip(".")
def ext_allowed(filename, allowed):
    return _safe_ext(filename) in allowed
def save_upload(file_storage, subdir, allowed_ext,
    upload_root):
    """保存上传文件，返回相对 static 的路径（如 uploads/patches/xxx.png）。
    文件名重命名为「时间戳_随机串.扩展名」，避免中文名与重名问题。
    """
    if file_storage is None or not file_storage.filename:
        return None, "未选择文件"
    ext = _safe_ext(file_storage.filename)
    if ext not in allowed_ext:
        return None, "不支持的文件类型：.%s" % ext
    target_dir = os.path.join(upload_root, subdir)
    os.makedirs(target_dir, exist_ok=True)
    # 仅保留安全字符，并加时间戳与随机串防重名
    base = secure_filename(os.path.splitext(file_storage.fil
    ename)[0]) or "file"
```

```
    base = re.sub(r"[^A-Za-z0-9_\-]", "", base)[:40] or
    "file"
    name = "%s_%s.%s" % (datetime.now().strftime("%Y%m%d%H%M
    %S"),
                         uuid.uuid4().hex[:6], ext)
    abs_path = os.path.join(target_dir, name)
    file_storage.save(abs_path)
    rel = os.path.join("uploads", subdir,
    name).replace("\\", "/")
    return rel, None
def remove_upload(rel_path, upload_root):
    """删除已保存的上传文件（相对 static 的路径）。"""
    if not rel_path:
        return
    abs_path = os.path.join(upload_root,
    rel_path.replace("uploads/", "", 1))
    try:
        if os.path.isfile(abs_path):
            os.remove(abs_path)
    except OSError:
        pass
# ==== app/utils/charts.py ====
# -*- coding: utf-8 -*-
"""轻量图表。
用纯 SVG 输出，不依赖任何绘图库或前端脚本：
- 服务端渲染，页面与打印（PDF）中都能稳定显示
- 每个图元都带数值标签，黑白打印时仍可读（不单靠颜色区分）
- 单色方案，符合界面「一个视图一个强调色」的约定
"""
from html import escape
ACCENT = "#1f3a5f"
ACCENT_SOFT = "#7f93ad"
INK = "#23262b"
MUTED = "#6b7280"
LINE = "#dfe3e8"
# 灰度梯度：黑白打印时相邻扇区 / 条柱仍可区分
GRAYS = ["#1f3a5f", "#456188", "#6d86a8", "#96a9c4",
    "#bcc9da", "#dbe3ec"]
def _fmt(v, digits=4):
    try:
        return ("%%.%df" % digits) % float(v)
    except (TypeError, ValueError):
        return str(v)
def hbars(items, width=620, bar_h=22, gap=12, label_w=150,
    value_w=76):
    """水平条形图。items: [(标签, 数值, 显示文本)]"""
    if not items:
        return '<p class="muted">暂无数据</p>'
    n = len(items)
    height = n * (bar_h + gap) + 10
```

```
    max_v = max((abs(float(v)) for _, v, _ in items),
    default=1) or 1
    body_w = width - label_w - value_w
    parts = ['<svg viewBox="0 0 %d %d" width="100%%"
    height="%d" '
             'role="img" aria-label="条形图" '
             'style="max-width:%dpx;font-family:inherit;">'
    % (width, height, height, width)]
    for i, (label, value, disp) in enumerate(items):
        y = i * (bar_h + gap) + 5
        frac = (abs(float(value)) / max_v) if max_v else 0
        w = max(2.0, body_w * frac)
        fill = GRAYS[i % len(GRAYS)]
        parts.append(
            '<text x="%d" y="%d" text-anchor="end"
    font-size="12" fill="%s">%s</text>'
            % (label_w - 10, y + bar_h * 0.72, INK,
    escape(str(label))))
        parts.append(
            '<rect x="%d" y="%d" width="%.1f" height="%d"
    rx="3" fill="%s"/>'
            % (label_w, y, w, bar_h, fill))
        parts.append(
            '<text x="%d" y="%d" font-size="12" fill="%s" '
            'style="font-variant-numeric:tabular-nums;">%s</
    text>'
            % (label_w + w + 8, y + bar_h * 0.72, MUTED,
    escape(str(disp))))
    parts.append('</svg>')
    return "".join(parts)
def donut(items, size=210, thickness=34):
    """环形图（权重分布）。items: [(标签, 数值, 显示文本)]
    图例中给出数值与占比，黑白打印时靠图例与灰度深浅区分。
    """
    vals = [(l, float(v), d) for l, v, d in items if
    float(v) > 0]
    if not vals:
        return '<p class="muted">暂无数据</p>'
    total = sum(v for _, v, _ in vals) or 1
    r = (size - thickness) / 2
    cx = cy = size / 2
    import math
    parts = ['<svg viewBox="0 0 %d %d" width="%d"
    height="%d" role="img" '
             'aria-label="权重分布环形图">' % (size, size, size,
    size)]
    angle = -math.pi / 2
    for i, (label, v, disp) in enumerate(vals):
        sweep = 2 * math.pi * (v / total)
        a0, a1 = angle, angle + sweep
```

```
        x0, y0 = cx + r * math.cos(a0), cy + r *                    901
    math.sin(a0)                                                    902
        x1, y1 = cx + r * math.cos(a1), cy + r *                    903
    math.sin(a1)                                                    904
        large = 1 if sweep > math.pi else 0                         905
        parts.append(                                               906
            '<path d="M %.2f %.2f A %.2f %.2f 0 %d 1 %.2f           907
    %.2f" '                                                         908
            'fill="none" stroke="%s" stroke-width="%d"/>'           909
           % (x0, y0, r, r, large, x1, y1, GRAYS[i %                910
    len(GRAYS)], thickness))                                        911
        angle = a1                                                  912
    parts.append('<text x="%d" y="%d" text-anchor="middle"          913
    font-size="12" '                                                914
                 'fill="%s">合计</text>' % (cx, cy - 2,             915
    MUTED))                                                         916
    parts.append('<text x="%d" y="%d" text-anchor="middle"          917
    font-size="15" '                                                918
                 'fill="%s" style="font-variant-numeric:tabu        919
    lar-nums;">%s</text>'                                           920
                 % (cx, cy + 18, INK, _fmt(total, 4)))              921
    parts.append('</svg>')                                          922
    legend = ['<table style="width:auto;margin-top:10px;"><t        923
    head><tr>'                                                      924
              '<th scope="col">项目</th><th                         925
    scope="col">数值</th>'                                          926
              '<th scope="col">占比</th></tr></thead><tbody>']      927
    for i, (label, v, disp) in enumerate(vals):                     928
        legend.append(                                              929
            '<tr><td style="text-align:left;">'                     930
            '<span style="display:inline-block;width:10px;he        931
    ight:10px;'                                                     932
            'border-radius:2px;background:%s;margin-right:6p        933
    x;"></span>%s</td>'                                             934
            '<td class="mono">%s</td><td                            935
    class="mono">%.1f%%</td></tr>'                                  936
           % (GRAYS[i % len(GRAYS)], escape(str(label)),            937
    escape(str(disp)),                                              938
               100.0 * v / total))                                  939
    legend.append('</tbody></table>')                               940
    return ('<div style="display:flex;flex-wrap:wrap;gap:20p        941
    x;align-items:center;">'                                        942
            + "".join(parts) + "".join(legend) + '</div>')          943
# ==== app/utils/browser.py ====                                    944
# -*- coding: utf-8 -*-                                             945
"""HTML 转 PDF（报告导出）。                                        946
复用本机已安装的 Chrome/Edge 无头模式打印，避免引入 reportlab、     947
weasyprint 等额外依赖。生成过程使用系统临时目录中的 ASCII 文件名，  948
避免中文路径在 file:// URL 下出问题。                               949
"""                                                                 950
```

```
import os
import shutil
import subprocess
import tempfile
import uuid
CANDIDATES = [
    r"C:\Program Files\Google\Chrome\Application\chrome.exe",
    r"C:\Program Files (x86)\Google\Chrome\Application\chrome.exe",
    r"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe",
    r"C:\Program Files\Microsoft\Edge\Application\msedge.exe",
]
# 允许通过环境变量覆盖，便于在其他机器上指定浏览器
ENV_KEY = "DSH_BROWSER_PATH"
def browser_path():
    """定位可用的浏览器可执行文件，找不到返回 None。"""
    override = os.environ.get(ENV_KEY)
    if override and os.path.isfile(override):
        return override
    for p in CANDIDATES:
        if os.path.isfile(p):
            return p
    for name in ("chrome", "msedge", "chromium"):
        found = shutil.which(name)
        if found:
            return found
    return None
def html_to_pdf(html, timeout=90):
    """把 HTML 字符串渲染为 PDF，返回字节；失败返回 (None, 原因)。"""
    exe = browser_path()
    if exe is None:
        return None, ("未找到可用的浏览器（Chrome/Edge），无法生成 PDF。"
                      "可设置环境变量 %s 指向浏览器可执行文件。" % ENV_KEY)
    tmpdir = tempfile.mkdtemp(prefix="dshpdf_")
    tag = uuid.uuid4().hex[:8]
    html_path = os.path.join(tmpdir, "r_%s.html" % tag)
    pdf_path = os.path.join(tmpdir, "r_%s.pdf" % tag)
    try:
        with open(html_path, "w", encoding="utf-8") as f:
            f.write(html)
        cmd = [
            exe, "--headless=new", "--disable-gpu",
    "--no-sandbox",
            "--no-first-run", "--no-default-browser-check",
            "--no-pdf-header-footer",
            "--virtual-time-budget=4000",
            "--print-to-pdf=%s" % pdf_path,
```

```
                "file:///" + html_path.replace("\\", "/"),
            ]
            p = subprocess.run(cmd, capture_output=True,
        text=True,
                               timeout=timeout,
        encoding="utf-8", errors="replace")
            if not os.path.isfile(pdf_path) or
        os.path.getsize(pdf_path) < 800:
                return None, "浏览器未能生成 PDF（%s）" % (p.stderr or
        "").strip()[:200]
            with open(pdf_path, "rb") as f:
                return f.read(), None
        except subprocess.TimeoutExpired:
            return None, "生成 PDF 超时，请稍后重试"
        except Exception as e:
            return None, "生成 PDF 失败：%s" % e
        finally:
            shutil.rmtree(tmpdir, ignore_errors=True)
# ==== app/services/ahp_service.py ====
# -*- coding: utf-8 -*-
"""AHP 权重引擎。
严格按论文 4.2.1 节实现：
- 式(2)(3)(4)：特征根法求权重（行元素乘积 → n 次方根 → 归一化）
- 式(5)(6)(7)：一致性检验（λmax、CI、CR）
- 论文要求 CR < 0.1；RI 按阶数查表
"""
import numpy as np
from app.utils.constants import RI_TABLE, CR_THRESHOLD
def matrix_from_upper(upper):
    """由判断矩阵上三角构造完整互反矩阵。
    参数
    ----
    upper : list[list[float]] 或 np.ndarray，上三角元素（不含对角线）。
        例：3 阶时传入 [[a12, a13], [a23]]。
        对角线为 1，下三角取倒数（a_ji = 1/a_ij）。
    """
    upper = np.asarray(upper, dtype=float)
    k = upper.shape[0]
    n = int(round((1 + (1 + 8 * k) ** 0.5) / 2))
    A = np.ones((n, n), dtype=float)
    idx = 0
    for i in range(n):
        for j in range(i + 1, n):
            A[i, j] = upper[idx]
            A[j, i] = 1.0 / upper[idx]
            idx += 1
    return A
def weights_and_consistency(A):
    """由判断矩阵 A 计算权重与一致性指标。
    返回
```

```
    ----                                                                    1051
    dict：{weights: np.ndarray, lambda_max, ci, cr, n,                      1052
    consistent: bool}                                                       1053
    """                                                                     1054
    A = np.asarray(A, dtype=float)                                          1055
    n = A.shape[0]                                                          1056
    # 式(2) 行元素乘积                                                      1057
    row_prod = np.prod(A, axis=1)                                           1058
    # 式(3) 开 n 次方根                                                     1059
    root = row_prod ** (1.0 / n)                                            1060
    # 式(4) 归一化                                                          1061
    weights = root / root.sum()                                             1062
    # 式(5) 最大特征值 λmax = (1/n) Σ (A·W)_i / W_i                         1063
    aw = A.dot(weights)                                                     1064
    lambda_max = float(np.sum(aw / weights) / n)                            1065
    # 式(6) 一致性指标 CI                                                   1066
    ci = (lambda_max - n) / (n - 1) if n > 1 else 0.0                       1067
    # 式(7) 一致性比例 CR = CI / RI                                         1068
    ri = RI_TABLE.get(n, 1.49)                                              1069
    cr = ci / ri if ri else 0.0                                             1070
    return {                                                                1071
        "weights": weights,                                                 1072
        "lambda_max": lambda_max,                                           1073
        "ci": ci,                                                           1074
        "cr": cr,                                                           1075
        "n": n,                                                             1076
        "consistent": cr < CR_THRESHOLD,                                    1077
    }                                                                       1078
def find_most_inconsistent(A):                                              1079
    """定位判断矩阵中最不一致的一对比较。                                  1080
    依据：完全一致矩阵应满足 a_ij = w_i / w_j。                             1081
    计算每个上三角元素与其理想值的相对偏差，返回偏差最大者。               1082
    返回                                                                    1083
    ----                                                                    1084
    dict：{i, j, value, expected, deviation}，偏差最大的 (i, j)             1085
    及对应取值。                                                            1086
    """                                                                     1087
    A = np.asarray(A, dtype=float)                                          1088
    r = weights_and_consistency(A)                                          1089
    w = r["weights"]                                                        1090
    n = A.shape[0]                                                          1091
    best = None                                                             1092
    for i in range(n):                                                      1093
        for j in range(i + 1, n):                                           1094
            expected = w[i] / w[j]                                          1095
            if expected == 0:                                               1096
                continue                                                    1097
            deviation = abs(A[i, j] - expected) / expected                  1098
            if best is None or deviation >                                  1099
    best["deviation"]:                                                      1100
```

```
                    best = {"i": i, "j": j, "value": A[i, j],
                            "expected": expected, "deviation":
    deviation}
    return best
def aggregate_matrices(matrices):
    """多专家判断矩阵聚合：几何平均（保持互反性）。
    参数
    ----
    matrices : list[np.ndarray]，多个同阶判断矩阵。
    """
    stack = np.stack([np.asarray(m, dtype=float) for m in
    matrices])
    return np.prod(stack, axis=0) ** (1.0 / len(stack))
def judge_from_weights(weights):
    """由权重向量构造完全一致判断矩阵并检验。
    用途：验证给定权重向量能否作为某个完全一致判断矩阵的主特征向量
    （A_ij = w_i / w_j 时 λmax = n、CR = 0）。
    """
    w = np.asarray(weights, dtype=float)
    n = w.shape[0]
    A = np.outer(w, 1.0 / w)
    r = weights_and_consistency(A)
    r["matrix"] = A
    return r
# ==== app/services/standardization.py ====
# -*- coding: utf-8 -*-
"""指标标准化（的一部分）。
按论文 4.2.2 节式(8) 实现：
    x' = (x - x_min) / (x_max - x_min)
按指标性质分为四类：越大越优 / 越小越优 / 适中 / 离散赋值。
所有理想极值取自 app.utils.constants，可配置。
"""
from app.utils.constants import (
    IDEAL_SOIL_RATIO, IDEAL_THICKNESS, IDEAL_PH_RANGE,
    IDEAL_ORGANIC,
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,
    HEAVY_METAL_EQUAL_WEIGHT, SLOPE_BREAK, ASPECT_SCORES,
    PH_TOL_LOW, PH_TOL_HIGH,
)
def std_benefit(value, ideal):
    """越大越优指标标准化：以理想值为上限，按比例给分，饱和于 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, value / ideal))
def std_cost(value, extreme):
    """越小越优指标标准化：以极值为参考，线性折减至 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, 1.0 - value / extreme))
def std_mid_ph(value, lo, hi, tol_lo=PH_TOL_LOW,
```

```
    tol_hi=PH_TOL_HIGH):
    """适中型指标标准化（pH）：理想区间内取 1，区间外线性衰减。"""
    if value is None:
        return None
    if lo <= value <= hi:
        return 1.0
    if value > hi:
        return max(0.0, (tol_hi - value) / (tol_hi - hi))
    return max(0.0, (value - tol_lo) / (lo - tol_lo))
def std_slope(value, break_point=SLOPE_BREAK, upper=90.0):
    """坡度标准化。
    论文 4.2.2 节：以 25° 为陡坡重要分界。25° 及以下记满分 1，
    超过 25° 线性折减，90° 记 0。
    """
    if value is None:
        return None
    if value <= break_point:
        return 1.0
    return max(0.0, (upper - value) / (upper - break_point))
def aspect_score(aspect):
    """坡向离散赋值：阳坡 0.2 / 阴阳坡 0.6 / 阴坡 1.0（论文 4.2.2 节）。"""
    if aspect is None:
        return None
    return ASPECT_SCORES.get(aspect)
def pollution_score(heavy_metals):
    """污染指标：8 种重金属等权（各 0.125）合成。
    论文注：未检测 Cr、Zn 的，从矿山类型看未受污染，按无污染（=1）处理。
    heavy_metals : dict，键为金属名，值为含量 mg/kg；缺失键视为未检测。
    """
    total = 0.0
    for metal, extreme in HEAVY_METAL_LIMIT.items():
        value = heavy_metals.get(metal)
        total += std_cost(value, extreme) if value is not
    None else 1.0
    return total / len(HEAVY_METAL_LIMIT)
def standardize_unit(unit_type, data, weights=None):
    """对单个微地质单元的全部指标做标准化。
    参数
    ----
    unit_type : str，岩质边坡 / 碎石边坡 / 平台
    data : dict，实测数据，字段见接口契约：
        soil_ratio, thickness, ph, organic_matter,
    heavy_metals(dict),
        coverage, shannon, slope, aspect
    weights : dict，该单元类型的权重（app.utils.constants.DEFAULT_WEIG
    HTS 结构），
                未提供时用默认值。
    返回
    ----
    dict：{ 一级指标： { 二级指标： （实测值， 标准化值， 权重， 贡献） } }
```

```
    标准化值可能为 None，表示该指标不适用或方法待定。
    """
    from app.utils.constants import DEFAULT_WEIGHTS
    if weights is None:
        weights = DEFAULT_WEIGHTS[unit_type]
    l1_enabled = weights["l1"].keys()
    l2w = weights["l2"]
    result = {}
    # ---- 土壤指标 ----
    if "土壤指标" in l1_enabled:
        result["土壤指标"] = {
            "土石比": (data.get("soil_ratio"),
                       std_benefit(data.get("soil_ratio"),
IDEAL_SOIL_RATIO),
                       l2w["土壤指标"]["土石比"]),
            "土层厚度": (data.get("thickness"),
                         std_benefit(data.get("thickness"),
IDEAL_THICKNESS),
                         l2w["土壤指标"]["土层厚度"]),
            "pH": (data.get("ph"),
                   std_mid_ph(data.get("ph"),
*IDEAL_PH_RANGE),
                   l2w["土壤指标"]["pH"]),
            "有机质": (data.get("organic_matter"),
                       std_benefit(data.get("organic_matter")
, IDEAL_ORGANIC),
                       l2w["土壤指标"]["有机质"]),
            "污染指标": (data.get("heavy_metals") or {},
                         pollution_score(data.get("heavy_meta
ls") or {}),
                         l2w["土壤指标"]["污染指标"]),
        }
    # ---- 植被指标 ----
    if "植被指标" in l1_enabled:
        result["植被指标"] = {
            "植被覆盖度": (data.get("coverage"),
                           std_benefit(data.get("coverage"),
IDEAL_COVERAGE),
                           l2w["植被指标"]["植被覆盖度"]),
            "生物多样性": (data.get("shannon"),
                           std_benefit(data.get("shannon"),
IDEAL_SHANNON),
                           l2w["植被指标"]["生物多样性"]),
        }
    # ---- 地形指标 ----
    if "地形指标" in l1_enabled:
        result["地形指标"] = {
            "坡度": (data.get("slope"),
                     std_slope(data.get("slope")),
                     l2w["地形指标"]["坡度"]),
```

```
                "坡向": (data.get("aspect"),                                    1251
                         aspect_score(data.get("aspect")),                      1252
                         l2w["地形指标"]["坡向"]),                               1253
            }                                                                   1254
    return result                                                               1255
# ==== app/services/weight_service.py ====                                      1256
# -*- coding: utf-8 -*-                                                         1257
"""权重方案的存取与生效权重解析。                                                1258
系统有两套权重来源：                                                            1259
1. 论文表 3 给出的默认权重（app.utils.constants.DEFAULT_WEIGHTS），             1260
   是**权威取值来源**，案例复算与报告结论以此为准；                             1261
2. 通过层次分析法在本系统中实际算出的权重，保存在                               1262
   weight_schemes / weight_items 两张表中，可标记为「生效」。                   1263
本模块负责把这两者合成为计算引擎直接可用的嵌套权重结构：                       1264
以论文表 3 为底，用生效方案中已保存的条目覆盖。                                1265
没有任何生效方案时，即完全使用论文表 3 的权重。                                1266
这样既保证默认行为与论文一致，又使 AHP 计算结果真正参与评价，                  1267
而不是算完即弃。                                                              1268
"""                                                                           1269
import copy                                                                   1270
from app.models import db, WeightScheme, WeightItem                           1271
from app.utils.constants import DEFAULT_WEIGHTS                               1272
from app.services.indicator_service import TREE                               1273
# 方案名称：每个微地质单元类型固定一个可生效的方案                              1274
ACTIVE_NAME = "自定义权重方案（AHP 计算结果）"                                  1275
def _code_maps():                                                             1276
    """由指标树导出 编码 → 名称 的映射，避免名称在两处各写一遍。"""             1277
    l1, l2 = {}, {}                                                           1278
    for row in TREE:                                                          1279
        code, name, level, parent = row[0], row[1], row[2],                   1280
    row[3]                                                                    1281
        if level == 1:                                                        1282
            l1[code] = name                                                   1283
        elif level == 2:                                                      1284
            l2[code] = (l1.get(parent, parent), name)                         1285
    return l1, l2                                                             1286
def effective_weights(unit_type):                                             1287
    """返回该单元类型当前生效的权重。                                          1288
    返回 (weights, source_label, scheme)                                      1289
        weights      嵌套权重结构，可直接交给标准化与评价引擎                 1290
        source_label 权重来源说明，用于界面与报告标注                         1291
        scheme       生效的权重方案对象，无自定义方案时为 None                 1292
    """                                                                       1293
    base = DEFAULT_WEIGHTS.get(unit_type)                                     1294
    if base is None:                                                          1295
        return None, "未定义该微地质单元类型的权重", None                      1296
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ                  1297
    e,                                                                        1298
                                                                              1299
    name=ACTIVE_NAME).first()                                                 1300
```

```
    if scheme is None:                                                  1301
        return copy.deepcopy(base), "论文表 3 默认权重", None           1302
    merged = copy.deepcopy(base)                                        1303
    l1_map, l2_map = _code_maps()                                       1304
    applied = 0                                                         1305
    for item in WeightItem.query.filter_by(scheme_id=scheme.            1306
    id).all():                                                          1307
        code, w = item.indicator_code, item.weight                      1308
        if w is None or w <= 0:                                         1309
            continue                                                    1310
        if code in l1_map and l1_map[code] in merged["l1"]:             1311
            merged["l1"][l1_map[code]] = w                              1312
            applied += 1                                                1313
        elif code in l2_map:                                            1314
            l1_name, l2_name = l2_map[code]                             1315
            if l1_name in merged["l2"] and l2_name in                   1316
    merged["l2"][l1_name]:                                              1317
                merged["l2"][l1_name][l2_name] = w                      1318
                applied += 1                                            1319
    if applied == 0:                                                    1320
        return copy.deepcopy(base), "论文表 3 默认权重", None           1321
    return merged, "自定义方案（%d 项覆盖论文表 3）" % applied, scheme    1322
def save_weights(unit_type, code_weight_pairs, note=None):              1323
    """把一组「指标编码 → 权重」写入生效方案（已存在则更新对应条目）。  1324
    仅覆盖本次传入的条目，其余条目保留，便于分层次（一级 / 二级）       1325
    分次录入而无需一次填完。                                            1326
    """                                                                 1327
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ            1328
    e,                                                                  1329
                                                                        1330
    name=ACTIVE_NAME).first()                                           1331
    if scheme is None:                                                  1332
        scheme = WeightScheme(unit_type=unit_type,                      1333
    name=ACTIVE_NAME,                                                   1334
                                  level=1, is_default=True)             1335
        db.session.add(scheme)                                          1336
        db.session.flush()                                              1337
    if note:                                                            1338
        scheme.level = scheme.level or 1                                1339
    if not scheme.is_default:                                           1340
        scheme.is_default = True                                        1341
    written = 0                                                         1342
    for code, w in code_weight_pairs:                                   1343
        if w is None or w <= 0:                                         1344
            continue                                                    1345
        item = WeightItem.query.filter_by(scheme_id=scheme.i            1346
    d,                                                                  1347
                                                                        1348
    indicator_code=code).first()                                        1349
        if item is None:                                                1350
```

```
                item = WeightItem(scheme_id=scheme.id,
    indicator_code=code)
                db.session.add(item)
            item.weight = float(w)
            written += 1
    db.session.commit()
    return scheme, written
def reset_weights(unit_type):
    """删除该单元类型的自定义方案，恢复为论文表 3 的默认权重。"""
    schemes = WeightScheme.query.filter_by(unit_type=unit_ty
    pe,

    name=ACTIVE_NAME).all()
    n = 0
    for s in schemes:
        WeightItem.query.filter_by(scheme_id=s.id).delete(
            synchronize_session=False)
        db.session.delete(s)
        n += 1
    db.session.commit()
    return n
def scheme_items(unit_type):
    """返回该单元类型已保存的自定义权重条目 {指标编码： 权重}。"""
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ
    e,

    name=ACTIVE_NAME).first()
    if scheme is None:
        return {}, None
    items = {i.indicator_code: i.weight
             for i in WeightItem.query.filter_by(scheme_id=s
    cheme.id).all()}
    return items, scheme
def summary():
    """各单元类型的权重来源概览，供权重首页展示。"""
    rows = []
    for unit_type in DEFAULT_WEIGHTS:
        weights, label, scheme = effective_weights(unit_type
    )
        rows.append({
            "unit_type": unit_type,
            "label": label,
            "custom": scheme is not None,
        })
    return rows
# ==== app/services/evaluation_service.py ====
# -*- coding: utf-8 -*-
"""适宜性评价引擎。
按论文式(９) 实现：
- 单元级：F' = Σ（二级指标权重 × 该二级指标下三级指标标准化值）
```

```
- 矿山级：F = Σ（Si × F'i），Si 为各微地质单元面积占比
- 等级判定按论文表 4。
"""
from app.utils.constants import DEFAULT_WEIGHTS, grade_of
from app.services import standardization as std
def score_unit(unit_type, data, weights=None):
    """计算单个微地质单元的评价因子 F'。
    返回
    ----
    dict：{ l1_scores: {一级指标: 得分}, unit_score: F', details:
    [...] }
    若某一级指标下存在"不适用"（标准化值为 None）的二级指标，
    则按其余项权重归一化后计算，保证单元得分始终有定义。
    """
    if weights is None:
        weights = DEFAULT_WEIGHTS[unit_type]
    std_result = std.standardize_unit(unit_type, data,
    weights)
    l1_weights = weights["l1"]
    l1_scores = {}
    details = []
    for l1_name, items in std_result.items():
        wsum = 0.0
        acc = 0.0
        for l2_name, (raw, std_value, w2) in items.items():
            if std_value is None:
                # 不适用项：不计入
                details.append({
                    "l1": l1_name, "l2": l2_name, "raw":
    raw,
                    "std_value": None, "weight": w2,
    "contribution": None,
                })
                continue
            contribution = std_value * w2
            acc += contribution
            wsum += w2
            details.append({
                "l1": l1_name, "l2": l2_name, "raw": raw,
                "std_value": std_value, "weight": w2,
                "contribution": contribution,
            })
        l1_scores[l1_name] = acc / wsum if wsum > 0 else 0.0
    # 一级指标权重按实际参与计算的指标归一化。
    # 论文表 3 的权重本身已归一（和为 1），此处为恒等变换；
    # 若某层指标缺失或自定义权重不完整，可避免单元得分被整体压低。
    lw_sum = sum(l1_weights[k] for k in l1_scores if k in
    l1_weights)
    if lw_sum > 0:
        unit_score = sum(l1_weights[k] * l1_scores[k]
```

```
                                for k in l1_scores if k in
    l1_weights) / lw_sum
    else:
        unit_score = 0.0
    return {"l1_scores": l1_scores, "unit_score":
    unit_score, "details": details}
def score_mine(units, weights_map=None):
    """计算整个矿山的评价因子 F 与适宜性等级。
    参数
    ----
    units : list[dict]，每个元素：
        {unit_type, ratio, data}
        ratio 为该单元面积占比（0~1），data 为实测数据 dict。
    weights_map : dict，{单元类型： 权重结构}。未提供的单元类型
        使用论文表 3 的默认权重。权重可来自 AHP 计算并经
        weight_service 保存生效，从而真正参与评价。
    返回
    ----
    dict:{ final_score, grade, grade_info, unit_results }
    """
    weights_map = weights_map or {}
    total = 0.0
    unit_results = []
    for u in units:
        r = score_unit(u["unit_type"], u["data"],
                       weights_map.get(u["unit_type"]))
        weighted = u["ratio"] * r["unit_score"]
        total += weighted
        unit_results.append({
            "unit_type": u["unit_type"],
            "ratio": u["ratio"],
            "unit_score": r["unit_score"],
            "weighted": weighted,
            "l1_scores": r["l1_scores"],
            "details": r["details"],
        })
    grade_info = grade_of(total)
    return {
        "final_score": total,
        "grade": grade_info["grade"],
        "grade_info": grade_info,
        "unit_results": unit_results,
    }
# ==== app/services/report_service.py ====
# -*- coding: utf-8 -*-
"""报告生成：生成 Word 评价报告。
依赖 python-docx。报告结构：
标题 → 图斑信息 → 评价结论 → 各微地质单元得分 → 完整得分明细 → 生成时间。
"""
import io
```

```
from datetime import datetime
from docx import Document
from docx.shared import Pt, Cm
from docx.oxml.ns import qn
from docx.enum.text import WD_ALIGN_PARAGRAPH
def _set_run(run, text=None, size=12, bold=False):
    if text is not None:
        run.text = text
    run.font.name = "Times New Roman"
    run.font.size = Pt(size)
    run.font.bold = bold
    # 中文字体
    run._element.rPr.rFonts.set(qn("w:eastAsia"), "宋体")
    return run
def _para(doc, text, size=12, bold=False, align=None,
    space_after=6):
    p = doc.add_paragraph()
    if align is not None:
        p.alignment = align
    p.paragraph_format.space_after = Pt(space_after)
    _set_run(p.add_run(), text, size, bold)
    return p
def _heading(doc, text, level=1):
    h = doc.add_heading(text, level=level)
    for run in h.runs:
        run.font.name = "Times New Roman"
        run._element.rPr.rFonts.set(qn("w:eastAsia"), "黑体")
    return h
def _table(doc, headers, rows, widths=None):
    t = doc.add_table(rows=1, cols=len(headers))
    t.style = "Table Grid"
    hdr = t.rows[0].cells
    for i, htext in enumerate(headers):
        _set_run(hdr[i].paragraphs[0].add_run(), htext,
    10.5, bold=True)
    for row in rows:
        cells = t.add_row().cells
        for i, val in enumerate(row):
            _set_run(cells[i].paragraphs[0].add_run(),
    str(val), 10.5)
    return t
def build_report(patch, mine_result):
    """生成评价报告 Word 文档，返回 BytesIO。
    参数
    ----
    patch : MinePatch，图斑对象
    mine_result : evaluation_service.score_mine 的返回 dict
    """
    doc = Document()
    # 页边距
```

```
    for section in doc.sections:
        section.top_margin = Cm(2.5)
        section.bottom_margin = Cm(2.5)
        section.left_margin = Cm(2.5)
        section.right_margin = Cm(2.5)
    _heading(doc, "历史遗留废弃矿山自然恢复适宜性评价报告", 0)
    _para(doc, "软件：历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1．0",
          size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER)
    _para(doc, "生成时间：%s" % datetime.now().strftime("%Y-%m-%d
%H:%M:%S"),
          size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER,
space_after=12)
    # 图斑信息
    _heading(doc, "一、图斑基本信息", 1)
    info = [
        ("图斑名称", patch.name),
        ("所在行政区", patch.admin_region or "—"),
        ("图斑面积（公顷）", str(patch.area_ha) if patch.area_ha is
not None else "—"),
        ("开采矿种", patch.mineral_type or "—"),
        ("开采方式", patch.mining_method or "—"),
    ]
    _table(doc, ["项目", "内容"], info)
    # 评价结论
    _heading(doc, "二、评价结论", 1)
    gi = mine_result["grade_info"]
    concl = [
        ("最终评价因子 F", "%.4f" % mine_result["final_score"]),
        ("适宜性等级", mine_result["grade"]),
        ("判定依据", gi["meaning"]),
        ("推荐措施", gi["measure"]),
    ]
    _table(doc, ["项目", "结果"], concl)
    # 各单元得分
    _heading(doc, "三、各微地质单元得分", 1)
    rows = []
    for u in mine_result["unit_results"]:
        rows.append((u["unit_type"], "%.2f%%" % (u["ratio"]
* 100),
                     "%.4f" % u["unit_score"], "%.4f" %
u["weighted"]))
    _table(doc, ["微地质单元", "面积占比", "单元评价因子 F'", "加权贡献"],
rows)
    # 完整得分明细
    _heading(doc, "四、得分明细", 1)
    for u in mine_result["unit_results"]:
        _para(doc, "%s（占比 %.2f%%）" % (u["unit_type"],
u["ratio"] * 100),
              size=11, bold=True, space_after=4)
        detail_rows = []
```

```
            for d in u["details"]:
                raw = "—" if d["raw"] is None else d["raw"]
                std = "—" if d["std_value"] is None else "%.4f"
        % d["std_value"]
                w = "%.4f" % d["weight"]
                c = "—" if d["contribution"] is None else
        "%.4f" % d["contribution"]
                detail_rows.append((d["l1"], d["l2"], raw, std,
        w, c))
            _table(doc, ["一级指标", "二级指标", "实测值", "标准化值", "权重",
        "贡献"],
                   detail_rows)
            # 一级指标得分
            l1_row = []
            for k, v in u["l1_scores"].items():
                l1_row.append("%s=%.4f" % (k, v))
            _para(doc, "一级指标得分：" + ",".join(l1_row), size=10.5,
        space_after=10)
        buf = io.BytesIO()
        doc.save(buf)
        buf.seek(0)
        return buf
# ==== app/services/indicator_service.py ====
# -*- coding: utf-8 -*-
"""指标定义维护。
把论文表 2/表 3 的指标体系写入 indicators 表，使指标名称、单位、
理想值与取值依据可在界面上查看维护，而不再只存在于代码常量中。
代码常量（app.utils.constants）仍是计算时的**权威取值来源**；
本表用于展示、核对与记录取值依据。二者由 seed_indicators() 保持同步。
"""
from app.models import db, Indicator
from app.utils.constants import (
    UNIT_TYPES, UNIT_ENABLED_INDICATORS, IDEAL_SOIL_RATIO,
    IDEAL_THICKNESS,
    IDEAL_PH_RANGE, IDEAL_ORGANIC, IDEAL_COVERAGE,
    IDEAL_SHANNON,
    HEAVY_METAL_LIMIT, SLOPE_BREAK, ASPECT_SCORES,
)
ALL_UNITS = ",".join(UNIT_TYPES)
# (code, name, level, parent, unit, ideal_value,
    ideal_type, enabled_units, desc, is_constraint)
TREE = [
    # ---- 一级指标 ----
    ("soil", "土壤指标", 1, None, None, None, None, "碎石边坡,平台",
    "论文表 2 评价指标之一级指标", False),
    ("veg", "植被指标", 1, None, None, None, None, ALL_UNITS,
    "论文表 2 评价指标之一级指标", False),
    ("terrain", "地形指标", 1, None, None, None, None,
    "岩质边坡,碎石边坡", "论文表 2 评价指标之一级指标", False),
    # ---- 二级指标：土壤 ----
```

```
        ("soil_ratio", "土石比", 2, "soil", "—",
        str(IDEAL_SOIL_RATIO), "benefit", "碎石边坡,平台",
          "论文 4.2.2 节:理想值 0.8,根据矿山修复实际情况确定", False),
        ("thickness", "土层厚度", 2, "soil", "cm",
        str(IDEAL_THICKNESS), "benefit", "碎石边坡,平台",
          "论文 4.2.2 节:理想值 20 cm", False),
        ("ph", "pH", 2, "soil", "—", "%g~%g" % IDEAL_PH_RANGE,
        "mid", "碎石边坡,平台",
          "论文 4.2.2 节:pH 用适中指标的标准化方法,理想区间 %g~%g" %
        IDEAL_PH_RANGE, False),
        ("organic_matter", "有机质", 2, "soil", "g/kg",
        str(IDEAL_ORGANIC), "benefit", "碎石边坡,平台",
          "《全国第二次土壤普查养分等级划分标准》二等(较丰富)最低值 30 g/kg", False),
        ("pollution", "污染指标", 2, "soil", "mg/kg", "见三级指标",
        "cost", "碎石边坡,平台",
          "同时作为限制性指标与评价指标;8 种重金属等权合成(各 0.125)", True),
        # ---- 二级指标:植被 ----
        ("coverage", "植被覆盖度", 2, "veg", "%",
        str(IDEAL_COVERAGE), "benefit", ALL_UNITS,
          "论文 4.2.2 节:参照生态系统取值 75%", False),
        ("shannon", "生物多样性", 2, "veg", "—", str(IDEAL_SHANNON),
        "benefit", ALL_UNITS,
          "Shannon-Wiener 多样性指数,参照生态系统取值 0.7", False),
        # ---- 二级指标:地形 ----
        ("slope", "坡度", 2, "terrain", "°", "%g° 分界" %
        SLOPE_BREAK, "cost", "岩质边坡,碎石边坡",
          "论文 4.2.2 节引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号:25°
        为陡坡重要分界", False),
        ("aspect", "坡向", 2, "terrain", "—",
        "阳坡0.2/阴阳坡0.6/阴坡1.0", "discrete", "岩质边坡,碎石边坡",
          "论文 4.2.2 节:阴坡及半阴坡水分条件更优,故赋值高于阳坡", False),
]
# 污染指标下的三级指标(8 种重金属)
for _m, _limit in HEAVY_METAL_LIMIT.items():
        TREE.append((
              "hm_" + _m, _m, 3, "pollution", "mg/kg",
        str(_limit), "cost",
              "碎石边坡,平台",
              "《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值 %g mg/kg" %
        _limit,
              False,
        ))
def seed_indicators():
        """写入/更新指标定义(幂等)。返回写入条数。"""
        code_map = {
              "benefit": "越大越优", "cost": "越小越优",
              "mid": "适中型", "discrete": "离散赋值",
        }
        n = 0
        for (code, name, level, parent, unit, ideal, itype,
```

```
    units, desc, is_con) in TREE:
        ind = Indicator.query.filter_by(code=code).first()
        if ind is None:
            ind = Indicator(code=code)
            db.session.add(ind)
        ind.name = name
        ind.level = level
        ind.parent_code = parent
        ind.unit = unit
        ind.ideal_value = ideal
        ind.ideal_type = itype
        ind.enabled_units = units
        ind.description = desc
        ind.is_constraint = is_con
        ind.is_evaluative = (level in (2, 3))
        n += 1
    db.session.commit()
    return n
def ideal_type_label(code):
    return {"benefit": "越大越优", "cost": "越小越优",
            "mid": "适中型", "discrete": "离散赋值"}.get(code,
    code or "—")
# ==== app/services/import_service.py ====
# -*- coding: utf-8 -*-
"""指标数据批量导入。
支持 CSV 与 Excel(.xlsx/.xls)。一行对应一个微地质单元,
按「单元类型」匹配图斑内已有的微地质单元后写入指标数据。
表头约定(缺失列按不适用处理):
    单元类型, 土石比, 土层厚度, pH, 有机质,
    Hg, As, Ni, Cu, Cd, Pb, Cr, Zn,
    植被覆盖度, Shannon, 坡度, 坡向
"""
import csv
import io
import json
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
    UNIT_TYPES
# 表头(顺序即导出模板顺序)
HEADERS = (["单元类型", "土石比", "土层厚度", "pH", "有机质"]
           + list(HEAVY_METALS)
           + ["植被覆盖度", "Shannon", "坡度", "坡向"])
FIELD_ALIASES = {
    "单元类型": "unit_type",
    "土石比": "soil_ratio",
    "土层厚度": "thickness",
    "pH": "ph", "PH": "ph", "ph": "ph",
    "有机质": "organic_matter",
    "植被覆盖度": "coverage",
    "Shannon": "shannon", "shannon": "shannon", "生物多样性":
    "shannon",
```

```
    "坡度": "slope",
    "坡向": "aspect",
}
for _m in HEAVY_METALS:
    FIELD_ALIASES[_m] = "hm_" + _m
def build_template_csv():
    """生成导入模板（含表头与示例行）。"""
    buf = io.StringIO()
    w = csv.writer(buf)
    w.writerow(HEADERS)
    # 示例：碎石边坡一行，说明每列含义
    row = {h: "" for h in HEADERS}
    row.update({"单元类型": "碎石边坡", "土石比": "0.34", "土层厚度": "10",
                "pH": "9.24", "有机质": "4.89", "Hg": "0.05",
    "As": "5.17",
                "Ni": "12.60", "Cu": "11.70", "Cd": "0.04",
    "Pb": "6.50",
                "植被覆盖度": "87.92", "Shannon": "0.60",
                "坡度": "28.75", "坡向": "阳坡"})
    w.writerow([row[h] for h in HEADERS])
    return buf.getvalue()
def _read_rows(filename, stream):
    """读取上传文件为「表头 -> 值」的字典列表。"""
    ext = filename.rsplit(".", 1)[-1].lower()
    if ext == "csv":
        text = stream.read().decode("utf-8-sig",
    errors="replace")
        reader = csv.DictReader(io.StringIO(text))
        return [dict(r) for r in reader]
    if ext in ("xlsx", "xls"):
        from openpyxl import load_workbook
        wb = load_workbook(io.BytesIO(stream.read()),
    data_only=True)
        ws = wb.active
        rows = list(ws.iter_rows(values_only=True))
        if not rows:
            return []
        header = [str(c).strip() if c is not None else ""
    for c in rows[0]]
        out = []
        for r in rows[1:]:
            if all(c is None or str(c).strip() == "" for c
    in r):
                continue
            out.append({header[i]: (r[i] if i < len(r) else
    None)
                        for i in range(len(header)) if
    header[i]})
        return out
    raise ValueError("不支持的文件类型：.%s" % ext)
```

```
def _num(value):
    if value is None:
        return None, None
    s = str(value).strip()
    if s == "":
        return None, None
    try:
        return float(s), None
    except ValueError:
        return None, "「%s」不是有效数值" % s
def parse_upload(filename, stream, unit_type_map):
    """解析上传文件。
    参数
    ----
    unit_type_map : dict，{单元类型: MicroUnit 对象}
    返回
    ----
    (parsed, errors)
      parsed : list[dict]，每项 {unit_type, fields{...},
    heavy_metals{}}
      errors : list[str]，逐行问题说明
    """
    rows = _read_rows(filename, stream)
    parsed, errors = [], []
    if not rows:
        return [], ["文件为空或没有数据行"]
    # 表头校验
    unknown = [h for h in (rows[0].keys() if rows else [])
    if h not in FIELD_ALIASES]
    if unknown:
        errors.append("存在无法识别的列，将被忽略：%s" %
    "、".join(map(str, unknown)))
    for idx, raw in enumerate(rows, start=2):       # 第 1 行是表头
        rec = {}
        for k, v in raw.items():
            key = FIELD_ALIASES.get(str(k).strip())
            if key:
                rec[key] = v
        unit_type = str(rec.get("unit_type") or "").strip()
        if not unit_type:
            errors.append("第 %d 行：缺少「单元类型」" % idx)
            continue
        if unit_type not in unit_type_map:
            errors.append("第 %d 行：图斑内不存在单元类型「%s」（现有：%s）"
                          % (idx, unit_type,
    "、".join(unit_type_map) or "无"))
            continue
        if unit_type not in UNIT_TYPES:
            errors.append("第 %d 行：单元类型「%s」不是系统支持的三种之一"
                          % (idx, unit_type))
```

```
            continue
        fields, heavy = {}, {}
        ok = True
        for key, val in rec.items():
            if key in ("unit_type",):
                continue
            if key.startswith("hm_"):
                n, err = _num(val)
                if err:
                    errors.append("第 %d 行:%s %s" % (idx,
key[3:], err))
                    ok = False
                elif n is not None:
                    heavy[key[3:]] = n
                continue
            if key == "aspect":
                s = str(val).strip() if val is not None
else ""
                if s == "":
                    fields["aspect"] = None
                elif s in ASPECT_TYPES:
                    fields["aspect"] = s
                else:
                    errors.append("第 %d 行:坡向「%s」无效（应为 %s）"
                                  % (idx, s,
"、".join(ASPECT_TYPES)))
                    ok = False
                continue
            n, err = _num(val)
            if err:
                errors.append("第 %d 行:%s %s" % (idx, key,
err))
                ok = False
            else:
                fields[key] = n
        if ok:
            parsed.append({"unit_type": unit_type,
"fields": fields,
                           "heavy_metals": heavy})
    return parsed, errors
# ==== app/services/workflow.py ====
# -*- coding: utf-8 -*-
"""评价流程前置条件（定量评价与报告输出的流程门禁）。
论文 4.1 节确定的评价流程存在严格的先后关系：
    建立图斑 → 限制性条件判定（通过） → 划分微地质单元并计算面积占比
    → 逐单元录入指标数据 → 计算评价因子并判定等级 → 输出报告
论文原文：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
废弃矿山，按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡三种
地质微单元类型，并计算各种地形面积占比。」限制性条件判定（一票否决）
在前，微地质单元划分在后，顺序不可颠倒。
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本模块把该顺序固化为可检查的状态。后续模块在入口处调用它，
前置步骤未完成时直接拦截并给出「缺什么、去哪补」的提示，
避免在条件不具备时产出看似合理的评价结论。
"""
from app.models import (db, MicroUnit, IndicatorData,
    ConstraintCheck,
                        EvaluationResult)
#  指标数据非空字段（用于判断某单元是否已录入数据）
DATA_FIELDS = ("soil_ratio", "thickness", "ph",
    "organic_matter",
              "coverage", "shannon", "slope", "aspect")
def _has_data(ind):
    if ind is None:
        return False
    if any(getattr(ind, f, None) is not None for f in
    DATA_FIELDS):
        return True
    return bool(ind.heavy_metals_dict())
def patch_progress(patch_id):
    """检查某图斑的评价流程完成情况。
    返回 dict：
        units       {done, count, ratio_sum, detail}
        constraint  {done, passed, detail}
        data        {done, ready, missing, detail}
        blocking    未满足的前置步骤列表（按流程顺序）
        ready       是否可进入定量评价
    """
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    ratio_sum = sum(u.ratio or 0.0 for u in units)
    #  ---- 步骤一：微地质单元 ----
    units_step = {
        "done": bool(units),
        "count": len(units),
        "ratio_sum": ratio_sum,
        "detail": ("已划分 %d 个微地质单元，面积占比合计 %.2f%%"
                   % (len(units), ratio_sum * 100))
                  if units else "尚未划分微地质单元",
    }
    #  ---- 步骤二：限制性条件判定 ----
    cc = ConstraintCheck.query.filter_by(patch_id=patch_id)
    \
        .order_by(ConstraintCheck.id.desc()).first()
    if cc is None:
        constraint_step = {"done": False, "passed": False,
                           "detail": "尚未执行限制性条件判定"}
    elif cc.passed:
        constraint_step = {"done": True, "passed": True,
                           "detail": "限制性条件判定已通过"}
    else:
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    constraint_step = {"done": True, "passed": False,
                       "detail": "限制性条件判定未通过（%s）" %
(cc.reason or "")}
# ---- 步骤三：指标数据 ----
missing = []
for u in units:
    ind = IndicatorData.query.filter_by(unit_id=u.id).fi
rst()
    if not _has_data(ind):
        missing.append(u.unit_type)
if not units:
    data_step = {"done": False, "ready": False,
"missing": [],
                 "detail": "尚无可录入数据的微地质单元"}
elif missing:
    data_step = {"done": False, "ready": False,
"missing": missing,
                 "detail": "以下微地质单元尚未录入指标数据：%s"
                           % "、".join(missing)}
else:
    data_step = {"done": True, "ready": True,
"missing": [],
                 "detail": "全部 %d 个微地质单元的指标数据已录入" %
len(units)}
ratio_ok = ratio_sum > 0.999 if units else False
if units and not ratio_ok:
    units_step["detail"] += "（占比合计不等于 １００%，请检查面积录入）"
# ---- 汇总未满足的前置条件（按论文 ４．１ 节的流程顺序：限制性判定优先） ----
blocking = []
if not constraint_step["done"]:
    blocking.append({
        "step": "constraint", "label": "执行限制性条件判定",
        "reason": "尚未执行限制性条件判定。论文流程要求先判定政策、区位约束、"
                  "土壤污染、地质灾害与耕地属性，通过后方可进入定量评价。",
        "fix_url": "/constraint/%d" % patch_id,
"fix_label": "去执行限制性判定",
    })
elif not constraint_step["passed"]:
    blocking.append({
        "step": "constraint", "label": "限制性条件判定未通过",
        "reason": "限制性条件判定未通过。按论文流程，此类矿山应直接判定为"
                  "「人工辅助或生态重建」，不进入定量评价。",
        "fix_url": "/constraint/%d" % patch_id,
"fix_label": "去复核判定结果",
    })
if not units_step["done"]:
    blocking.append({
        "step": "units", "label": "划分微地质单元",
        "reason": "该图斑尚未划分微地质单元，无法确定各单元的面积权重。",
        "fix_url": "/unit/%d" % patch_id, "fix_label":
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        "去划分微地质单元",
            })
        elif not ratio_ok:
            blocking.append({
                "step": "units", "label": "补齐单元面积",
                "reason": "各微地质单元面积占比合计不等于 100%，加权结果会失真。",
                "fix_url": "/unit/%d" % patch_id, "fix_label":
        "去核对单元面积",
            })
        if units_step["done"] and not data_step["ready"]:
            blocking.append({
                "step": "data", "label": "录入指标数据",
                "reason": data_step["detail"] +
        "。缺少数据时计算会把空值按不适用跳过，"
                          "得出的结论不能反映实际情况。",
                "fix_url": "/indicator/", "fix_label":
        "去录入指标数据",
            })
        return {
            "units": units_step,
            "constraint": constraint_step,
            "data": data_step,
            "blocking": blocking,
            "ready": not blocking,
        }
    def latest_result(patch_id):
        """取该图斑最近一次的评价记录。"""
        return EvaluationResult.query.filter_by(patch_id=patch_i
        d) \
            .order_by(EvaluationResult.id.desc()).first()
    def report_progress(patch_id):
        """报告输出的前置条件：评价流程齐备，且已有评价记录。
        报告本质上是评价结论的呈现，因此必须先真实执行过定量评价，
        导出时读取已保存的评价结果，而不是在导出环节重新计算。
        否则评价完成后再修改指标数据，报告会与系统记录的评价结论不一致，
        同一份评价出现两个数字，结论无法追溯。
        返回 dict：{ready, blocking, result, flow}
            result 为最近一次评价记录（EvaluationResult），无记录时为空。
        """
        flow = patch_progress(patch_id)
        blocking = list(flow["blocking"])
        record = latest_result(patch_id)
        if not blocking and record is None:
            blocking.append({
                "step": "evaluate", "label": "执行适宜性评价",
                "reason": "该图斑尚无评价记录。报告是评价结论的呈现，"
                          "需先执行定量评价后才能生成。",
                "fix_url": "/evaluate/%d" % patch_id,
                "fix_label": "去执行适宜性评价",
            })
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    return {
        "ready": not blocking,
        "blocking": blocking,
        "result": record,
        "flow": flow,
    }
# ==== app/services/dashboard.py ====
# -*- coding: utf-8 -*-
"""工作台数据：统计数字、评价流程步骤状态、侧边栏待办角标。
首页（工作台）需要把「现在有哪些数据」「流程走到哪一步」「哪一步还差活」
一次算清楚，供模板直接渲染。集中在此，避免在视图与模板里各算一遍。
"""
from flask import g
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    Standard)
from app.services import workflow as wf
#  评价流程的八个步骤（顺序即论文 4.1 节的作业顺序）
#    key          步骤标识
#    no           步骤编号（01~08）
#    title        名称
#    icon         图标名（见 templates/_macros.html 的 ICONS）
#    desc         一句话说明
#    href         入口地址
STEPS = [
    dict(key="project", no="01", title="项目与矿山档案",
    icon="folder",
         href="/project/",
         desc="建立评价项目、录入矿山图斑、挂接遥感影像、设置参照生态系统"),
    dict(key="standard", no="02", title="政策与标准知识库",
    icon="book",
         href="/standard/",
         desc="收录地方标准与政策文件，结构化判定规则，支持省际口径比对"),
    dict(key="constraint", no="03", title="限制性指标判定",
    icon="shield",
         href="/constraint/",
         desc="政策、区位约束、土壤污染、地质灾害、耕地属性五项一票否决"),
    dict(key="unit", no="04", title="微地质单元", icon="layers",
         href="/unit/",
         desc="划分岩质边坡、碎石边坡、平台三类单元，自动计算面积占比"),
    dict(key="ahp", no="05", title="指标体系与权重",
    icon="sliders",
         href="/ahp/",
         desc="指标定义维护，判断矩阵录入、权重计算与一致性检验、多专家汇总"),
    dict(key="indicator", no="06", title="指标数据采集",
    icon="database",
         href="/indicator/",
         desc="按微地质单元录入土壤、植被、地形指标，支持 CSV / Excel 批量导入"),
    dict(key="evaluate", no="07", title="适宜性定量评价",
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    icon="target",
        href="/evaluate/",
        desc="指标标准化、分单元赋分、面积加权合成，判定适宜性等级"),
    dict(key="report", no="08", title="报告与结果输出",
    icon="file",
        href="/report/",
        desc="结果图表与逐层得分明细，导出 PDF 与 Word 评价报告"),
]
def _has_data(ind):
    return wf._has_data(ind)
def snapshot():
    """一次算出工作台需要的全部数据（同一请求内只算一次）。
    返回 dict：
      stats      {projects, patches, unit_types, standards}
      steps      八个步骤，附 available / pending / note
      badges     {key: 待办数量}，仅含有待办的项
      progress   {patches, units, data_units, ready,
    evaluated}
    """
    cached = getattr(g, "_dash_snapshot", None)
    if cached is None:
        cached = _compute()
        g._dash_snapshot = cached
    return cached
def _compute():
    projects = Project.query.count()
    patches = MinePatch.query.all()
    patch_ids = [p.id for p in patches]
    units = MicroUnit.query.filter(
        MicroUnit.patch_id.in_(patch_ids)).all() if
    patch_ids else []
    unit_by_patch = {}
    for u in units:
        unit_by_patch.setdefault(u.patch_id, []).append(u)
    unit_ids = [u.id for u in units]
    data_unit_ids = set()
    if unit_ids:
        for ind in IndicatorData.query.filter(
                IndicatorData.unit_id.in_(unit_ids)).all():
            if _has_data(ind):
                data_unit_ids.add(ind.unit_id)
    passed_constraint = set()
    if patch_ids:
        for c in ConstraintCheck.query.filter(
                ConstraintCheck.patch_id.in_(patch_ids)).all
    ():
            if c.passed:
                passed_constraint.add(c.patch_id)
    evaluated = set()
    if patch_ids:
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    for r in EvaluationResult.query.filter(
            EvaluationResult.patch_id.in_(patch_ids)).all():
        evaluated.add(r.patch_id)
# ---- 各步骤的待办数量 ----
pending = {
    "constraint": sum(1 for p in patches if p.id not in passed_constraint),
    "unit": sum(1 for p in patches if not unit_by_patch.get(p.id)),
    "indicator": 0,
    "evaluate": 0,
}
ready_patches = 0
for p in patches:
    us = unit_by_patch.get(p.id, [])
    if us and any(u.id not in data_unit_ids for u in us):
        pending["indicator"] += 1
    pr = wf.patch_progress(p.id)
    if pr["ready"]:
        ready_patches += 1
        if p.id not in evaluated:
            pending["evaluate"] += 1
# ---- 步骤可用性：前置数据不具备时置灰 ----
avail = {
    "project": True,
    "standard": True,
    "constraint": bool(patches),
    "unit": bool(patches),
    "ahp": True,
    "indicator": bool(units),
    "evaluate": ready_patches > 0,
    "report": bool(evaluated),
}
notes = {
    "constraint": "需先建立矿山图斑",
    "unit": "需先建立矿山图斑",
    "indicator": "需先划分微地质单元",
    "evaluate": "需前置步骤齐备的图斑",
    "report": "需先执行适宜性评价",
}
steps = []
for s in STEPS:
    steps.append(dict(
        s,
        available=avail[s["key"]],
        pending=pending.get(s["key"], 0),
        note=None if avail[s["key"]] else
notes.get(s["key"], ""),
```

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        ))                                                                  2201
    unit_types = len({u.unit_type for u in units})                          2202
    stats = {                                                               2203
        "projects": projects,                                               2204
        "patches": len(patches),                                            2205
        "unit_types": unit_types,                                           2206
        "standards": Standard.query.count(),                                2207
    }                                                                       2208
    return {                                                                2209
        "stats": stats,                                                     2210
        "steps": steps,                                                     2211
        "badges": {k: v for k, v in pending.items() if v},                  2212
        "progress": {                                                       2213
            "patches": len(patches),                                        2214
            "units": len(units),                                            2215
            "data_units": len(data_unit_ids),                               2216
            "ready": ready_patches,                                         2217
            "evaluated": len(evaluated),                                    2218
        },                                                                  2219
    }                                                                       2220
def nav_badges():                                                           2221
    """侧边栏待办角标：{入口地址前缀： 数量}，只含有待办的项。"""                2222
    snap = snapshot()                                                       2223
    return {                                                                2224
        "/constraint/": snap["badges"].get("constraint", 0),               2225
        "/unit/": snap["badges"].get("unit", 0),                            2226
        "/indicator/": snap["badges"].get("indicator", 0),                  2227
        "/evaluate/": snap["badges"].get("evaluate", 0),                    2228
    }                                                                       2229
# ==== app/services/audit.py ====                                           2230
# -*- coding: utf-8 -*-                                                     2231
"""操作审计。                                                               2232
在应用工厂中以 after_request 钩子统一埋点：凡是成功的写操作（POST/PUT/DELETE）  2233
且已登录，即写入 operation_logs 表，无需在每个视图里手工调用。               2234
动作名称由「端点 → 中文动作」映射表给出，未收录的端点回退为方法 + 路径。       2235
"""                                                                         2236
from flask import request                                                   2237
from flask_login import current_user                                        2238
from app.models import db, OperationLog                                     2239
# 端点 → 可读动作名                                                         2240
ACTION_NAMES = {                                                            2241
    "auth.login": "登录系统",                                               2242
    "auth.logout": "退出登录",                                              2243
    "auth.add_user": "新增用户",                                            2244
    "auth.edit_user": "编辑用户",                                           2245
    "auth.delete_user": "删除用户",                                         2246
    "auth.reset_password": "重置用户密码",                                  2247
    "project.add": "新建项目",                                              2248
    "project.edit": "编辑项目",                                             2249
    "project.delete": "删除项目",                                           2250
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    "project.patch_add": "新建矿山图斑",
    "project.patch_edit": "编辑矿山图斑",
    "project.patch_delete": "删除矿山图斑",
    "project.upload_image": "上传遥感影像",
    "unit.add": "添加微地质单元",
    "unit.edit": "编辑微地质单元",
    "unit.delete": "删除微地质单元",
    "standard.add": "添加标准/政策文件",
    "standard.edit": "编辑标准/政策文件",
    "standard.delete": "删除标准/政策文件",
    "standard.rules": "维护标准规则",
    "standard.rule_delete": "删除标准规则",
    "constraint.check": "执行限制性条件判定",
    "ahp.save_expert": "录入专家判断矩阵",
    "ahp.delete_expert": "删除专家判断矩阵",
    "ahp.aggregate": "汇总多专家矩阵",
    "indicator.edit": "保存指标数据",
    "indicator.import_data": "批量导入指标数据",
    "indicator.update_meta": "维护指标定义",
    "evaluate.run": "执行适宜性评价",
}
# 无需记录的操作（查看类或纯导航）
SKIP_ENDPOINTS = {"static", "auth.index"}
def action_name(endpoint):
    if endpoint in ACTION_NAMES:
        return ACTION_NAMES[endpoint]
    return "写操作 %s %s" % (request.method, request.path)
def extract_target_id():
    """从视图参数中取出被操作对象的主键，取不到则返回 None。"""
    args = request.view_args or {}
    for key in ("pid", "patch_id", "unit_id", "sid", "rid",
    "result_id", "user_id"):
        if key in args:
            try:
                return int(args[key])
            except (TypeError, ValueError):
                return None
    return None
def record(endpoint, status_code):
    """记录一次写操作。失败（4xx/5xx）也记录，便于追溯。"""
    if endpoint in SKIP_ENDPOINTS:
        return
    try:
        detail = "路径 %s" % request.path
        if status_code >= 400:
            detail += ";结果 HTTP %d" % status_code
        db.session.add(OperationLog(
            user_id=current_user.id if
    current_user.is_authenticated else None,
            action=action_name(endpoint),
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                target_type=(endpoint or "").split(".")[0] or
    None,
                target_id=extract_target_id(),
                detail=detail,
            ))
            db.session.commit()
    except Exception:
            # 审计失败不应影响业务响应
            db.session.rollback()
# ==== app/views/auth.py ====
# -*- coding: utf-8 -*-
"""用户与权限管理 —— 登录、注销、用户维护、操作日志。"""
from flask import (Blueprint, render_template, redirect,
    url_for, request,
                       flash, abort)
from flask_login import login_user, logout_user,
    login_required, current_user
from werkzeug.security import check_password_hash,
    generate_password_hash
from app.forms import LoginForm, EmptyForm
from app.models import db, User, OperationLog
from app.utils.constants import ROLE_LABELS
bp = Blueprint("auth", __name__)
def require_admin():
    """非管理员直接 403，避免误操作基础数据。"""
    if current_user.role != "admin":
        abort(403)
@bp.route("/")
def index():
    if current_user.is_authenticated:
        from app.services import dashboard as dash
        return render_template("auth/home.html",
    dash=dash.snapshot())
    return redirect(url_for("auth.login"))
@bp.route("/login", methods=["GET", "POST"])
def login():
    if current_user.is_authenticated:
        return redirect(url_for("auth.index"))
    form = LoginForm()
    if form.validate_on_submit():
        user = User.query.filter_by(username=form.username.d
    ata).first()
        if user and check_password_hash(user.password_hash,
    form.password.data):
            if not user.active:
                flash("该账号已停用，请联系管理员", "error")
                return render_template("auth/login.html",
    form=form)
            login_user(user, remember=form.remember.data)
            # 用 toast 类别：渲染为页面正中的浮层，淡入停留后淡出。
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            # 登录成功只是告知性反馈，不需要用户停下来阅读。
            flash("登录成功", "toast")
            return redirect(url_for("auth.index"))
        flash("用户名或密码错误", "error")
    return render_template("auth/login.html", form=form)
@bp.route("/logout")
@login_required
def logout():
    logout_user()
    flash("已退出登录", "ok")
    return redirect(url_for("auth.login"))
# ---------------------------- 用户维护 ----------------------------
@bp.route("/users")
@login_required
def users():
    all_users = User.query.order_by(User.id).all()
    return render_template("auth/users.html",
    users=all_users,
                           role_labels=ROLE_LABELS,
    form=EmptyForm())
@bp.route("/users/add", methods=["GET", "POST"])
@login_required
def add_user():
    require_admin()
    if request.method == "POST":
        username = request.form.get("username", "").strip()
        password = request.form.get("password", "")
        role = request.form.get("role", "viewer")
        if not username or not password:
            flash("用户名和密码不能为空", "error")
            return redirect(url_for("auth.add_user"))
        if User.query.filter_by(username=username).first():
            flash("用户名已存在", "error")
            return redirect(url_for("auth.add_user"))
        db.session.add(User(username=username, role=role,
                            password_hash=generate_password_
    hash(password)))
        db.session.commit()
        flash("用户已添加", "ok")
        return redirect(url_for("auth.users"))
    return render_template("auth/add_user.html")
@bp.route("/users/<int:user_id>/edit", methods=["GET",
    "POST"])
@login_required
def edit_user(user_id):
    require_admin()
    user = db.get_or_404(User, user_id)
    if request.method == "POST":
        role = request.form.get("role", user.role)
```

```
        active = request.form.get("active") == "1"
        # 保护：不能把最后一个管理员降级或停用，否则无人能管理系统
        if user.role == "admin" and (role != "admin" or not
    active):
            others = User.query.filter(User.role == "admin",
                                       User.id != user.id,
    User.active.is_(True)).count()
            if others == 0:
                flash("系统至少需保留一名启用状态的管理员", "error")
                return redirect(url_for("auth.edit_user",
    user_id=user.id))
        user.role = role
        user.active = active
        new_password = request.form.get("password",
    "").strip()
        if new_password:
            user.password_hash = generate_password_hash(new_
    password)
        db.session.commit()
        flash("用户信息已保存", "ok")
        return redirect(url_for("auth.users"))
    return render_template("auth/edit_user.html", user=user,
                           role_labels=ROLE_LABELS,
    form=EmptyForm())
@bp.route("/users/<int:user_id>/delete", methods=["POST"])
@login_required
def delete_user(user_id):
    require_admin()
    user = db.get_or_404(User, user_id)
    if user.id == current_user.id:
        flash("不能删除当前登录的账号", "error")
        return redirect(url_for("auth.users"))
    if user.role == "admin":
        others = User.query.filter(User.role == "admin",
                                   User.id != user.id,
                                   User.active.is_(True)).co
    unt()
        if others == 0:
            flash("系统至少需保留一名启用状态的管理员", "error")
            return redirect(url_for("auth.users"))
    db.session.delete(user)
    db.session.commit()
    flash("用户「%s」已删除" % user.username, "ok")
    return redirect(url_for("auth.users"))
# ------------------------------ 操作日志 ------------------------------
    ----
@bp.route("/logs")
@login_required
def logs():
```

```
    page = request.args.get("page", 1, type=int)
    per_page = 50
    q = OperationLog.query.order_by(OperationLog.id.desc())
    total = q.count()
    records = q.offset((page - 1) * per_page).limit(per_page
    ).all()
    users_map = {u.id: u.username for u in User.query.all()}
    pages = max(1, (total + per_page - 1) // per_page)
    return render_template("auth/logs.html",
    records=records, page=page,
                           pages=pages, total=total,
    users_map=users_map)
# ==== app/views/project.py ====
# -*- coding: utf-8 -*-
"""项目与矿山档案 —— 项目、图斑、参照生态系统、遥感影像挂接。"""
from datetime import datetime
from flask import (Blueprint, render_template, request,
    redirect, url_for,
                   flash, current_app)
from flask_login import login_required, current_user
from app.forms import EmptyForm
from app.models import db, Project, MinePatch,
    ReferenceEcosystem
bp = Blueprint("project", __name__, url_prefix="/project")
@bp.route("/")
@login_required
def index():
    projects = Project.query.order_by(Project.id.desc()).all
    ()
    return render_template("project/index.html",
    projects=projects,
                           form=EmptyForm())
@bp.route("/add", methods=["GET", "POST"])
@login_required
def add():
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        p = Project(
            name=request.form.get("name", "").strip(),
            code=request.form.get("code", "").strip() or
    None,
            description=request.form.get("description",
    "").strip() or None,
            created_by=current_user.id,
        )
        if not p.name:
            flash("项目名称不能为空", "error")
        else:
            db.session.add(p)
```

```
            db.session.commit()                                          2501
            flash("项目已创建", "ok")                                    2502
            return redirect(url_for("project.index"))                    2503
    return render_template("project/edit.html", form=form,               2504
    project=None)                                                        2505
@bp.route("/<int:pid>/edit", methods=["GET", "POST"])                    2506
@login_required                                                          2507
def edit(pid):                                                           2508
    p = db.get_or_404(Project, pid)                                      2509
    form = EmptyForm()                                                   2510
    if request.method == "POST" and form.validate_on_submit(             2511
    ):                                                                   2512
        p.name = request.form.get("name", "").strip()                    2513
        p.code = request.form.get("code", "").strip() or                 2514
    None                                                                 2515
        p.description = request.form.get("description",                  2516
    "").strip() or None                                                  2517
        p.status = request.form.get("status", "进行中")                  2518
        db.session.commit()                                              2519
        flash("项目已保存", "ok")                                        2520
        return redirect(url_for("project.index"))                        2521
    return render_template("project/edit.html", form=form,               2522
    project=p)                                                           2523
@bp.route("/<int:pid>/delete", methods=["POST"])                         2524
@login_required                                                          2525
def delete(pid):                                                         2526
    p = db.get_or_404(Project, pid)                                      2527
    db.session.delete(p)                                                 2528
    db.session.commit()                                                  2529
    flash("项目已删除", "ok")                                            2530
    return redirect(url_for("project.index"))                            2531
# ------------------------------ 矿山图斑 ----------------------------2532
    ----                                                                 2533
@bp.route("/<int:pid>/patches")                                          2534
@login_required                                                          2535
def patches(pid):                                                        2536
    p = db.get_or_404(Project, pid)                                      2537
    return render_template("project/patches.html",                       2538
    project=p,                                                           2539
                                 form=EmptyForm())                       2540
@bp.route("/patch/<int:patch_id>/delete", methods=["POST"])              2541
@login_required                                                          2542
def patch_delete(patch_id):                                              2543
    """删除矿山图斑，及其微地质单元与指标数据。"""                       2544
    from app.models import MicroUnit, IndicatorData,                     2545
    EvaluationResult, \                                                  2546
        ScoreDetail, ConstraintCheck, ReferenceEcosystem                 2547
    patch = db.get_or_404(MinePatch, patch_id)                           2548
    pid = patch.project_id                                               2549
    unit_ids = [u.id for u in MicroUnit.query.filter_by(patc             2550
```

```
    h_id=patch.id).all()]
    if unit_ids:
        IndicatorData.query.filter(IndicatorData.unit_id.in_
    (unit_ids)).delete(
            synchronize_session=False)
    for r in EvaluationResult.query.filter_by(patch_id=patch
    .id).all():
        ScoreDetail.query.filter_by(result_id=r.id).delete(
            synchronize_session=False)
    EvaluationResult.query.filter_by(patch_id=patch.id).dele
    te(
        synchronize_session=False)
    ConstraintCheck.query.filter_by(patch_id=patch.id).delet
    e(
        synchronize_session=False)
    MicroUnit.query.filter_by(patch_id=patch.id).delete(sync
    hronize_session=False)
    ReferenceEcosystem.query.filter_by(patch_id=patch.id).de
    lete(
        synchronize_session=False)
    db.session.delete(patch)
    db.session.commit()
    flash("图斑「%s」及其数据已删除" % patch.name, "ok")
    return redirect(url_for("project.patches", pid=pid))
@bp.route("/<int:pid>/patch/add", methods=["GET", "POST"])
@login_required
def patch_add(pid):
    p = db.get_or_404(Project, pid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        def fv(name, cast=str):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        patch = MinePatch(
            project_id=p.id,
            name=request.form.get("name", "").strip(),
            admin_region=fv("admin_region"),
            area_ha=fv("area_ha", float),
            mineral_type=fv("mineral_type"),
            mining_method=fv("mining_method"),
            stop_years=fv("stop_years", float),
            responsible_entity_missing=bool(request.form.get
    ("responsible_entity_missing")),
        )
        if not patch.name:
            flash("图斑名称不能为空", "error")
        else:
            db.session.add(patch)
            db.session.flush()
```

```
            # 参照生态系统默认取论文值                                    2601
            db.session.add(ReferenceEcosystem(                          2602
                patch_id=patch.id, coverage_benchmark=75.0,             2603
    shannon_benchmark=0.7))                                             2604
            db.session.commit()                                         2605
            flash("图斑已创建", "ok")                                    2606
            return redirect(url_for("project.patches",                  2607
    pid=p.id))                                                          2608
    return render_template("project/patch_edit.html",                   2609
    form=form, project=p, patch=None)                                   2610
@bp.route("/patch/<int:patch_id>/edit", methods=["GET",                 2611
    "POST"])                                                            2612
@login_required                                                         2613
def patch_edit(patch_id):                                               2614
    patch = db.get_or_404(MinePatch, patch_id)                          2615
    form = EmptyForm()                                                  2616
    if request.method == "POST" and form.validate_on_submit(            2617
    ):                                                                  2618
        def fv(name, cast=str):                                         2619
            v = request.form.get(name, "").strip()                      2620
            return cast(v) if v != "" else None                         2621
        patch.name = request.form.get("name", "").strip()               2622
        patch.admin_region = fv("admin_region")                         2623
        patch.area_ha = fv("area_ha", float)                            2624
        patch.mineral_type = fv("mineral_type")                         2625
        patch.mining_method = fv("mining_method")                       2626
        patch.stop_years = fv("stop_years", float)                      2627
        patch.responsible_entity_missing =                              2628
    bool(request.form.get("responsible_entity_missing"))                2629
        # 参照生态系统                                                  2630
        ref = ReferenceEcosystem.query.filter_by(patch_id=pa            2631
    tch.id).first()                                                     2632
        if ref is None:                                                 2633
            ref = ReferenceEcosystem(patch_id=patch.id)                 2634
            db.session.add(ref)                                         2635
        ref.coverage_benchmark = fv("coverage_benchmark",               2636
    float) or 75.0                                                      2637
        ref.shannon_benchmark = fv("shannon_benchmark",                 2638
    float) or 0.7                                                       2639
        ref.dominant_species = fv("dominant_species")                   2640
        patch.image_note = fv("image_note")                             2641
        db.session.commit()                                             2642
        flash("图斑已保存", "ok")                                        2643
        return redirect(url_for("project.patches",                      2644
    pid=patch.project_id))                                              2645
    return render_template("project/patch_edit.html",                   2646
    form=form,                                                          2647
                           project=patch.project,                       2648
    patch=patch,                                                        2649
                           reference=ReferenceEcosystem.quer            2650
```

```
    y.filter_by(                                                    2651
                                        patch_id=patch.id).first())  2652
# ---------------------------- 遥感影像挂接                          2653
    ----------------------------                                    2654
@bp.route("/patch/<int:patch_id>/image", methods=["POST"])          2655
@login_required                                                     2656
def upload_image(patch_id):                                         2657
    """上传遥感底图 / 无人机正射影像并挂接到图斑。"""                  2658
    from app.utils import uploads as up                             2659
    patch = db.get_or_404(MinePatch, patch_id)                      2660
    form = EmptyForm()                                              2661
    if not form.validate_on_submit():                               2662
        flash("表单校验失败，请重试", "error")                      2663
        return redirect(url_for("project.patch_edit",               2664
    patch_id=patch.id))                                             2665
    rel, err = up.save_upload(request.files.get("image"),           2666
    "patches",                                                      2667
                                    current_app.config["ALLOWED_IM2668
    AGE_EXT"],                                                      2669
                                    current_app.config["UPLOAD_FOL2670
    DER"])                                                          2671
    if err:                                                         2672
        flash(err, "error")                                         2673
        return redirect(url_for("project.patch_edit",               2674
    patch_id=patch.id))                                             2675
    # 替换旧影像时清理原文件，避免残留                              2676
    if patch.image_path:                                            2677
        up.remove_upload(patch.image_path,                          2678
    current_app.config["UPLOAD_FOLDER"])                            2679
    patch.image_path = rel                                          2680
    patch.image_uploaded_at = datetime.utcnow()                     2681
    note = request.form.get("image_note", "").strip()               2682
    if note:                                                        2683
        patch.image_note = note                                     2684
    db.session.commit()                                             2685
    flash("遥感影像已挂接", "ok")                                   2686
    return redirect(url_for("project.patch_edit",                   2687
    patch_id=patch.id))                                             2688
@bp.route("/patch/<int:patch_id>/image/delete",                     2689
    methods=["POST"])                                               2690
@login_required                                                     2691
def delete_image(patch_id):                                         2692
    from app.utils import uploads as up                             2693
    patch = db.get_or_404(MinePatch, patch_id)                      2694
    form = EmptyForm()                                              2695
    if form.validate_on_submit():                                   2696
        up.remove_upload(patch.image_path,                          2697
    current_app.config["UPLOAD_FOLDER"])                            2698
        patch.image_path = None                                     2699
        patch.image_uploaded_at = None                              2700
```

```
            db.session.commit()                                          2701
            flash("遥感影像已移除", "ok")                                  2702
        return redirect(url_for("project.patch_edit",                    2703
        patch_id=patch.id))                                              2704
# ==== app/views/unit.py ====                                            2705
# -*- coding: utf-8 -*-                                                  2706
"""微地质单元管理 —— 岩质边坡 / 碎石边坡 / 平台的划分与面积占比。"""          2707
from flask import Blueprint, render_template, request,                   2708
    redirect, url_for, flash                                             2709
from flask_login import login_required                                   2710
from app.forms import EmptyForm                                          2711
from app.models import db, MinePatch, MicroUnit,                         2712
    IndicatorData                                                        2713
from app.utils.constants import UNIT_TYPES                               2714
bp = Blueprint("unit", __name__, url_prefix="/unit")                     2715
def recalc_ratio(patch_id):                                              2716
    """按面积重算该图斑各单元面积占比。"""                                  2717
    units = MicroUnit.query.filter_by(patch_id=patch_id).all             2718
    ()                                                                   2719
    total = sum(u.area or 0.0 for u in units)                            2720
    if total > 0:                                                        2721
        for u in units:                                                  2722
            u.ratio = round((u.area or 0.0) / total, 6)                  2723
    return total                                                         2724
@bp.route("/")                                                           2725
@login_required                                                          2726
def list_patches():                                                      2727
    """微地质单元入口：列出各图斑及其单元划分情况。"""                      2728
    patches = MinePatch.query.order_by(MinePatch.id.desc()).             2729
    all()                                                                2730
    rows = []                                                            2731
    for p in patches:                                                    2732
        units = MicroUnit.query.filter_by(patch_id=p.id).all             2733
    ()                                                                   2734
        rows.append({                                                    2735
            "patch": p,                                                  2736
            "count": len(units),                                         2737
            "ratio_sum": sum(u.ratio or 0.0 for u in units),             2738
        })                                                               2739
    return render_template("unit/list.html", rows=rows)                  2740
@bp.route("/<int:patch_id>")                                             2741
@login_required                                                          2742
def index(patch_id):                                                     2743
    patch = db.get_or_404(MinePatch, patch_id)                           2744
    units = MicroUnit.query.filter_by(patch_id=patch_id).all             2745
    ()                                                                   2746
    total = sum(u.area or 0.0 for u in units)                            2747
    ratio_sum = sum(u.ratio or 0.0 for u in units)                       2748
    return render_template("unit/index.html", patch=patch,               2749
    units=units,                                                         2750
```

```
                                total_area=total,
    ratio_sum=ratio_sum,
                                unit_types=UNIT_TYPES,
    form=EmptyForm())
@bp.route("/<int:patch_id>/add", methods=["POST"])
@login_required
def add(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if form.validate_on_submit():
        unit_type = request.form.get("unit_type")
        if unit_type not in UNIT_TYPES:
            flash("微地质单元类型不正确", "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        # 论文 4.1 节将图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
        # 允许重复会让面积占比与按单元类型组织的指标数据对不上号。
        if MicroUnit.query.filter_by(patch_id=patch.id,
                                     unit_type=unit_type).fi
    rst():
            flash("该图斑已存在「%s」，同一类型只能划分一个" % unit_type,
    "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        area = request.form.get("area", "").strip()
        u = MicroUnit(patch_id=patch.id,
    unit_type=unit_type,
                      area=float(area) if area else None,
                      description=request.form.get("descript
    ion", "").strip() or None)
        db.session.add(u)
        db.session.flush()
        recalc_ratio(patch.id)
        db.session.commit()
        flash("微地质单元已添加", "ok")
    return redirect(url_for("unit.index",
    patch_id=patch_id))
@bp.route("/edit/<int:unit_id>", methods=["GET", "POST"])
@login_required
def edit(unit_id):
    """单个微地质单元的编辑页。"""
    u = db.get_or_404(MicroUnit, unit_id)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        new_type = request.form.get("unit_type",
    u.unit_type)
        if new_type not in UNIT_TYPES:
            flash("微地质单元类型不正确", "error")
            return redirect(url_for("unit.edit",
```

```
    unit_id=u.id))                                                    2801
        # 改类型时同样不允许与本图斑已有的其他单元重复                  2802
        dup = MicroUnit.query.filter_by(patch_id=u.patch_id,          2803
                                                                      2804
    unit_type=new_type).first()                                       2805
        if dup is not None and dup.id != u.id:                        2806
            flash("该图斑已存在「%s」，同一类型只能划分一个" % new_type,  2807
    "error")                                                          2808
            return redirect(url_for("unit.edit",                      2809
    unit_id=u.id))                                                    2810
        u.unit_type = new_type                                        2811
        area = request.form.get("area", "").strip()                   2812
        u.area = float(area) if area else None                        2813
        u.description = request.form.get("description",               2814
    "").strip() or None                                               2815
        recalc_ratio(u.patch_id)                                      2816
        db.session.commit()                                           2817
        flash("微地质单元已保存", "ok")                               2818
        return redirect(url_for("unit.index",                         2819
    patch_id=u.patch_id))                                             2820
    return render_template("unit/edit.html", unit=u,                  2821
    patch=u.patch,                                                    2822
                                unit_types=UNIT_TYPES, form=form)     2823
@bp.route("/delete/<int:unit_id>", methods=["POST"])                  2824
@login_required                                                       2825
def delete(unit_id):                                                  2826
    u = db.get_or_404(MicroUnit, unit_id)                             2827
    pid = u.patch_id                                                  2828
    name = u.unit_type                                                2829
    # 一并删除该单元的指标数据，避免残留                             2830
    IndicatorData.query.filter_by(unit_id=u.id).delete(synch          2831
    ronize_session=False)                                             2832
    db.session.delete(u)                                              2833
    db.session.flush()                                                2834
    recalc_ratio(pid)                                                 2835
    db.session.commit()                                               2836
    flash("微地质单元「%s」已删除" % name, "ok")                      2837
    return redirect(url_for("unit.index", patch_id=pid))              2838
# ==== app/views/standard.py ====                                     2839
# -*- coding: utf-8 -*-                                               2840
"""政策与标准知识库 —— 标准 / 政策文件的结构化规则与省际口径比对。"""  2841
from flask import Blueprint, render_template, request,                2842
    redirect, url_for, flash                                          2843
from flask_login import login_required                                2844
from app.forms import EmptyForm                                       2845
from app.models import db, Standard, StandardRule                     2846
bp = Blueprint("standard", __name__, url_prefix="/standard")          2847
@bp.route("/")                                                        2848
@login_required                                                       2849
def index():                                                          2850
```

```
    standards = Standard.query.order_by(Standard.category,   2851
    Standard.province).all()                                 2852
    local = [s for s in standards if s.category == "地方标准"]  2853
    policy = [s for s in standards if s.category != "地方标准"] 2854
    return render_template("standard/index.html",            2855
    local=local, policy=policy,                              2856
                                form=EmptyForm())            2857
@bp.route("/add", methods=["GET", "POST"])                   2858
@login_required                                              2859
def add():                                                   2860
    form = EmptyForm()                                       2861
    if request.method == "POST" and form.validate_on_submit(2862
    ):                                                       2863
        s = Standard(                                        2864
            province=request.form.get("province",            2865
    "").strip(),                                             2866
            name=request.form.get("name", "").strip(),       2867
            doc_number=request.form.get("doc_number",        2868
    "").strip() or None,                                     2869
            publish_date=request.form.get("publish_date",    2870
    "").strip() or None,                                     2871
            implement_date=request.form.get("implement_date"2872
    , "").strip() or None,                                   2873
            category=request.form.get("category", "地方标准"),  2874
            summary=request.form.get("summary", "").strip()  2875
    or None,                                                 2876
        )                                                    2877
        if not s.province or not s.name:                     2878
            flash("省份与名称不能为空", "error")                2879
        else:                                                2880
            db.session.add(s)                                2881
            db.session.commit()                              2882
            flash("标准已添加", "ok")                          2883
            return redirect(url_for("standard.index"))       2884
    return render_template("standard/edit.html", form=form,  2885
    standard=None)                                           2886
@bp.route("/<int:sid>/edit", methods=["GET", "POST"])        2887
@login_required                                              2888
def edit(sid):                                               2889
    s = db.get_or_404(Standard, sid)                         2890
    form = EmptyForm()                                       2891
    if request.method == "POST" and form.validate_on_submit(2892
    ):                                                       2893
        s.province = request.form.get("province",            2894
    "").strip()                                              2895
        s.name = request.form.get("name", "").strip()        2896
        s.doc_number = request.form.get("doc_number",        2897
    "").strip() or None                                      2898
        s.publish_date = request.form.get("publish_date",    2899
    "").strip() or None                                      2900
```

```
            s.implement_date = request.form.get("implement_date"
    , "").strip() or None
            s.category = request.form.get("category", "地方标准")
            s.summary = request.form.get("summary", "").strip()
    or None
            db.session.commit()
            flash("已保存", "ok")
            return redirect(url_for("standard.rules", sid=s.id))
    return render_template("standard/edit.html", form=form,
    standard=s)
@bp.route("/<int:sid>/delete", methods=["POST"])
@login_required
def delete(sid):
    s = db.get_or_404(Standard, sid)
    StandardRule.query.filter_by(standard_id=sid).delete()
    db.session.delete(s)
    db.session.commit()
    flash("已删除", "ok")
    return redirect(url_for("standard.index"))
@bp.route("/<int:sid>/rules", methods=["GET", "POST"])
@login_required
def rules(sid):
    s = db.get_or_404(Standard, sid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        r = StandardRule(
            standard_id=s.id,
            item=request.form.get("item", "").strip(),
            operator=request.form.get("operator", ">="),
            threshold=request.form.get("threshold",
    "").strip() or None,
            condition_type=request.form.get("condition_type"
    , "必要条件"),
            region=request.form.get("region", "").strip()
    or None,
            note=request.form.get("note", "").strip() or
    None,
        )
        if not r.item:
            flash("判定项不能为空", "error")
        else:
            db.session.add(r)
            db.session.commit()
            flash("规则已添加", "ok")
        return redirect(url_for("standard.rules", sid=s.id))
    rs = StandardRule.query.filter_by(standard_id=s.id).all(
    )
    return render_template("standard/rules.html",
    form=form, standard=s, rules=rs)
```

```
@bp.route("/rule/<int:rid>/delete", methods=["POST"])
@login_required
def rule_delete(rid):
    r = db.get_or_404(StandardRule, rid)
    sid = r.standard_id
    db.session.delete(r)
    db.session.commit()
    flash("规则已删除", "ok")
    return redirect(url_for("standard.rules", sid=sid))
@bp.route("/compare")
@login_required
def compare():
    """省际口径比对：同一指标在不同省份的阈值横向对比。"""
    rules = StandardRule.query.join(Standard).all()
    groups = {}
    for r in rules:
        groups.setdefault(r.item, []).append(r)
    return render_template("standard/compare.html",
    groups=groups)
# ==== app/views/constraint.py ====
# -*- coding: utf-8 -*-
"""限制性指标判定 —— 一票否决判定。
按论文表 2 与 4.1 节流程，共 5 项：政策、区位约束、土壤污染、地质灾害、耕地属性。
任一项不通过即判定为"人工辅助或生态重建"，不再进入定量评价。
"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, MinePatch, ConstraintCheck
from app.utils.constants import CONSTRAINT_ITEMS
bp = Blueprint("constraint", __name__,
    url_prefix="/constraint")
# 字段名 -> 中文名
FIELDS = [
    ("policy_pass", "政策约束"),
    ("location_pass", "区位约束"),
    ("pollution_pass", "土壤污染"),
    ("geo_pass", "地质灾害"),
    ("farmland_pass", "耕地属性"),
]
@bp.route("/")
@login_required
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    latest = {}
    for p in patches:
        latest[p.id] = ConstraintCheck.query.filter_by(patch
    _id=p.id) \
```

```
                .order_by(ConstraintCheck.id.desc()).first()
    return render_template("constraint/index.html",
    patches=patches, latest=latest)
@bp.route("/<int:patch_id>", methods=["GET", "POST"])
@login_required
def check(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    record = ConstraintCheck.query.filter_by(patch_id=patch_
    id) \
        .order_by(ConstraintCheck.id.desc()).first()
    if request.method == "POST" and form.validate_on_submit(
    ):
        if record is None:
            record = ConstraintCheck(patch_id=patch.id)
            db.session.add(record)
        for field, _ in FIELDS:
            setattr(record, field, request.form.get(field)
    == "1")
        record.passed = all(getattr(record, f) for f, _ in
    FIELDS)
        failed = [name for f, name in FIELDS if not
    getattr(record, f)]
        record.reason = ("全部通过" if record.passed
                         else "未通过：" + "、".join(failed))
        db.session.commit()
        flash("判定结果已保存", "ok")
        return redirect(url_for("constraint.check",
    patch_id=patch.id))
    # 预处理各条件的当前取值，供模板直接使用（Jinja 无 getattr）
    field_states = [(f, name, None if record is None else
    getattr(record, f))
                    for f, name in FIELDS]
    return render_template("constraint/check.html",
    patch=patch, form=form,
                           fields=FIELDS,
    field_states=field_states,
                           record=record)
# ==== app/views/ahp.py ====
# -*- coding: utf-8 -*-
"""指标体系与 AHP 权重 —— 路由与页面逻辑。"""
import json
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from app.forms import AHPSelectForm, EmptyForm
from app.models import db, Indicator, ExpertMatrix,
    WeightScheme
from app.utils.constants import DEFAULT_WEIGHTS, UNIT_TYPES
from app.services import ahp_service as ahp
from app.services import indicator_service as ind_svc
```

```
from app.services import weight_service as ws
bp = Blueprint("ahp", __name__, url_prefix="/ahp")
#  一次计算的权重结果对应的指标编码：名称 → 编码
_L1_CODE = {"土壤指标": "soil", "植被指标": "veg", "地形指标":
    "terrain"}
_L2_CODE = {
    ("土壤指标", "土石比"): "soil_ratio",
    ("土壤指标", "土层厚度"): "thickness",
    ("土壤指标", "pH"): "ph",
    ("土壤指标", "有机质"): "organic_matter",
    ("土壤指标", "污染指标"): "pollution",
    ("植被指标", "植被覆盖度"): "coverage",
    ("植被指标", "生物多样性"): "shannon",
    ("地形指标", "坡度"): "slope",
    ("地形指标", "坡向"): "aspect",
}
def to_code_pairs(unit_type, level, l1, names, weights):
    """把一次 AHP 计算的权重结果转成「指标编码 → 权重」，便于落库。"""
    pairs = []
    for name, w in zip(names, weights):
        if level == 1:
            code = _L1_CODE.get(name)
        else:
            code = _L2_CODE.get((l1, name))
        if code:
            pairs.append((code, w))
    return pairs
def indicator_list(unit_type, level, l1=None):
    """返回 (指标名称列表, 默认权重列表)。
    level=1 时返回一级指标；level=2 时返回指定一级指标下的二级指标。
    """
    weights = DEFAULT_WEIGHTS[unit_type]
    if level == 1:
        names = list(weights["l1"].keys())
        vec = [weights["l1"][n] for n in names]
        return names, vec
    # level == 2
    l2 = weights["l2"][l1]
    names = list(l2.keys())
    vec = [l2[n] for n in names]
    return names, vec
def l1_choices(unit_type):
    """某单元类型的一级指标名称列表（用于二级指标录入时选择父指标）。"""
    return list(DEFAULT_WEIGHTS[unit_type]["l1"].keys())
@bp.route("/", methods=["GET", "POST"])
def index():
    form = AHPSelectForm()
    # 先根据请求中的单元类型填充 l1 选项，再进行表单校验
    unit = request.form.get("unit_type") or
    form.unit_type.choices[0][0]
```

```
    form.l1.choices = [(n, n) for n in l1_choices(unit)]
    if form.validate_on_submit():
        unit_type = form.unit_type.data
        level = int(form.level.data)
        if level == 1:
            return redirect(url_for("ahp.matrix",
unit_type=unit_type, level=1))
        if not form.l1.data:
            flash("选择「二级指标」时，请同时选择所属的一级指标", "error")
            return render_template("ahp/index.html",
form=form,
                                   weight_rows=ws.summary())
        return redirect(url_for("ahp.matrix",
unit_type=unit_type,
                                level=2, l1=form.l1.data))
    return render_template("ahp/index.html", form=form,
                           weight_rows=ws.summary())
@bp.route("/matrix", methods=["GET", "POST"])
def matrix():
    unit_type = request.values.get("unit_type")
    level = int(request.values.get("level", 1))
    l1 = request.values.get("l1")
    if unit_type not in DEFAULT_WEIGHTS:
        flash("无效的微地质单元类型", "error")
        return redirect(url_for("ahp.index"))
    names, vec = indicator_list(unit_type, level, l1)
    n = len(names)
    form = EmptyForm()
    # 预填默认值：论文表 3 权重对应的完全一致矩阵 a_ij = w_i / w_j
    default = [[round(vec[i] / vec[j], 4) for j in
range(n)] for i in range(n)]
    def compute(upper):
        A = ahp.matrix_from_upper(upper)
        r = ahp.weights_and_consistency(A)
        bad = ahp.find_most_inconsistent(A)
        return {
            "weights": [round(v, 4) for v in r["weights"]],
            "lambda_max": round(r["lambda_max"], 4),
            "ci": r["ci"],
            "cr": r["cr"],
            "consistent": r["consistent"],
            "matrix": [[round(A[i, j], 4) for j in
range(n)] for i in range(n)],
            "most_inconsistent": bad,
        }
    result = None
    if request.method == "POST" and form.validate_on_submit(
    ):
        action = request.form.get("action", "calc")
        try:
```

```
            upper = [float(request.form.get("m_%d_%d" % (i,
    j), "1"))
                     for i in range(n) for j in range(i +
    1, n)]
            result = compute(upper)
        except (ValueError, ZeroDivisionError):
            flash("矩阵输入有误：请填入正数（对角线自动为 1，下三角自动取倒数）",
    "error")
            return redirect(url_for("ahp.matrix",
    unit_type=unit_type,
                                    level=level, l1=l1))
        if action == "save":
            if not result["consistent"]:
                flash("一致性比例 CR=%.4f ≥ 0.1，判断矩阵不满足一致性要求，"
                      "请调整后再保存" % result["cr"], "error")
            else:
                pairs = to_code_pairs(unit_type, level, l1,
    names,
                                      result["weights"])
                scheme, cnt = ws.save_weights(unit_type,
    pairs)
                flash("已保存并为「%s」生效 %d 项权重（CR=%.4f）。"
                      "此后该单元类型的评价将以此权重计算。"
                      % (unit_type, cnt, result["cr"]),
    "ok")
                return redirect(url_for("ahp.matrix",
    unit_type=unit_type,
                                        level=level, l1=l1))
    elif request.values.get("calc") == "1":
        # 按默认（论文表 3）矩阵直接试算，便于一键查看权重与一致性
        result = compute([default[i][j] for i in range(n)
    for j in range(i + 1, n)])
    _, source_label, custom_scheme = ws.effective_weights(un
    it_type)
    return render_template(
        "ahp/matrix.html",
        form=form, unit_type=unit_type, level=level, l1=l1,
        names=names, vec=vec, default=default,
    result=result,
        source_label=source_label, custom_scheme=custom_sche
    me,
    )
@bp.route("/weights/reset", methods=["POST"])
def weights_reset():
    """删除自定义权重方案，恢复为该单元类型的论文表 3 默认权重。"""
    unit_type = request.form.get("unit_type")
    form = EmptyForm()
    if form.validate_on_submit():
        if unit_type not in DEFAULT_WEIGHTS:
            flash("无效的微地质单元类型", "error")
```

```
        else:                                                          3201
            n = ws.reset_weights(unit_type)                            3202
            if n:                                                      3203
                flash("已恢复「%s」为论文表 3 默认权重" % unit_type,          3204
    "ok")                                                              3205
            else:                                                      3206
                flash("「%s」本就是论文表 3 默认权重，无需恢复" % unit_type,   3207
    "ok")                                                              3208
    return redirect(url_for("ahp.index"))                              3209
# ---------------------------- 指标维护 ----------------------------   3210
    ----                                                               3211
@bp.route("/indicators")                                               3212
def indicators():                                                      3213
    """指标定义维护：查看指标树、单位、理想值与取值依据。"""                    3214
    if Indicator.query.count() == 0:                                   3215
        ind_svc.seed_indicators()                                      3216
    all_ind = Indicator.query.order_by(Indicator.level,                3217
    Indicator.id).all()                                                3218
    by_parent = {}                                                     3219
    for i in all_ind:                                                  3220
        by_parent.setdefault(i.parent_code, []).append(i)              3221
    return render_template("ahp/indicators.html",                      3222
    by_parent=by_parent,                                               3223
                                  all_ind=all_ind,                     3224
    unit_types=UNIT_TYPES,                                             3225
                                  type_label=ind_svc.ideal_type_lab    3226
    el,                                                                3227
                                  form=EmptyForm())                    3228
@bp.route("/indicators/<code>/edit", methods=["GET",                   3229
    "POST"])                                                           3230
def indicator_edit(code):                                              3231
    ind = Indicator.query.filter_by(code=code).first_or_404(           3232
    )                                                                  3233
    form = EmptyForm()                                                 3234
    if request.method == "POST" and form.validate_on_submit(           3235
    ):                                                                 3236
        name = request.form.get("name", "").strip()                    3237
        if not name:                                                   3238
            flash("指标名称不能为空", "error")                            3239
            return redirect(url_for("ahp.indicator_edit",              3240
    code=code))                                                        3241
        ind.name = name                                                3242
        ind.unit = request.form.get("unit", "").strip() or             3243
    None                                                               3244
        ind.ideal_value = request.form.get("ideal_value",              3245
    "").strip() or None                                                3246
        ind.ideal_type = request.form.get("ideal_type") or             3247
    ind.ideal_type                                                     3248
        units = request.form.getlist("enabled_units")                  3249
        ind.enabled_units = ",".join(units) if units else              3250
```

```
    None                                                                3251
        ind.is_constraint = request.form.get("is_constraint"           3252
    ) == "1"                                                            3253
        ind.description = request.form.get("description",               3254
    "").strip() or None                                                 3255
        db.session.commit()                                             3256
        flash("指标「%s」已保存" % ind.name, "ok")                          3257
        return redirect(url_for("ahp.indicators"))                      3258
    return render_template("ahp/indicator_edit.html",                   3259
    ind=ind,                                                            3260
                           unit_types=UNIT_TYPES,                       3261
                           type_label=ind_svc.ideal_type_lab            3262
    el, form=form)                                                      3263
@bp.route("/indicators/reset", methods=["POST"])                        3264
def indicators_reset():                                                 3265
    """按论文取值重置指标定义。"""                                          3266
    form = EmptyForm()                                                  3267
    if form.validate_on_submit():                                       3268
        n = ind_svc.seed_indicators()                                   3269
        flash("已按论文取值重置 %d 项指标定义" % n, "ok")                     3270
    return redirect(url_for("ahp.indicators"))                          3271
# ---------------------------- 多专家矩阵 ---------------------------    3272
    -----                                                               3273
def expert_scope():                                                     3274
    unit_type = request.values.get("unit_type") or                      3275
    UNIT_TYPES[0]                                                       3276
    level = int(request.values.get("level", 1))                         3277
    l1 = request.values.get("l1") or None                               3278
    return unit_type, level, l1                                         3279
def expert_query(unit_type, level, l1):                                 3280
    q = ExpertMatrix.query.filter_by(unit_type=unit_type,               3281
    level=level)                                                        3282
    if level == 2:                                                      3283
        q = q.filter_by(l1_name=l1)                                     3284
    else:                                                               3285
        q = q.filter(ExpertMatrix.l1_name.is_(None))                    3286
    return q.order_by(ExpertMatrix.id)                                  3287
def default_scheme(unit_type, level, l1=None):                          3288
    """取（或建）该范围的权重方案，专家矩阵统一挂到它下面。"""                3289
    name = "%s · %s%s" % (unit_type,                                    3290
                          "一级指标" if level == 1 else "二级指标",       3291
                          "(%s)" % l1 if l1 else "")                    3292
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ            3293
    e,                                                                  3294
                                          level=level,                  3295
    name=name).first()                                                  3296
    if scheme is None:                                                  3297
        scheme = WeightScheme(unit_type=unit_type,                      3298
    level=level, name=name,                                             3299
                          is_default=False)                             3300
```

```
        db.session.add(scheme)
        db.session.flush()
    return scheme
@bp.route("/experts", methods=["GET", "POST"])
def experts():
    """多专家判断矩阵录入与汇总。"""
    unit_type, level, l1 = expert_scope()
    if unit_type not in DEFAULT_WEIGHTS:
        flash("无效的微地质单元类型", "error")
        return redirect(url_for("ahp.index"))
    names, vec = indicator_list(unit_type, level, l1)
    n = len(names)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        action = request.form.get("action")
        if action == "add":
            expert_name = request.form.get("expert_name",
    "").strip() or "未具名专家"
            try:
                upper = [float(request.form.get("m_%d_%d" %
    (i, j), "1"))
                         for i in range(n) for j in range(i
    + 1, n)]
            except ValueError:
                flash("矩阵输入有误，请填入正数", "error")
                return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                                        level=level, l1=l1))
            A = ahp.matrix_from_upper(upper)
            r = ahp.weights_and_consistency(A)
            scheme = default_scheme(unit_type, level, l1)
            db.session.add(ExpertMatrix(
                scheme_id=scheme.id,
                unit_type=unit_type, l1_name=(l1 if level
    == 2 else None),
                expert_name=expert_name, level=level,
                matrix_json=json.dumps([list(map(float,
    row)) for row in A]),
                cr=round(r["cr"], 6)))
            db.session.commit()
            flash("已录入专家「%s」的判断矩阵（CR=%.4f）" % (expert_name,
    r["cr"]), "ok")
        elif action == "save_agg":
            # 把汇总权重落库并生效，使多专家结果真正参与评价计算
            weights = _aggregate(unit_type, level, l1)
            if weights is None:
                flash("暂无可汇总的专家矩阵", "error")
            elif not weights["consistent"]:
                flash("汇总矩阵一致性比例 CR=%.4f ≥ 0.1，不满足要求，"
```

```
                              "请先检查或调整专家矩阵" % weights["cr"],
    "error")
            else:
                names_, _ = indicator_list(unit_type,
    level, l1)
                pairs = to_code_pairs(unit_type, level, l1,
    names_,
                                      weights["weights"])
                _, cnt = ws.save_weights(unit_type, pairs)
                flash("已保存并为「%s」生效 %d 项汇总权重（CR=%.4f）"
                      % (unit_type, cnt, weights["cr"]),
    "ok")
        return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                                level=level, l1=l1))
    rows = expert_query(unit_type, level, l1).all()
    default = [[round(vec[i] / vec[j], 4) for j in
    range(n)] for i in range(n)]
    return render_template("ahp/experts.html",
    unit_type=unit_type, level=level,
                           l1=l1, names=names, rows=rows,
    default=default,
                           n=n, form=form)
@bp.route("/experts/<int:mid>/delete", methods=["POST"])
def expert_delete(mid):
    row = db.get_or_404(ExpertMatrix, mid)
    unit_type, level, l1 = row.unit_type, row.level,
    row.l1_name
    form = EmptyForm()
    if form.validate_on_submit():
        db.session.delete(row)
        db.session.commit()
        flash("已删除专家矩阵", "ok")
    return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                            level=level, l1=l1))
def _aggregate(unit_type, level, l1):
    """按几何平均汇总多专家矩阵并做一致性检验。无矩阵时返回 None。"""
    rows = expert_query(unit_type, level, l1).all()
    mats = []
    for r in rows:
        try:
            mats.append([list(map(float, line))
                         for line in json.loads(r.matrix_jso
    n)])
        except Exception:
            continue
    if not mats:
        return None
    agg = ahp.aggregate_matrices(mats)
```

```
    res = ahp.weights_and_consistency(agg)                          3401
    return {                                                        3402
        "count": len(mats),                                         3403
        "weights": res["weights"],                                  3404
        "lambda_max": res["lambda_max"],                            3405
        "ci": res["ci"], "cr": res["cr"],                           3406
        "consistent": res["consistent"],                            3407
        "most_inconsistent": ahp.find_most_inconsistent(agg)        3408
    ,                                                               3409
        "expert_crs": [r.cr for r in rows if r.cr is not            3410
    None],                                                          3411
    }                                                               3412
@bp.route("/experts/aggregate")                                     3413
def experts_aggregate():                                            3414
    """按几何平均汇总多专家矩阵，并做一致性检验。"""                    3415
    unit_type, level, l1 = expert_scope()                           3416
    if unit_type not in DEFAULT_WEIGHTS:                            3417
        flash("无效的微地质单元类型", "error")                        3418
        return redirect(url_for("ahp.index"))                       3419
    names, vec = indicator_list(unit_type, level, l1)               3420
    rows = expert_query(unit_type, level, l1).all()                 3421
    result = _aggregate(unit_type, level, l1)                       3422
    if result is not None:                                          3423
        result["weights"] = [round(v, 4) for v in                   3424
    result["weights"]]                                              3425
        result["lambda_max"] = round(result["lambda_max"],          3426
    4)                                                              3427
    return render_template("ahp/experts_aggregate.html",            3428
    unit_type=unit_type,                                            3429
                                level=level, l1=l1, names=names,    3430
    rows=rows,                                                      3431
                                result=result, form=EmptyForm())    3432
# ==== app/views/indicator.py ====                                  3433
# -*- coding: utf-8 -*-                                             3434
"""指标数据采集与管理 —— 路由与页面逻辑。                             3435
按微地质单元组织录入；系统按单元类型自动决定应录入哪些指标                3436
（岩质边坡无土壤指标、平台无地形指标）。                               3437
"""                                                                 3438
import io                                                           3439
import json                                                         3440
from flask import (Blueprint, render_template, request,             3441
    redirect, url_for,                                              3442
                  flash, send_file)                                 3443
from app.forms import EmptyForm                                     3444
from app.models import db, MinePatch, MicroUnit,                    3445
    IndicatorData                                                   3446
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,         3447
    UNIT_ENABLED_INDICATORS                                         3448
from app.services import import_service as imp                      3449
bp = Blueprint("indicator", __name__,                               3450
```

```
    url_prefix="/indicator")
# 指标字段定义：字段名 -> (中文名, 单位, 是否可选)
SOIL_FIELDS = [
    ("soil_ratio", "土石比", ""),
    ("thickness", "土层厚度", "cm"),
    ("ph", "pH", ""),
    ("organic_matter", "有机质", "g/kg"),
]
VEG_FIELDS = [
    ("coverage", "植被覆盖度", "%"),
    ("shannon", "Shannon-Wiener 多样性指数", ""),
]
TERRAIN_FIELDS = [
    ("slope", "坡度", "°"),
    ("aspect", "坡向", ""),
]
def unit_field_config(unit_type):
    """根据单元类型返回需录入的字段分组。"""
    enabled = UNIT_ENABLED_INDICATORS.get(unit_type, ())
    groups = []
    if "土壤指标" in enabled:
        groups.append(("土壤指标", SOIL_FIELDS, True))  # True = 含重金属
    if "植被指标" in enabled:
        groups.append(("植被指标", VEG_FIELDS, False))
    if "地形指标" in enabled:
        groups.append(("地形指标", TERRAIN_FIELDS, False))
    return groups
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).all()
    return render_template("indicator/index.html", patches=patches)
@bp.route("/<int:unit_id>", methods=["GET", "POST"])
def edit(unit_id):
    unit = db.get_or_404(MicroUnit, unit_id)
    ind = IndicatorData.query.filter_by(unit_id=unit.id).first()
    form = EmptyForm()
    groups = unit_field_config(unit.unit_type)
    if request.method == "POST" and form.validate_on_submit():
        def f(name, cast=float):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        heavy = {}
        for m in HEAVY_METALS:
            v = f("hm_" + m)
            if v is not None:
```

```
                heavy[m] = v
        if ind is None:
            ind = IndicatorData(unit_id=unit.id)
            db.session.add(ind)
        ind.soil_ratio = f("soil_ratio")
        ind.thickness = f("thickness")
        ind.ph = f("ph")
        ind.organic_matter = f("organic_matter")
        ind.heavy_metals = json.dumps(heavy,
    ensure_ascii=False) if heavy else None
        ind.coverage = f("coverage")
        ind.shannon = f("shannon")
        ind.slope = f("slope")
        ind.aspect = request.form.get("aspect", "").strip()
    or None
        db.session.commit()
        flash("指标数据已保存", "ok")
        return redirect(url_for("indicator.index"))
    # 组装当前值供模板回显
    current = {}
    if ind:
        current = {
            "soil_ratio": ind.soil_ratio, "thickness":
    ind.thickness,
            "ph": ind.ph, "organic_matter":
    ind.organic_matter,
            "coverage": ind.coverage, "shannon":
    ind.shannon,
            "slope": ind.slope, "aspect": ind.aspect,
        }
        current["heavy"] = ind.heavy_metals_dict()
    return render_template("indicator/edit.html",
                           unit=unit, form=form,
    groups=groups,
                           heavy_metals=HEAVY_METALS,
                           aspect_types=ASPECT_TYPES,
    current=current)
# ------------------------------ 批量导入 ------------------------------
    ----
@bp.route("/import", methods=["GET", "POST"])
def import_data():
    """批量导入指标数据（CSV / Excel）。"""
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    form = EmptyForm()
    report = None
    if request.method == "POST" and form.validate_on_submit(
    ):
        patch_id = request.form.get("patch_id", type=int)
        patch = db.session.get(MinePatch, patch_id) if
```

```
patch_id else None
    if patch is None:
        flash("请选择矿山图斑", "error")
        return redirect(url_for("indicator.import_data")
)
    file = request.files.get("datafile")
    if file is None or not file.filename:
        flash("请选择要导入的文件", "error")
        return redirect(url_for("indicator.import_data")
)
    ext = file.filename.rsplit(".", 1)[-1].lower()
    if ext not in ("csv", "xlsx", "xls"):
        flash("仅支持 CSV 或 Excel(.xlsx/.xls)文件", "error")
        return redirect(url_for("indicator.import_data")
)
    units = MicroUnit.query.filter_by(patch_id=patch.id)
.all()
    unit_map = {}
    for u in units:
        unit_map.setdefault(u.unit_type, u)
    if not unit_map:
        flash("该图斑尚未划分微地质单元，请先划分后再导入", "error")
        return redirect(url_for("indicator.import_data")
)
    try:
        parsed, errors = imp.parse_upload(file.filename,
file.stream, unit_map)
    except Exception as e:
        flash("文件解析失败：%s" % e, "error")
        return redirect(url_for("indicator.import_data")
)
    saved = 0
    for rec in parsed:
        unit = unit_map[rec["unit_type"]]
        ind = IndicatorData.query.filter_by(unit_id=unit
.id).first()
        if ind is None:
            ind = IndicatorData(unit_id=unit.id)
            db.session.add(ind)
        f = rec["fields"]
        for key in ("soil_ratio", "thickness", "ph",
"organic_matter",
                    "coverage", "shannon", "slope",
"aspect"):
            if key in f:
                setattr(ind, key, f[key])
        if rec["heavy_metals"]:
            ind.heavy_metals = json.dumps(rec["heavy_met
als"],
```

```
    ensure_ascii=False)                                                    3601
            saved += 1                                                     3602
        db.session.commit()                                                3603
        report = {"patch": patch.name, "saved": saved,                     3604
    "errors": errors,                                                      3605
                  "total": len(parsed) + len(                              3606
                      [e for e in errors if e.startswith("第               3607
    ")])}                                                                  3608
        if saved:                                                          3609
            flash("已导入 %d 个微地质单元的指标数据" % saved, "ok")          3610
        else:                                                              3611
            flash("没有可导入的数据行，请检查文件内容", "error")            3612
    return render_template("indicator/import.html",                        3613
    patches=patches,                                                       3614
                                  form=form, report=report,                3615
    headers=imp.HEADERS)                                                   3616
@bp.route("/import/template")                                              3617
def import_template():                                                     3618
    """下载导入模板（CSV，含一行示例）。"""                                3619
    csv_text = imp.build_template_csv()                                    3620
    data = csv_text.encode("utf-8-sig")       # 带 BOM，便于               3621
    Excel 正确识别中文                                                     3622
    return send_file(io.BytesIO(data), as_attachment=True,                 3623
                     download_name="指标数据导入模板.csv",                 3624
                     mimetype="text/csv")                                  3625
# ==== app/views/evaluate.py ====                                          3626
# -*- coding: utf-8 -*-                                                    3627
"""适宜性定量评价引擎 —— 路由与页面逻辑。                                  3628
从数据库加载某图斑的微地质单元与指标数据，调用评价引擎计算，              3629
展示各单元得分、最终评价因子 F、适宜性等级与推荐措施。                    3630
入口处按论文 4.1 节的流程顺序检查前置条件（见 app/services/workflow.py），3631
前置步骤未完成时不予计算，避免在条件不具备时产出评价结论。                3632
"""                                                                        3633
import json                                                                3634
from flask import Blueprint, render_template, request,                     3635
    redirect, url_for, flash                                               3636
from app.forms import EmptyForm                                            3637
from app.models import (db, MinePatch, MicroUnit,                          3638
    IndicatorData,                                                         3639
                        EvaluationResult, ScoreDetail)                     3640
from app.services import evaluation_service as ev                          3641
from app.services import workflow as wf                                    3642
bp = Blueprint("evaluate", __name__, url_prefix="/evaluate")               3643
def load_patch_input(patch_id):                                            3644
    """从数据库组装评价引擎输入。"""                                       3645
    units = MicroUnit.query.filter_by(patch_id=patch_id).all               3646
    ()                                                                     3647
    result = []                                                            3648
    for u in units:                                                        3649
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi               3650
```

```
    rst()
        data = {
            "soil_ratio": ind.soil_ratio if ind else None,
            "thickness": ind.thickness if ind else None,
            "ph": ind.ph if ind else None,
            "organic_matter": ind.organic_matter if ind
    else None,
            "heavy_metals": ind.heavy_metals_dict() if ind
    else {},
            "coverage": ind.coverage if ind else None,
            "shannon": ind.shannon if ind else None,
            "slope": ind.slope if ind else None,
            "aspect": ind.aspect if ind else None,
        }
        result.append({
            "unit_id": u.id,
            "unit_type": u.unit_type,
            "ratio": u.ratio if u.ratio is not None else
    0.0,
            "data": data,
        })
    return result
def _round(v, n=6):
    return None if v is None else round(v, n)
def load_weights(units):
    """解析各单元类型当前生效的权重与来源说明。
    权重可来自 AHP 计算并保存的生效方案；未配置时用论文表 3 的默认值。
    来源一并记录，便于报告标注与事后复核。
    """
    from app.services import weight_service as ws
    weights, sources = {}, {}
    for u in units:
        ut = u["unit_type"]
        if ut not in weights:
            w, label, _ = ws.effective_weights(ut)
            weights[ut] = w
            sources[ut] = label
    return weights, sources
def pack_units(unit_results, sources=None):
    """把评价结果整理为可持久化的 JSON。
    完整保留各单元得分、一级指标得分与逐条二级指标明细，
    使结果页与报告能够复现「已记录的那一次评价」，
    而不是每次打开时重新推算。
    """
    sources = sources or {}
    out = []
    for u in unit_results:
        out.append({
            "unit_type": u["unit_type"],
            "ratio": _round(u["ratio"]),
```

```
                "unit_score": _round(u["unit_score"]),
                "weighted": _round(u["weighted"]),
                "weight_source": sources.get(u["unit_type"]),
                "l1_scores": {k: _round(v) for k, v in
    u["l1_scores"].items()},
                "details": [{
                    "l1": d["l1"], "l2": d["l2"],
                    "raw": None if d["raw"] is None else
    str(d["raw"]),
                    "std_value": _round(d["std_value"]),
                    "weight": _round(d["weight"]),
                    "contribution": _round(d["contribution"]),
                } for d in u["details"]],
            })
    return out
def save_evaluation(patch_id, units):
    """执行评价并把结果与逐层明细落库，返回评价记录。
    权重取各单元类型当前生效的方案（AHP 计算结果），未配置时用
    论文表 3 的默认权重；来源一并记录。结果 JSON 保留各单元得分、
    一级指标得分与逐条二级指标明细，结果页与报告据此复现
    记录下来的那次评价，不再重新推算。
    """
    weights, sources = load_weights(units)
    res = ev.score_mine(units, weights)
    saved = EvaluationResult(
        patch_id=patch_id,
        final_score=round(res["final_score"], 4),
        grade=res["grade"],
        measures=res["grade_info"]["measure"],
        unit_scores=json.dumps(pack_units(res["unit_results"
    ], sources),
                               ensure_ascii=False),
    )
    db.session.add(saved)
    db.session.flush()        # 先取得 result.id，再写明细
    unit_ids = {u["unit_type"]: u["unit_id"] for u in units}
    for u in res["unit_results"]:
        for d in u["details"]:
            db.session.add(ScoreDetail(
                result_id=saved.id,
                unit_id=unit_ids.get(u["unit_type"]),
                l1_name=d["l1"], l2_name=d["l2"],
                raw_value=None if d["raw"] is None else
    str(d["raw"]),
                std_value=d["std_value"],
    weight=d["weight"],
                contribution=d["contribution"]))
    db.session.commit()
    return saved
def block_reasons(progress):
```

```
    """把未满足的前置条件转成可读的拦截原因。"""
    return ["%s：%s" % (b["label"], b["reason"]) for b in
    progress["blocking"]]
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    latest = {}
    progress = {}
    for p in patches:
        latest[p.id] = EvaluationResult.query.filter_by(patc
    h_id=p.id) \
            .order_by(EvaluationResult.id.desc()).first()
        progress[p.id] = wf.patch_progress(p.id)
    return render_template("evaluate/index.html",
    patches=patches,
                           latest=latest, progress=progress)
@bp.route("/<int:patch_id>", methods=["GET", "POST"])
def run(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    progress = wf.patch_progress(patch_id)
    if request.method == "POST" and form.validate_on_submit(
    ):
        if not progress["ready"]:
            for reason in block_reasons(progress):
                flash("无法执行评价 —— " + reason, "error")
            return redirect(url_for("evaluate.run",
    patch_id=patch_id))
        units = load_patch_input(patch_id)
        saved = save_evaluation(patch.id, units)
        return redirect(url_for("evaluate.result",
    result_id=saved.id))
    # 展示将采用的权重来源，便于在执行前确认
    units_now = load_patch_input(patch_id)
    _, sources = load_weights(units_now) if units_now else
    ({}, {})
    return render_template("evaluate/run.html",
    patch=patch, form=form,
                           progress=progress,
    weight_sources=sources)
@bp.route("/result/<int:result_id>")
def result(result_id):
    r = db.get_or_404(EvaluationResult, result_id)
    patch = db.session.get(MinePatch, r.patch_id)
    unit_scores = json.loads(r.unit_scores) if
    r.unit_scores else []
    return render_template("evaluate/result.html", r=r,
    patch=patch,
                           unit_scores=unit_scores)
```

```
# ==== app/views/report.py ====
# -*- coding: utf-8 -*-
"""报告与结果输出 —— 得分明细、图表、Word / PDF 导出。"""
import io
import json
from datetime import datetime
from flask import Blueprint, render_template, send_file,
    flash, redirect, url_for
from app.models import db, MinePatch, ReferenceEcosystem
from app.services import report_service as rs
from app.services import workflow as wf
from app.utils import charts
from app.utils import browser
from app.utils.constants import grade_of
bp = Blueprint("report", __name__, url_prefix="/report")
def _prepare(patch_id):
    """加载图斑并取出已记录的评价结果，供明细页与导出复用。
    报告是评价结论的呈现，因此这里读取最近一次保存的评价记录，
    而不是重新计算：评价完成后再改动指标数据，也不会让报告与
    系统记录的评价结论对不上。没有任何评价记录时，返回流程检查
    结果，由调用方拦截并提示先去执行评价。
    """
    patch = db.get_or_404(MinePatch, patch_id)
    prog = wf.report_progress(patch_id)
    if not prog["ready"]:
        return patch, None, None, None, prog
    rec = prog["result"]
    result = {
        "final_score": rec.final_score,
        "grade": rec.grade,
        "grade_info": grade_of(rec.final_score),
        "unit_results": json.loads(rec.unit_scores) if
    rec.unit_scores else [],
    }
    reference = ReferenceEcosystem.query.filter_by(patch_id=
    patch_id).first()
    return patch, reference, result, rec, prog
def _deny(progress, patch_id, endpoint="report.detail"):
    """前置条件未满足时的统一处理：逐条提示并退回报告列表。"""
    for b in progress["blocking"]:
        flash("暂不能生成报告 —— %s：%s" % (b["label"],
    b["reason"]), "error")
    if progress["blocking"]:
        flash("请先完成上述步骤后再生成报告。", "error")
    return redirect(url_for("report.index"))
def _charts(result):
    """生成两张图表：单元得分对比 + 一级指标平均得分分布。"""
    units = [(u["unit_type"], u["unit_score"],
              "%.4f" % u["unit_score"]) for u in
    result["unit_results"]]
```

```
    chart_units = charts.hbars(units)
    # 各一级指标在各单元上的平均得分，反映其对评价结果的实际影响
    acc, cnt = {}, {}
    for u in result["unit_results"]:
        for k, v in (u.get("l1_scores") or {}).items():
            acc[k] = acc.get(k, 0.0) + v
            cnt[k] = cnt.get(k, 0) + 1
    items = [(k, acc[k] / cnt[k], "%.4f" % (acc[k] /
    cnt[k])) for k in acc]
    items.sort(key=lambda x: -x[1])
    chart_weight = charts.donut(items)
    return chart_units, chart_weight
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    progress = {p.id: wf.report_progress(p.id) for p in
    patches}
    return render_template("report/index.html",
    patches=patches,
                           progress=progress)
@bp.route("/<int:patch_id>")
def detail(patch_id):
    patch, reference, result, record, prog =
    _prepare(patch_id)
    if result is None:
        return _deny(prog, patch_id)
    chart_units, chart_weight = _charts(result)
    return render_template("report/detail.html",
    patch=patch, result=result,
                           record=record,
    reference=reference,
                           chart_units=chart_units,
    chart_weight=chart_weight)
@bp.route("/<int:patch_id>/word")
def word(patch_id):
    patch, reference, result, record, prog =
    _prepare(patch_id)
    if result is None:
        return _deny(prog, patch_id)
    buf = rs.build_report(patch, result)
    filename = "自然恢复适宜性评价报告_%s.docx" % patch.name
    return send_file(buf, as_attachment=True,
    download_name=filename,
                     mimetype="application/vnd.openxmlformat
    s-officedocument."
                              "wordprocessingml.document")
@bp.route("/<int:patch_id>/pdf")
def pdf(patch_id):
    """导出 PDF 报告：服务端渲染打印模板，再交浏览器无头打印。"""
```

```
    patch, reference, result, record, prog =
    _prepare(patch_id)
    if result is None:
        return _deny(prog, patch_id)
    chart_units, chart_weight = _charts(result)
    html = render_template(
        "report/print.html", patch=patch,
    reference=reference, result=result,
        chart_units=chart_units, chart_weight=chart_weight,
        generated_at=datetime.now().strftime("%Y-%m-%d
    %H:%M"))
    data, err = browser.html_to_pdf(html)
    if err:
        flash("PDF 导出失败：%s" % err, "error")
        return redirect(url_for("report.detail",
    patch_id=patch_id))
    return send_file(io.BytesIO(data), as_attachment=True,
                     download_name="自然恢复适宜性评价报告_%s.pdf" %
    patch.name,
                     mimetype="application/pdf")
# ==== app/views/search.py ====
# -*- coding: utf-8 -*-
"""全局搜索：在评价项目、矿山图斑、标准与政策文件中查找关键词。
顶栏的搜索框指向本模块。只做关键词包含匹配，不做分词，
但覆盖名称、编号、行政区、矿种、文号、摘要等常用检索字段。
"""
from sqlalchemy import or_
from flask import Blueprint, render_template, request
from app.models import db, Project, MinePatch, Standard
bp = Blueprint("search", __name__)
LIMIT = 50              # 每类结果上限，避免一次渲染过多
@bp.route("/search")
def index():
    q = (request.args.get("q") or "").strip()
    projects, patches, standards = [], [], []
    if q:
        like = "%" + q + "%"
        projects = Project.query.filter(or_(
            Project.name.like(like),
            Project.code.like(like),
            Project.description.like(like),
        )).order_by(Project.id.desc()).limit(LIMIT).all()
        patches = MinePatch.query.filter(or_(
            MinePatch.name.like(like),
            MinePatch.admin_region.like(like),
            MinePatch.mineral_type.like(like),
        )).order_by(MinePatch.id.desc()).limit(LIMIT).all()
        standards = Standard.query.filter(or_(
            Standard.name.like(like),
            Standard.doc_number.like(like),
```

```
                Standard.province.like(like),                                    3951
                Standard.summary.like(like),                                     3952
            )).order_by(Standard.id).limit(LIMIT).all()                          3953
    total = len(projects) + len(patches) + len(standards)                        3954
    return render_template("search.html", q=q,                                   3955
    projects=projects,                                                           3956
                                patches=patches,                                 3957
    standards=standards, total=total)                                            3958
# ==== app/templates/base.html ====                                              3959
{% import "_macros.html" as ui %}<!DOCTYPE html>                                 3960
<html lang="zh-CN">                                                              3961
<head>                                                                           3962
  <meta charset="UTF-8">                                                         3963
  <meta name="viewport" content="width=device-width,                             3964
    initial-scale=1.0">                                                          3965
  <link rel="icon" href="{{ url_for('static',                                    3966
    filename='favicon.ico') }}">                                                 3967
  <title>{% block title %}历史遗留废弃矿山自然恢复适宜性评价系统{% endblock      3968
    %}</title>                                                                   3969
  <style>                                                                        3970
    /* ---------- 设计变量：绿色主色 + 中性灰阶 ---------- */                    3971
    :root {                                                                      3972
      --accent: #1f6b45;                  /* 主色：深绿，呼应自然恢复主题 */    3973
      --accent-hover: #185738;                                                   3974
      --accent-soft: #e9f4ee;                                                    3975
      --side-1: #17503a;                  /* 侧边栏渐变起止色 */                3976
      --side-2: #113c2b;                                                         3977
      --danger: #b3261e;                                                         3978
      --danger-soft: #fdecea;                                                    3979
      --ok: #1e6b3a;                                                             3980
      --ok-soft: #eaf6ef;                                                        3981
      --todo: #d98324;                    /* 待办角标 */                        3982
      --ink: #23262b;                                                            3983
      --ink-soft: #6b7280;                                                       3984
      --line: #e3e7ea;                                                           3985
      --surface: #ffffff;                                                        3986
      --canvas: #f4f7f5;                                                         3987
      --radius: 10px;                                                            3988
      --shadow: 0 1px 2px rgba(0, 0, 0, .05);                                    3989
      --header-h: 60px;                                                          3990
      --nav-w: 226px;                                                            3991
    }                                                                            3992
    * { box-sizing: border-box; }                                                3993
    html, body { margin: 0; padding: 0; }                                        3994
    body {                                                                       3995
      font-family: "Microsoft YaHei", "PingFang SC",                             3996
    "SimSun", sans-serif;                                                        3997
      font-size: 14px;                                                           3998
      line-height: 1.6;                                                          3999
      color: var(--ink);                                                         4000
```

```
  background: var(--canvas);
}
/* ---------- 跳到主要内容（键盘可达性） ---------- */
.skip-link {
  position: absolute;
  left: 12px;
  top: -60px;
  z-index: 30;
  padding: 8px 14px;
  background: var(--accent);
  color: #fff;
  border-radius: var(--radius);
  text-decoration: none;
  transition: top .15s ease-out;
}
.skip-link:focus { top: 12px; }
/* ---------- 整体骨架：左侧深绿导航 + 右侧内容 ---------- */
.app-shell {
  display: flex;
  align-items: stretch;
  min-height: 100dvh;
}
/* ---------- 侧边栏 ---------- */
.app-side {
  flex: 0 0 var(--nav-w);
  width: var(--nav-w);
  display: flex;
  flex-direction: column;
  background: linear-gradient(180deg, var(--side-1) 0%,
var(--side-2) 100%);
  color: #fff;
}
.app-brand {
  display: flex;
  align-items: center;
  gap: 11px;
  padding: 17px 16px 15px;
  border-bottom: 1px solid rgba(255, 255, 255, .10);
}
.app-brand__mark {
  flex: none;
  display: grid;
  place-items: center;
  width: 36px;
  height: 36px;
  border-radius: 10px;
  background: rgba(255, 255, 255, .15);
}
.app-brand__name {
  font-size: 14.5px;
```

```
    font-weight: 600;
    line-height: 1.35;
    letter-spacing: .2px;
}
.app-brand__sub {
    font-size: 11.5px;
    line-height: 1.35;
    color: rgba(255, 255, 255, .58);
}
.app-nav { flex: 1; padding: 6px 12px 14px; overflow-y:
auto; }
.app-nav__group {
   margin: 12px 0 5px;
    padding: 0 10px;
    font-size: 11.5px;
    letter-spacing: .05em;
    color: rgba(255, 255, 255, .48);
}
.app-nav ul { margin: 0; padding: 0; list-style: none; }
.app-nav a {
    display: flex;
    align-items: center;
    gap: 10px;
    padding: 9px 11px;
   margin-bottom: 2px;
    border-radius: 8px;
    color: rgba(255, 255, 255, .80);
    text-decoration: none;
    transition: background-color .16s ease, color .16s
ease;
}
.app-nav a:hover { background: rgba(255, 255, 255,
.10); color: #fff; }
.app-nav a:focus-visible {
    outline: 2px solid rgba(255, 255, 255, .60);
    outline-offset: 1px;
}
.app-nav a[aria-current="page"] {
    background: rgba(255, 255, 255, .17);
    color: #fff;
    font-weight: 600;
}
.app-nav .ico { flex: none; opacity: .92; }
.app-nav__badge {
    margin-left: auto;
    min-width: 20px;
    padding: 0 6px;
    font-size: 11px;
    font-weight: 600;
    line-height: 18px;
```

```
    text-align: center;
    border-radius: 9px;
    background: var(--todo);
    color: #fff;
    font-variant-numeric: tabular-nums;
}
.app-user {
    display: flex;
    align-items: center;
    gap: 10px;
    padding: 12px 16px;
    border-top: 1px solid rgba(255, 255, 255, .10);
}
.app-user__avatar {
    flex: none;
    display: grid;
    place-items: center;
    width: 32px;
    height: 32px;
    border-radius: 50%;
    background: rgba(255, 255, 255, .16);
    font-size: 13px;
    font-weight: 600;
}
.app-user__name { font-size: 13px; line-height: 1.3; }
.app-user__state {
    display: flex;
    align-items: center;
    gap: 5px;
    font-size: 11.5px;
    line-height: 1.3;
    color: rgba(255, 255, 255, .60);
}
.app-user__dot {
    width: 6px;
    height: 6px;
    border-radius: 50%;
    background: #6ed49a;
}
.app-user a {
    margin-left: auto;
    padding: 5px;
    border-radius: 6px;
    color: rgba(255, 255, 255, .68);
    text-decoration: none;
}
.app-user a:hover { color: #fff; background: rgba(255,
255, 255, .12); }
/* ---------- 右侧内容区 ---------- */
.app-content {
```

```
    flex: 1;
    min-width: 0;
    display: flex;
    flex-direction: column;
}
/* ---------- 顶栏：面包屑 + 搜索 ---------- */
.app-header {
    display: flex;
    align-items: center;
    gap: 14px;
    height: var(--header-h);
    padding: 0 22px;
    background: var(--surface);
    border-bottom: 1px solid var(--line);
}
.crumb {
    display: flex;
    align-items: center;
    gap: 7px;
    font-size: 13px;
    color: var(--ink-soft);
    white-space: nowrap;
}
.crumb a { color: var(--ink-soft); text-decoration:
none; }
.crumb a:hover { color: var(--accent); }
.crumb strong { color: var(--ink); font-weight: 600; }
.crumb .ico { color: #c3c9d0; }
.app-header__spacer { flex: 1; }
.search-box {
    display: flex;
    align-items: center;
    gap: 8px;
    width: 268px;
    padding: 7px 12px;
    background: var(--canvas);
    border: 1px solid var(--line);
    border-radius: 9px;
    color: var(--ink-soft);
    transition: border-color .16s ease, background-color
.16s ease;
}
.search-box:focus-within { border-color: var(--accent);
background: #fff; }
.search-box input {
    flex: 1;
    min-width: 0;
    border: none;
    outline: none;
    background: transparent;
```

```
    font-family: inherit;
    font-size: 13px;
    color: var(--ink);
}
.search-box button {
    display: grid;
    place-items: center;
    padding: 0;
    border: none;
    background: transparent;
    color: inherit;
    cursor: pointer;
}
.icon-btn {
    display: grid;
    place-items: center;
    width: 34px;
    height: 34px;
    border-radius: 9px;
    border: 1px solid transparent;
    color: var(--ink-soft);
    text-decoration: none;
    transition: background-color .16s ease, color .16s ease,
                border-color .16s ease;
}
.icon-btn:hover {
    color: var(--accent);
    background: var(--accent-soft);
    border-color: rgba(31, 107, 69, .18);
}
.app-main {
    flex: 1;
    min-width: 0;
    padding: 22px 26px 26px;
}
.container { max-width: 1120px; }
/* ---------- 页脚 ---------- */
.app-foot {
    display: flex;
    align-items: center;
    justify-content: space-between;
    gap: 14px;
    padding: 0 26px 18px;
    font-size: 12px;
    color: var(--ink-soft);
}
/* ---------- 未登录时的裸页面（登录页） ---------- */
.login-shell {
    display: flex;
```

```
    flex-direction: column;
    min-height: 100dvh;
}
/* ---------- 卡片 ---------- */
.card {
    background: var(--surface);
    border: 1px solid var(--line);
    border-radius: var(--radius);
    box-shadow: var(--shadow);
    padding: 22px 24px;
    margin-bottom: 18px;
}
.card > h2 {
    margin: 0 0 14px;
    font-size: 16px;
    font-weight: 600;
    color: var(--accent);
    text-wrap: balance;
}
.card > h3 {
    margin: 20px 0 10px;
    font-size: 14px;
    font-weight: 600;
    color: var(--accent);
}
/* ---------- 表格 ---------- */
table {
    width: 100%;
    border-collapse: collapse;
    margin: 10px 0 16px;
    font-variant-numeric: tabular-nums;
}
caption {
    text-align: left;
    color: var(--ink-soft);
    font-size: 12px;
    padding-bottom: 6px;
}
th, td {
    border: 1px solid var(--line);
    padding: 8px 12px;
    text-align: center;
    vertical-align: middle;
}
th {
    background: var(--accent-soft);
    font-weight: 600;
    white-space: nowrap;
}
tbody tr:hover { background: #fafbfc; }
```

```
/* ---------- 表单控件 ---------- */
label { display: inline-block; }
.field { margin-bottom: 12px; }
.field__label {
  display: block;
  margin-bottom: 4px;
  color: var(--ink-soft);
  font-size: 13px;
}
input[type="text"], input[type="number"],
input[type="password"],
select, textarea {
  padding: 6px 9px;
  font-family: inherit;
  font-size: 14px;
  color: var(--ink);
  background: #fff;
  border: 1px solid #c6ccd4;
  border-radius: 4px;
  max-width: 100%;
}
input[type="number"] {
  width: 104px;
  text-align: center;
  font-variant-numeric: tabular-nums;
}
textarea { width: 100%; min-height: 64px; resize:
vertical; }
input:focus-visible, select:focus-visible,
textarea:focus-visible {
  outline: 2px solid var(--accent);
  outline-offset: 1px;
  border-color: var(--accent);
}
.checkbox {
  display: inline-flex;
  align-items: center;
  gap: 7px;
  cursor: pointer;
  user-select: none;
}
.checkbox input { width: 16px; height: 16px; margin: 0;
cursor: pointer; }
/* ---------- 按钮 ----------
   统一为圆角矩形：默认半透明底色，鼠标悬停时填充为实色。 */
.btn {
  display: inline-flex;
  align-items: center;
  justify-content: center;
  gap: 6px;
```

```
  padding: 8px 18px;
  font-family: inherit;
  font-size: 14px;
  line-height: 1.4;
  color: #fff;
  background: rgba(31, 107, 69, .80);
  border: 1px solid rgba(31, 107, 69, .30);
  border-radius: 8px;
  cursor: pointer;
  text-decoration: none;
  transition: background-color .16s ease, border-color .16s ease,
              color .16s ease, box-shadow .16s ease, transform .16s ease;
}
.btn:hover {
  background: var(--accent);
  border-color: var(--accent);
  box-shadow: 0 3px 12px rgba(31, 107, 69, .22);
}
.btn:active { transform: translateY(1px); box-shadow: none; }
.btn:focus-visible { outline: 2px solid var(--accent); outline-offset: 2px; }
.btn--ghost {
  color: var(--accent);
  background: rgba(31, 107, 69, .06);
  border-color: rgba(31, 107, 69, .22);
}
.btn--ghost:hover {
  color: #fff;
  background: var(--accent);
  border-color: var(--accent);
}
.btn--danger {
  color: var(--danger);
  background: rgba(179, 38, 30, .07);
  border-color: rgba(179, 38, 30, .26);
}
.btn--danger:hover {
  color: #fff;
  background: var(--danger);
  border-color: var(--danger);
  box-shadow: 0 3px 12px rgba(179, 38, 30, .20);
}
.btn--sm { padding: 5px 13px; font-size: 13px; border-radius: 7px; }
.btn:disabled { opacity: .45; cursor: not-allowed; }
.btn:disabled:hover {
  background: rgba(31, 107, 69, .80);
```

```
    border-color: rgba(31, 107, 69, .30);                                4401
    box-shadow: none;                                                    4402
}                                                                        4403
.btn--ghost:disabled:hover { background: rgba(31, 107,                   4404
69, .06); color: var(--accent); }                                        4405
/* ---------- 表格操作列的行内小按钮 ----------                           4406
   取代原来「文字 | 文字 | 文字」的写法：                                 4407
   圆角矩形 + 半透明底色，悬停时变实色。 */                              4408
.row-actions {                                                           4409
    display: inline-flex;                                                4410
    flex-wrap: nowrap;            /* 按钮不折行；宁可让旁边长名称换行 */  4411
    align-items: center;                                                 4412
    gap: 6px;                                                            4413
    white-space: nowrap;                                                 4414
}                                                                        4415
.row-actions form { display: inline-flex; margin: 0; }                   4416
.link-btn {                                                              4417
    display: inline-flex;                                                4418
    align-items: center;                                                 4419
    padding: 4px 11px;                                                   4420
    font-family: inherit;                                                4421
    font-size: 13px;                                                     4422
    line-height: 1.5;                                                    4423
    color: var(--accent);                                                4424
    background: rgba(31, 107, 69, .07);                                  4425
    border: 1px solid rgba(31, 107, 69, .20);                            4426
    border-radius: 7px;                                                  4427
    cursor: pointer;                                                     4428
    text-decoration: none;                                               4429
    white-space: nowrap;                                                 4430
    transition: background-color .16s ease, border-color                 4431
.16s ease,                                                               4432
                color .16s ease, box-shadow .16s ease,                   4433
transform .16s ease;                                                     4434
}                                                                        4435
.link-btn:hover {                                                        4436
    color: #fff;                                                         4437
    background: var(--accent);                                           4438
    border-color: var(--accent);                                         4439
    box-shadow: 0 2px 8px rgba(31, 107, 69, .20);                        4440
}                                                                        4441
.link-btn:active { transform: translateY(1px);                           4442
box-shadow: none; }                                                      4443
.link-btn:focus-visible { outline: 2px solid                             4444
var(--accent); outline-offset: 2px; }                                    4445
.link-btn--danger {                                                      4446
    color: var(--danger);                                                4447
    background: rgba(179, 38, 30, .07);                                  4448
    border-color: rgba(179, 38, 30, .24);                                4449
}                                                                        4450
```

```
.link-btn--danger:hover {
  color: #fff;
  background: var(--danger);
  border-color: var(--danger);
  box-shadow: 0 2px 8px rgba(179, 38, 30, .22);
}
.actions { display: flex; flex-wrap: wrap; align-items: center; gap: 8px; }
.actions form { display: inline; }
/* ---------- 提示条 ---------- */
.flash {
  padding: 10px 14px;
  margin-bottom: 14px;
  border-radius: var(--radius);
  border-left: 4px solid;
}
.flash.error {
  color: var(--danger);
  background: var(--danger-soft);
  border-left-color: var(--danger);
}
.flash.ok {
  color: var(--ok);
  background: var(--ok-soft);
  border-left-color: var(--ok);
}
/* ---------- 居中浮层提示（登录成功、各类操作完成） ---------- */
   外层 toast-stack 固定在页面正中心并纵向排列，
   内层每条各自淡入 → 停留 → 淡出，多条提示会依次排开而不重叠。
   半透明底 + 背景模糊做出毛玻璃质感，不做成实心块；
   pointer-events: none 让它不拦截鼠标，出现期间也能正常操作。 */
.toast-stack {
  position: fixed;
  top: 50%;
  left: 50%;
  z-index: 100;
  display: flex;
  flex-direction: column;
  align-items: center;
  gap: 10px;
  transform: translate(-50%, -50%);
  pointer-events: none;
}
.toast-center {
  max-width: min(560px, 78vw);
  padding: 13px 28px;
  border-radius: 999px;
  background: rgba(31, 107, 69, .62);          /* 半透明，非实体 */
  color: #fff;
```

```
    font-size: 16px;
    font-weight: 600;
    line-height: 1.65;
    letter-spacing: .03em;
    text-align: center;
    text-wrap: pretty;
    border: 1px solid rgba(255, 255, 255, .28);
    backdrop-filter: blur(9px) saturate(1.3);
    -webkit-backdrop-filter: blur(9px) saturate(1.3);
    box-shadow: 0 10px 34px rgba(19, 36, 58, .18);
    animation: toast-fade 2.6s ease-in-out forwards;
}
@keyframes toast-fade {
    0%          { opacity: 0; transform: scale(.90); }
    14%, 74%    { opacity: 1; transform: scale(1); }
    100%        { opacity: 0; transform: scale(1); }
}
/* 系统设置里要求「减少动态效果」时，去掉缩放，只做淡入淡出 */
@media (prefers-reduced-motion: reduce) {
    .toast-center { animation: toast-fade-plain 2.6s
linear forwards; }
    @keyframes toast-fade-plain {
        0%          { opacity: 0; }
        14%, 74%    { opacity: 1; }
        100%        { opacity: 0; }
    }
}
/* 不支持背景模糊时，加深底色保证文字仍然清楚 */
@supports not (backdrop-filter: blur(1px)) {
    .toast-center { background: rgba(31, 107, 69, .88); }
}
/* ---------- 文字工具类 ---------- */
.muted { color: var(--ink-soft); font-size: 12.5px; }
.mono { font-family: Consolas, "Courier New",
monospace; font-variant-numeric: tabular-nums; }
.num { font-variant-numeric: tabular-nums; }
.warn { color: var(--danger); font-weight: 600; }
.ok { color: var(--ok); font-weight: 600; }
.nowrap { white-space: nowrap; }
/* 表格操作列里常把删除按钮放在 <form> 中。form 默认是块级元素，
   会把「删除」挤到下一行，导致操作列参差不齐，这里让它留在行内。 */
.nowrap form { display: inline; margin: 0; }
.stack { display: flex; flex-direction: column; gap:
14px; }
/* 状态标签：不依赖颜色单独表意 */
.tag {
    display: inline-block;
    padding: 1px 9px;
    font-size: 12.5px;
    border-radius: 10px;
```

```
  border: 1px solid;
  white-space: nowrap;
}
.tag--ok { color: var(--ok); background:
var(--ok-soft); border-color: #b8ddc6; }
.tag--warn { color: var(--danger); background:
var(--danger-soft); border-color: #eec4c0; }
.tag--muted { color: var(--ink-soft); background:
#f1f3f6; border-color: #dde1e6; }
/* ---------- 登录页 ---------- */
/* 背景：淡雅远山（SVG）+ 渐变兜底；卡片垂直水平都居中 */
.body--login {
  background:
    url("{{ url_for('static', filename='login-bg.svg')
}}")
    center / cover no-repeat fixed,
    linear-gradient(180deg, #f6f9fc 0%, #e6eef5 100%);
}
.body--login .app-main {
  display: flex;
  align-items: center;
  justify-content: center;
  padding: 28px 20px;
}
.body--login .container { max-width: none; width: 100%;
}
/* 仅供读屏软件的隐藏文本 */
.visually-hidden {
  position: absolute !important;
  width: 1px;
  height: 1px;
  margin: -1px;
  padding: 0;
  overflow: hidden;
  clip: rect(0 0 0 0);
  white-space: nowrap;
  border: 0;
}
.login-wrap {
  display: flex;
  justify-content: center;
  width: 100%;
}
.login-card {
  width: 100%;
  max-width: 400px;
  padding: 30px 32px 26px;
  border-radius: 10px;
  box-shadow: 0 14px 44px rgba(19, 36, 58, .13);
}
```

```
.login-card h2 {
  margin: 0 0 6px;
  text-align: center;
  font-size: 20px;
}
.login-card__sub {
  margin: 0 0 22px;
  text-align: center;
  font-size: 12.5px;
  color: var(--ink-soft);
}
.login-card .field__label { font-size: 14px; color:
var(--ink); }
/* ---------- 工作台：页首横幅 ---------- */
.hero {
  position: relative;
  overflow: hidden;
  min-height: 176px;
  padding: 26px 28px;
  margin-bottom: 16px;
  border-radius: 12px;
  /* 左侧压深保证文字可读，右侧透出远山插画；
     背景按底部对齐，避免远山被居中裁掉 */
  background:
    linear-gradient(100deg, rgba(9, 40, 27, .92) 0%,
                            rgba(11, 48, 33, .68) 50%,
                            rgba(11, 48, 33, .06) 100%),
    url("{{ url_for('static', filename='hero.svg') }}")
    center bottom / cover no-repeat;
  color: #fff;
}
.hero__body { max-width: 720px; }
.hero__title {
  margin: 0 0 10px;
  font-size: 21px;
  font-weight: 700;
  line-height: 1.4;
  letter-spacing: .3px;
}
.hero__desc {
  margin: 0;
  max-width: 640px;
  font-size: 12.5px;
  line-height: 1.85;
  color: rgba(255, 255, 255, .86);
}
.hero__actions {
  display: flex;
  flex-wrap: wrap;
  gap: 10px;
```

```
  margin-top: 18px;                                                    4651
}                                                                      4652
.hero-btn {                                                            4653
  display: inline-flex;                                                4654
  align-items: center;                                                 4655
  gap: 6px;                                                            4656
  padding: 7px 15px;                                                   4657
  font-size: 13px;                                                     4658
  border-radius: 8px;                                                  4659
  text-decoration: none;                                               4660
  transition: background-color .16s ease, border-color                 4661
.16s ease,                                                             4662
              color .16s ease;                                         4663
}                                                                      4664
.hero-btn--solid {                                                     4665
  color: #14432f;                                                      4666
  background: rgba(255, 255, 255, .92);                                4667
  border: 1px solid rgba(255, 255, 255, .92);                          4668
}                                                                      4669
.hero-btn--solid:hover { background: #fff; }                           4670
.hero-btn--ghost {                                                     4671
  color: #fff;                                                         4672
  background: rgba(255, 255, 255, .14);                                4673
  border: 1px solid rgba(255, 255, 255, .45);                          4674
}                                                                      4675
.hero-btn--ghost:hover { background: rgba(255, 255,                    4676
255, .26); }                                                           4677
.hero-btn:focus-visible { outline: 2px solid #fff;                     4678
outline-offset: 2px; }                                                 4679
/* ---------- 工作台：数据概览 ---------- */                             4680
.stat-cards {                                                          4681
  display: grid;                                                       4682
  grid-template-columns: repeat(auto-fit, minmax(190px,                4683
1fr));                                                                 4684
  gap: 12px;                                                           4685
  margin-bottom: 22px;                                                 4686
}                                                                      4687
.stat-card {                                                           4688
  display: block;                                                      4689
  padding: 15px 17px;                                                  4690
  background: var(--surface);                                          4691
  border: 1px solid var(--line);                                       4692
  border-radius: var(--radius);                                        4693
  box-shadow: var(--shadow);                                           4694
  color: var(--ink);                                                   4695
  text-decoration: none;                                               4696
  transition: border-color .16s ease, box-shadow .16s                  4697
ease;                                                                  4698
}                                                                      4699
.stat-card:hover {                                                     4700
```

```
    border-color: rgba(31, 107, 69, .45);
    box-shadow: 0 3px 12px rgba(19, 60, 43, .08);
}
.stat-card__label {
    display: block;
    font-size: 12.5px;
    color: var(--ink-soft);
}
.stat-card__value {
    display: block;
    margin: 3px 0 2px;
    font-size: 26px;
    font-weight: 700;
    line-height: 1.25;
    color: var(--accent);
    font-variant-numeric: tabular-nums;
}
.stat-card__unit {
    margin-left: 3px;
    font-size: 13px;
    font-weight: 500;
    color: var(--ink-soft);
}
.stat-card__note {
    display: block;
    font-size: 12px;
    color: var(--ink-soft);
}
/* ---------- 工作台：分区标题 ---------- */
.section { margin-bottom: 22px; }
.section__head {
    display: flex;
    align-items: baseline;
    flex-wrap: wrap;
    gap: 6px 14px;
    margin-bottom: 12px;
}
.section__title { margin: 0; font-size: 16px; }
.section__desc { margin: 0; font-size: 12.5px; color:
var(--ink-soft); }
/* ---------- 工作台：流程步骤卡 ---------- */
.step-grid {
    display: grid;
    grid-template-columns: repeat(auto-fill,
minmax(238px, 1fr));
    gap: 12px;
}
.step-card {
    position: relative;
    display: block;
```

```
    padding: 16px 16px 15px;                                    4751
    background: var(--surface);                                 4752
    border: 1px solid var(--line);                              4753
    border-radius: var(--radius);                               4754
    box-shadow: var(--shadow);                                  4755
    color: var(--ink);                                          4756
    text-decoration: none;                                      4757
    transition: border-color .16s ease, box-shadow .16s         4758
ease,                                                           4759
                transform .16s ease;                            4760
}                                                               4761
.step-card:hover {                                              4762
    border-color: rgba(31, 107, 69, .55);                       4763
    box-shadow: 0 5px 16px rgba(19, 60, 43, .10);               4764
    transform: translateY(-1px);                                4765
}                                                               4766
.step-card:focus-visible { outline: 2px solid                   4767
var(--accent); outline-offset: 2px; }                           4768
.step-card__no {                                                4769
    position: absolute;                                         4770
    top: 13px;                                                  4771
    right: 15px;                                                4772
    font-size: 17px;                                            4773
    font-weight: 700;                                           4774
    color: #d7e3db;                                             4775
    font-variant-numeric: tabular-nums;                         4776
}                                                               4777
.step-card__icon {                                              4778
    display: grid;                                              4779
    place-items: center;                                        4780
    width: 32px;                                                4781
    height: 32px;                                               4782
    margin-bottom: 10px;                                        4783
    border-radius: 9px;                                         4784
    background: var(--accent-soft);                             4785
    color: var(--accent);                                       4786
}                                                               4787
.step-card__title {                                             4788
    display: block;                                             4789
    margin-bottom: 5px;                                         4790
    font-size: 14px;                                            4791
    font-weight: 600;                                           4792
    color: var(--accent);                                       4793
}                                                               4794
.step-card__desc {                                              4795
    display: block;                                             4796
    font-size: 12.5px;                                          4797
    line-height: 1.65;                                          4798
    color: var(--ink-soft);                                     4799
    text-wrap: pretty;                                          4800
```

```
}
.step-card__badge {
  display: inline-block;
  margin-top: 10px;
  padding: 1px 9px;
  font-size: 11.5px;
  border-radius: 9px;
  background: #fdf1e2;
  color: #9a5a12;
  border: 1px solid #f0d3ab;
}
/* 前置数据不具备时置灰，并说明缺什么 */
.step-card--locked {
  background: #fafbfa;
  border-color: #eceff0;
  box-shadow: none;
  cursor: default;
}
.step-card--locked .step-card__no { color: #e6e9e8; }
.step-card--locked .step-card__icon {
  background: #f1f3f2;
  color: #a9b2ae;
}
.step-card--locked .step-card__title { color: #9aa3a0; }
.step-card--locked .step-card__desc { color: #b3bab7; }
.step-card__lock {
  display: inline-block;
  margin-top: 10px;
  padding: 1px 9px;
  font-size: 11.5px;
  border-radius: 9px;
  background: #f1f3f2;
  color: #8b948f;
}
/* ---------- 工作台：整行入口 ---------- */
.row-card {
  display: flex;
  align-items: center;
  gap: 14px;
  padding: 15px 18px;
  background: var(--surface);
  border: 1px solid var(--line);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  color: var(--ink);
  text-decoration: none;
  transition: border-color .16s ease, box-shadow .16s
ease;
}
.row-card:hover {
```

```
    border-color: rgba(31, 107, 69, .45);
    box-shadow: 0 3px 12px rgba(19, 60, 43, .08);
}
.row-card__icon {
    display: grid;
    place-items: center;
    width: 36px;
    height: 36px;
    flex: none;
    border-radius: 10px;
    background: var(--accent-soft);
    color: var(--accent);
}
.row-card__main { flex: 1; min-width: 0; }
.row-card__title {
    display: block;
    font-size: 14px;
    font-weight: 600;
    color: var(--accent);
}
.row-card__desc {
    display: block;
    font-size: 12.5px;
    color: var(--ink-soft);
}
.row-card__arrow { color: #c3c9d0; flex: none; }
/* ---------- 结果摘要 ---------- */
.stat-row {
    display: flex;
    flex-wrap: wrap;
    gap: 28px;
    margin: 4px 0 6px;
}
.stat__label { font-size: 12.5px; color:
var(--ink-soft); }
.stat__value { font-size: 20px; font-weight: 700;
font-variant-numeric: tabular-nums; }
/* ---------- 流程步骤条 ---------- */
.steps { display: flex; flex-wrap: wrap; gap: 8px;
margin: 4px 0 18px; }
.step {
    display: flex; align-items: center; gap: 7px;
    padding: 8px 12px; font-size: 13px;
    background: #fff; border: 1px solid var(--line);
    border-radius: var(--radius);
}
.step__no {
    display: inline-flex; align-items: center;
justify-content: center;
    width: 20px; height: 20px; border-radius: 50%;
```

```
  background: #eef1f5; color: var(--ink-soft);
  font-size: 12px; font-variant-numeric: tabular-nums;
}
.step__name { font-weight: 600; }
.step__state { font-size: 12.5px; color:
var(--ink-soft); }
.step--done { border-color: #b8ddc6; background:
var(--ok-soft); }
.step--done .step__no { background: var(--ok); color:
#fff; }
.step--done .step__state { color: var(--ok); }
.step--fail { border-color: #eec4c0; background:
var(--danger-soft); }
.step--fail .step__no { background: var(--danger);
color: #fff; }
.step--fail .step__state { color: var(--danger);
font-weight: 600; }
.step--next { border-color: var(--accent); }
.step--next .step__no { background: var(--accent);
color: #fff; }
/* ---------- 提示块 ---------- */
.notice {
  padding: 14px 16px; margin: 0 0 16px;
  border-radius: var(--radius); border-left: 4px solid;
}
.notice--warn {
  background: var(--danger-soft);
  border-left-color: var(--danger);
  color: #7a1c17;
}
.notice__title { margin: 0 0 8px; font-weight: 600; }
.notice__list { margin: 0; padding-left: 20px;
line-height: 1.9; }
.notice__list a { color: var(--accent); }
.notice__hint { margin: 8px 0 0; font-size: 12.5px;
color: var(--ink-soft); }
/* ---------- 图斑面板（按图斑分块展示，如指标数据采集） ---------- */
.patch-panel {
  border: 1px solid var(--line);
  border-radius: var(--radius);
  background: var(--surface);
  padding: 14px 16px;
  margin: 0 0 14px;
}
.patch-panel:last-child { margin-bottom: 0; }
.patch-panel__head {
  display: flex;
  align-items: baseline;
  flex-wrap: wrap;
  gap: 6px 14px;
```

```
        margin: 0 0 12px;                                                    4951
        padding-bottom: 9px;                                                 4952
        border-bottom: 1px solid var(--line);                                4953
      }                                                                      4954
      .patch-panel__name {                                                   4955
        margin: 0;                                                           4956
        font-size: 15px;                                                     4957
        font-weight: 600;                                                    4958
      }                                                                      4959
      .patch-panel__meta {                                                   4960
        font-size: 12.5px;                                                   4961
        color: var(--ink-soft);                                              4962
      }                                                                      4963
      .patch-panel__meta a { color: var(--accent); }                         4964
      .patch-panel__tail { margin-left: auto; }                              4965
      .patch-panel .notice { margin: 0; }                                    4966
      .patch-panel table { margin: 0; }                                      4967
      /* ---------- 窄屏适配 ---------- */                                   4968
      @media (max-width: 960px) {                                            4969
        .app-shell { flex-direction: column; }                               4970
        .app-side { width: auto; flex: none; }                               4971
        .app-brand { padding: 12px 14px; }                                   4972
        .app-nav { padding: 4px 10px 10px; }                                 4973
        .app-nav ul { display: flex; flex-wrap: wrap; gap:                   4974
      4px; }                                                                 4975
        .app-nav a { margin-bottom: 0; }                                     4976
        .app-nav__group { margin: 8px 0 4px; }                               4977
        .app-user { padding: 10px 14px; }                                    4978
        .app-header { padding: 0 14px; gap: 10px; }                          4979
        .search-box { width: 180px; }                                        4980
        .app-main { padding: 16px 14px 20px; }                               4981
        .app-foot { padding: 0 14px 14px; }                                  4982
      }                                                                      4983
      @media (max-width: 640px) {                                            4984
        .crumb { display: none; }                                            4985
        .search-box { width: auto; flex: 1; }                                4986
      }                                                                      4987
    </style>                                                                 4988
</head>                                                                      4989
<body class="{% if not current_user.is_authenticated                         4990
    %}body--login{% endif %}">                                               4991
  <a class="skip-link" href="#main">跳到主要内容</a>                          4992
  {# 取一次即可（同一请求内重复调用返回相同内容）。                          4993
     类别为 toast 的渲染成居中浮层，其余仍是页面顶部的提示条 ——             4994
     错误提示需要停下来细读，不适合做成会消失的浮层。 #}                    4995
  {% set flash_msgs = get_flashed_messages(with_categories=t                 4996
    rue) %}                                                                  4997
  {# 类别为 toast / ok 的渲染成页面正中的浮层（操作完成的即时反馈）；       4998
     其余（尤其 error）仍是页面顶部的提示条 ——                              4999
     错误提示需要停下来细读，不适合做成会消失的浮层。                       5000
```

```
    没有浮层时这个容器是空的，不占任何位置。 #}                                5001
<div class="toast-stack">                                                      5002
  {% for category, message in flash_msgs if category in                        5003
  ("toast", "ok") %}                                                           5004
  <div class="toast-center" role="status"                                      5005
  aria-live="polite">{{ message }}</div>                                       5006
  {% endfor %}                                                                 5007
</div>                                                                         5008
{% if current_user.is_authenticated %}                                         5009
<div class="app-shell">                                                        5010
  <aside class="app-side">                                                     5011
    <div class="app-brand">                                                    5012
      <span class="app-brand__mark">{{                                        5013
  ui.icon("mountain", 21) }}</span>                                            5014
      <span>                                                                   5015
        <span class="app-brand__name">矿山自然恢复评价</span><br>              5016
        <span class="app-brand__sub">适宜性评价系统 Ｖ１．０</span>             5017
      </span>                                                                  5018
    </div>                                                                     5019
    <nav class="app-nav" aria-label="功能导航">                                5020
      <ul>                                                                     5021
        <li>                                                                   5022
          <a href="/"{% if home_active %}                                      5023
  aria-current="page"{% endif %}>                                              5024
            {{ ui.icon("home") }}<span>工作台</span>                           5025
          </a>                                                                 5026
        </li>                                                                  5027
      </ul>                                                                    5028
      {% for g in nav_groups %}                                                5029
      <p class="app-nav__group">{{ g.title }}</p>                              5030
      <ul>                                                                     5031
        {% for item in g.links %}                                              5032
        <li>                                                                   5033
          <a href="{{ item.href }}"{% if item.active %}                        5034
  aria-current="page"{% endif %}>                                              5035
            {{ ui.icon(item.icon) }}<span>{{ item.label                        5036
  }}</span>                                                                    5037
            {% if item.badge %}<span                                           5038
  class="app-nav__badge">{{ item.badge }}</span>{% endif                       5039
  %}                                                                           5040
          </a>                                                                 5041
        </li>                                                                  5042
        {% endfor %}                                                           5043
      </ul>                                                                    5044
      {% endfor %}                                                             5045
    </nav>                                                                     5046
    <div class="app-user">                                                     5047
      <span class="app-user__avatar">{{                                        5048
  current_user.username[0]|upper }}</span>                                     5049
      <span>                                                                   5050
```

```
          <span class="app-user__name">
            {{ role_labels.get(current_user.role,
  current_user.role) }} · {{ current_user.username }}
          </span><br>
          <span class="app-user__state">
            <span class="app-user__dot"
  aria-hidden="true"></span>在线
          </span>
        </span>
        <a href="/logout" title="退出登录" aria-label="退出登录">{{
  ui.icon("logout", 17) }}</a>
      </div>
  </aside>
  <div class="app-content">
    <header class="app-header">
      <nav class="crumb" aria-label="当前位置">
        <a href="/">首页</a>
        {{ ui.icon("chevron", 13) }}
        <strong>{{ page_title }}</strong>
      </nav>
      <span class="app-header__spacer"></span>
      <form class="search-box" method="get"
  action="/search" role="search">
        {{ ui.icon("search", 16) }}
        <label for="q" class="visually-hidden">搜索项目、图斑、标准</
  label>
        <input id="q" type="search" name="q"
  autocomplete="off"
               placeholder="搜索项目、图斑、标准" value="{{
  search_q }}">
        <button type="submit" aria-label="搜索">{{
  ui.icon("chevron", 15) }}</button>
      </form>
      <a class="icon-btn" href="/logs" title="操作日志"
  aria-label="操作日志">
        {{ ui.icon("bell", 18) }}
      </a>
    </header>
{% else %}
<div class="login-shell">
{% endif %}
    <main id="main" class="app-main">
      <div class="container">
        {% for category, message in flash_msgs if category
  not in ("toast", "ok") %}
        <p class="flash {{ category }}" role="status">{{
  message }}</p>
        {% endfor %}
        {% block content %}{% endblock %}
      </div>
```

```
      </main>                                                                  5101
    {% if current_user.is_authenticated %}                                     5102
        <footer class="app-foot">                                              5103
          <span>历史遗留废弃矿山自然恢复适宜性评价系统 V1.0</span>                5104
          <span>数据截止 {{ today }}</span>                                    5105
        </footer>                                                              5106
      </div>                                                                   5107
    </div>                                                                     5108
    {% else %}                                                                 5109
    </div>                                                                     5110
    {% endif %}                                                                5111
</body>                                                                        5112
</html>                                                                        5113
# ==== app/templates/_macros.html ====                                         5114
{# 共用宏：图标、流程状态面板 #}                                               5115
{# ============================================================================5116
   ==================                                                          5117
   图标：内联 SVG 线性图标（24×24 视图框，随文字颜色走）                       5118
   不引图标字体或外部图片，避免额外请求与资源依赖；                            5119
   stroke 用 currentColor，在深色侧边栏与浅色内容区都能自适应。                5120
   ============================================================================5121
   ================== #}                                                      5122
{% set ICONS = {                                                               5123
  "dot": '<circle cx="12" cy="12" r="3"/>',                                    5124
  "home": '<path d="M3 9.6 12 3l9 6.6V20a2 2 0 0 1-2 2H5a2                     5125
    2 0 0 1-2-2z"/><path d="M9.5 22v-7h5v7"/>',                                5126
  "folder": '<path d="M3 7.5A2 2 0 0 1 5 5.5h3.6l2 2.2H19a2                   5127
    2 0 0 1 2 2v7.8a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2z"/>',                        5128
  "book": '<path d="M4 4.8A2.8 2.8 0 0 1 6.8                                   5129
    2H20v17.5H6.8A2.8 2.8 0 0 0 4 22z"/><path d="M4                            5130
    19.5A2.8 2.8 0 0 1 6.8 16.7H20"/>',                                        5131
  "shield": '<path d="M12 22s8-4 8-10V5.2L12 2 4 5.2V12c0 6                    5132
    8 10 8 10z"/><path d="m9 12 2.2 2.2L15.5 10"/>',                           5133
  "layers": '<path d="m12 2.8 9 4.7-9 4.7-9-4.7z"/><path                       5134
    d="m3 12.5 9 4.7 9-4.7"/><path d="m3 17 9 4.7 9-4.7"/>',                   5135
  "sliders": '<path d="M4 21v-6M4 11V3M12 21v-9M12 8V3M20                      5136
    21v-4M20 13V3"/><path d="M1.5 15h5M9.5 12h5M17.5                           5137
    17h5"/>',                                                                  5138
  "database": '<ellipse cx="12" cy="5.5" rx="8"                                5139
    ry="3.2"/><path d="M4 5.5v13c0 1.8 3.6 3.2 8 3.2s8-1.4                     5140
    8-3.2v-13"/><path d="M4 12c0 1.8 3.6 3.2 8 3.2s8-1.4                       5141
    8-3.2"/>',                                                                 5142
  "target": '<circle cx="12" cy="12" r="9"/><circle cx="12"                    5143
    cy="12" r="4.6"/><circle cx="12" cy="12" r="1"/>',                         5144
  "file": '<path d="M14 2.8H7a2 2 0 0 0-2 2v14.4a2 2 0 0 0                     5145
    2 2h10a2 2 0 0 0 2-2V7.8z"/><path d="M14                                   5146
    2.8V7.8h5"/><path d="M8.8 13h6.4M8.8 17h6.4"/>',                           5147
  "users": '<circle cx="9.5" cy="8" r="3.6"/><path d="M3.5                     5148
    20.5v-1.4A3.6 3.6 0 0 1 7.1 15.5h4.8a3.6 3.6 0 0 1 3.6                     5149
    3.6v1.4"/><path d="M17 4.9a3.6 3.6 0 0 1 0 6.9"/><path                     5150
```

```
      d="M19 15.8a3.6 3.6 0 0 1 2.4 3.4v1.3"/>',
   "search": '<circle cx="11" cy="11" r="7"/><path d="m20.5
      20.5-4-4"/>',
   "bell": '<path d="M18 9.5a6 6 0 1 0-12 0c0 5-2.2 6.2-2.2
      6.2h16.4S18 14.5 18 9.5"/><path d="M13.8 19.5a2 2 0 0
      1-3.6 0"/>',
   "plus": '<path d="M12 5v14M5 12h14"/>',
   "upload": '<path d="M12 16V4"/><path d="m7.5 8.5 4.5-4.5
      4.5 4.5"/><path d="M4 16.5v2A2.5 2.5 0 0 0 6.5
      21h11a2.5 2.5 0 0 0 2.5-2.5v-2"/>',
   "logout": '<path d="M15 4h3.5A2.5 2.5 0 0 1 21 6.5v11A2.5
      2.5 0 0 1 18.5 20H15"/><path d="M10 16.5 5.5 12 10
      7.5"/><path d="M5.5 12H15"/>',
   "chevron": '<path d="m9.5 5.5 6.5 6.5-6.5 6.5"/>',
   "warn": '<path d="M12 3.5 21.5 20H2.5z"/><path d="M12
      10v4.2"/><path d="M12 17.4h.01"/>',
   "mountain": '<path d="m3 20 6.5-11L14 17l2.5-4L21
      20z"/><circle cx="17" cy="6" r="2.2"/>',
} %}
{% macro icon(name, size=18) %}
<svg class="ico" width="{{ size }}" height="{{ size }}"
     viewBox="0 0 24 24"
      fill="none" stroke="currentColor" stroke-width="1.7"
      stroke-linecap="round" stroke-linejoin="round"
      aria-hidden="true" focusable="false">{{
     ICONS.get(name, ICONS.dot)|safe }}</svg>
{% endmacro %}
{# 评价流程步骤条：按论文 4.1 节顺序显示各前置步骤的完成情况
   顺序为：限制性条件判定 → 划分微地质单元 → 录入指标数据 → 执行评价。
   论文原文：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
   废弃矿山，按照地形地貌划分……三种地质微单元类型」，故判定在前。 #}
{% macro workflow_steps(progress) %}
<div class="steps">
   <div class="step {% if progress.constraint.passed
      %}step--done{% elif progress.constraint.done
      %}step--fail{% else %}step--todo{% endif %}">
      <span class="step__no">1</span>
      <span class="step__name">限制性条件判定</span>
      <span class="step__state">
         {% if progress.constraint.passed %}已通过
         {% elif progress.constraint.done %}未通过
         {% else %}待完成{% endif %}
      </span>
   </div>
   <div class="step {% if progress.units.done %}step--done{%
      else %}step--todo{% endif %}">
      <span class="step__no">2</span>
      <span class="step__name">划分微地质单元</span>
      <span class="step__state">
         {% if progress.units.done %}已完成{% else %}待完成{% endif
```

```
      %}                                                                    5201
      </span>                                                               5202
   </div>                                                                   5203
   <div class="step {% if progress.data.ready %}step--done{%               5204
      else %}step--todo{% endif %}">                                        5205
      <span class="step__no">3</span>                                       5206
      <span class="step__name">录入指标数据</span>                          5207
      <span class="step__state">                                            5208
         {% if progress.data.ready %}已完成{% else %}待完成{% endif         5209
      %}                                                                    5210
      </span>                                                               5211
   </div>                                                                   5212
   <div class="step step--next">                                            5213
      <span class="step__no">4</span>                                       5214
      <span class="step__name">执行适宜性评价</span>                        5215
      <span class="step__state">当前步骤</span>                             5216
   </div>                                                                   5217
</div>                                                                      5218
{% endmacro %}                                                              5219
{# 前置条件未满足时的提示块，逐条给出原因与去处理入口 #}                   5220
{% macro workflow_blocking(progress) %}                                     5221
{% if progress.blocking %}                                                  5222
<div class="notice notice--warn">                                           5223
   <p class="notice__title">尚不能执行本步骤，请先完成以下前置工作：</p>   5224
   <ol class="notice__list">                                                5225
      {% for b in progress.blocking %}                                      5226
      <li>                                                                  5227
         <strong>{{ b.label }}</strong> —— {{ b.reason }}                   5228
         <a class="link-btn" href="{{ b.fix_url }}">{{                      5229
      b.fix_label }}</a>                                                    5230
      </li>                                                                 5231
      {% endfor %}                                                          5232
   </ol>                                                                    5233
   <p class="notice__hint">                                                 5234
      说明：本系统按论文 4.1 节的评价流程设置前置检查，                     5235
      避免在前置条件不具备时产出不合理的评价结论。                         5236
   </p>                                                                     5237
</div>                                                                      5238
{% endif %}                                                                 5239
{% endmacro %}                                                              5240
{# 一行式状态摘要，用于列表页 #}                                           5241
{% macro workflow_badge(progress) %}                                        5242
{% if progress.ready %}                                                     5243
<span class="tag tag--ok">可执行评价</span>                                 5244
{% elif progress.blocking %}                                                5245
<span class="tag tag--warn">{{ progress.blocking[0].label                   5246
   }}</span>                                                                5247
<span class="muted">{{ progress.blocking[0].reason }}</span>                5248
{% endif %}                                                                 5249
{% endmacro %}                                                              5250
```

```
{#  层级引导：本系统是「项目  →  图斑」两级结构，评价针对的是图斑而非项目。
    下游各页按图斑组织数据，故只要存在「还没建图斑的项目」，就提示下一步该做什么。  #}
{% macro patch_guide(projects_without_patch) %}
{% if projects_without_patch %}
<div class="notice notice--warn">
  <p class="notice__title">有 {{ projects_without_patch|lengt
    h }} 个项目下还没有图斑，本页不会出现它们。</p>
  <p>
    本系统是「项目 → 矿山图斑」两级结构：项目是归档单位，而
    <strong>限制性条件判定、微地质单元划分、指标数据与适宜性评价都是针对图斑进行的</strong>。
  </p>
  <ul class="notice__list">
    {% for p in projects_without_patch %}
    <li>
      {{ p.name }}
      —— <a class="link-btn" href="/project/{{ p.id
    }}/patch/add">为它新建图斑</a>
    </li>
    {% endfor %}
  </ul>
  <p class="notice__hint">
    也可以先进入 <a href="/project/">项目与矿山档案</a>，
    点该项目右侧的「＋ 新建图斑」，建好之后本页就会出现该图斑。
  </p>
</div>
{% endif %}
{% endmacro %}
# ==== app/templates/error.html ====
{% extends "base.html" %}
{% block title %}{{ code }} · {{ message }}{% endblock %}
{% block content %}
<div class="card" style="max-width:560px;">
  <h2>{{ code }} · {{ message }}</h2>
  <p class="muted">
    请返回上一页重试；若确认该操作应当可用，请检查当前账号的角色权限。
  </p>
  <p class="actions" style="margin-top:16px;">
    <a class="btn" href="/">返回首页</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/auth/login.html ====
{% extends "base.html" %}
{% block title %}登录 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<div class="login-wrap">
  <div class="card login-card">
    <h2>用户登录</h2>
    <p class="login-card__sub">历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ１．０</p>
    <form method="post" novalidate>
```

```
        {{ form.hidden_tag() }}                                          5301
        <div class="field">                                              5302
          <label class="field__label" for="{{                            5303
    form.username.id }}">用户名</label>                                  5304
          {{ form.username(id=form.username.id,                          5305
    autocomplete="username") }}                                          5306
        </div>                                                           5307
        <div class="field">                                              5308
          <label class="field__label" for="{{                            5309
    form.password.id }}">密码</label>                                    5310
          {{ form.password(id=form.password.id,                          5311
    autocomplete="current-password") }}                                  5312
        </div>                                                           5313
        <div class="field">                                              5314
          <label class="checkbox" for="{{ form.remember.id               5315
    }}">                                                                 5316
            {{ form.remember(id=form.remember.id) }}                     5317
            <span>记住我（下次自动登录，一个月内有效）</span>              5318
          </label>                                                       5319
        </div>                                                           5320
        <p style="margin:18px 0 0;">                                     5321
          {{ form.submit(class_="btn", style="width:100%;") }}5322
        </p>                                                             5323
      </form>                                                            5324
      <p class="muted" style="margin:14px 0 0;">演示账号：admin /        5325
      admin123</p>                                                       5326
    </div>                                                               5327
</div>                                                                   5328
{% endblock %}                                                           5329
# ==== app/templates/auth/home.html ====                                 5330
{% extends "base.html" %}                                                5331
{% from "_macros.html" import icon %}                                    5332
{% block title %}工作台 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}    5333
{% block content %}                                                      5334
{# ---------- 页首横幅 ---------- #}                                     5335
<section class="hero">                                                   5336
  <div class="hero__body">                                               5337
    <h1 class="hero__title">历史遗留废弃矿山自然恢复适宜性评价系统</h1>  5338
    <p class="hero__desc">                                               5339
      以论文《历史遗留废弃矿山自然恢复适宜性评价指标体系建设及典型案例》  5340
      建立的评价方法为算法内核。下方按论文 4.1 节的评价流程顺序排列，    5341
      系统会在每一步检查前置条件，前置步骤未完成时不会放行后续计算。      5342
    </p>                                                                 5343
    <div class="hero__actions">                                          5344
      <a class="hero-btn hero-btn--ghost"                                5345
    href="/indicator/import">                                            5346
        {{ icon("upload", 16) }}批量导入                                 5347
      </a>                                                               5348
      <a class="hero-btn hero-btn--solid"                                5349
    href="/project/add">                                                 5350
```

```
            {{ icon("plus", 16) }}新建评价项目                    5351
          </a>                                                    5352
        </div>                                                    5353
      </div>                                                      5354
</section>                                                        5355
{# ---------- 数据概览 ---------- #}                             5356
<section class="stat-cards">                                      5357
      <a class="stat-card" href="/project/">                      5358
        <span class="stat-card__label">评价项目</span>            5359
        <span class="stat-card__value">{{ dash.stats.projects     5360
        }}<span class="stat-card__unit">个</span></span>          5361
        <span class="stat-card__note">                            5362
          {%- if projects_without_patch -%}                       5363
            其中 {{ projects_without_patch|length }} 个项目尚未建立图斑  5364
          {%- elif dash.progress.patches -%}                      5365
            已建立 {{ dash.progress.patches }} 个矿山图斑         5366
          {%- else -%}                                            5367
            尚未建立矿山图斑                                      5368
          {%- endif -%}                                           5369
        </span>                                                   5370
      </a>                                                        5371
      <a class="stat-card" href="/project/">                      5372
        <span class="stat-card__label">矿山图斑</span>            5373
        <span class="stat-card__value">{{ dash.stats.patches      5374
        }}<span class="stat-card__unit">个</span></span>          5375
        <span class="stat-card__note">                            5376
          {% if dash.progress.ready %}其中 {{ dash.progress.ready 5377
        }} 个具备评价条件{% else %}暂无具备评价条件的图斑{% endif %}  5378
        </span>                                                   5379
      </a>                                                        5380
      <a class="stat-card" href="/unit/">                         5381
        <span class="stat-card__label">微地质单元</span>          5382
        <span class="stat-card__value">{{ dash.stats.unit_types   5383
        }}<span class="stat-card__unit">类</span></span>          5384
        <span class="stat-card__note">                            5385
          {% if dash.progress.units %}共划分 {{                   5386
        dash.progress.units }} 个单元{% else %}尚未划分单元{% endif %}  5387
        </span>                                                   5388
      </a>                                                        5389
      <a class="stat-card" href="/standard/">                     5390
        <span class="stat-card__label">标准与政策</span>          5391
        <span class="stat-card__value">{{ dash.stats.standards    5392
        }}<span class="stat-card__unit">项</span></span>          5393
        <span class="stat-card__note">地方标准与政策文件</span>   5394
      </a>                                                        5395
</section>                                                        5396
{# ---------- 评价流程 ---------- #}                             5397
<section class="section">                                         5398
      <div class="section__head">                                 5399
        <h2 class="section__title">评价流程</h2>                  5400
```

```
      <p class="section__desc">
        按论文 4.1 节顺序排列，前置步骤未完成时不会执行后续计算
      </p>
    </div>
    <div class="step-grid">
      {% for s in dash.steps %}
      {% if s.available %}
      <a class="step-card" href="{{ s.href }}">
        <span class="step-card__no">{{ s.no }}</span>
        <span class="step-card__icon">{{ icon(s.icon, 18)
      }}</span>
        <span class="step-card__title">{{ s.title }}</span>
        <span class="step-card__desc">{{ s.desc }}</span>
        {% if s.pending %}
        <span class="step-card__badge">待处理 {{ s.pending
      }}</span>
        {% endif %}
      </a>
      {% else %}
      <div class="step-card step-card--locked"
      aria-disabled="true">
        <span class="step-card__no">{{ s.no }}</span>
        <span class="step-card__icon">{{ icon(s.icon, 18)
      }}</span>
        <span class="step-card__title">{{ s.title }}</span>
        <span class="step-card__desc">{{ s.desc }}</span>
        <span class="step-card__lock">{{ s.note }}</span>
      </div>
      {% endif %}
      {% endfor %}
    </div>
</section>
{# ---------- 系统管理 ---------- #}
<section class="section">
    <div class="section__head">
      <h2 class="section__title">系统管理</h2>
    </div>
    <a class="row-card" href="/users">
      <span class="row-card__icon">{{ icon("users", 19)
      }}</span>
      <span class="row-card__main">
        <span class="row-card__title">用户与操作日志</span>
        <span class="row-card__desc">用户维护与角色权限；全部写操作自动留痕，可查询追溯
      </span>
      </span>
      <span class="row-card__arrow">{{ icon("chevron", 17)
      }}</span>
    </a>
</section>
{% endblock %}
```

```
# ==== app/templates/auth/users.html ====
{% extends "base.html" %}
{% block title %}用户与权限管理{% endblock %}
{% block content %}
<div class="card">
  <h2>用户管理</h2>
  <p class="muted">
    系统设有管理员、评价员、查看者三种角色：管理员可维护用户与基础数据，
    评价员可建项目、录数据、执行评价，查看者仅可查阅。
  </p>
  <p class="actions" style="margin-bottom:12px;">
    {% if current_user.role == "admin" %}
    <a class="btn btn--sm" href="/users/add">新增用户</a>
    {% endif %}
    <a class="btn btn--sm btn--ghost"
    href="/logs">查看操作日志</a>
  </p>
  {% if current_user.role != "admin" %}
  <p class="muted">
    当前角色为「{{ role_labels.get(current_user.role,
    current_user.role) }}」，
    仅管理员可新增、编辑或删除用户。
  </p>
  {% endif %}
  <table>
    <caption class="muted">共 {{ users|length }}
    个用户</caption>
    <thead>
      <tr>
        <th scope="col">ID</th>
        <th scope="col">用户名</th>
        <th scope="col">角色</th>
        <th scope="col">状态</th>
        {% if current_user.role == "admin" %}<th
    scope="col">操作</th>{% endif %}
      </tr>
    </thead>
    <tbody>
      {% for u in users %}
      <tr>
        <td class="num">{{ u.id }}</td>
        <td>
          {{ u.username }}
          {% if u.id == current_user.id %}<span
    class="muted">（当前登录）</span>{% endif %}
        </td>
        <td>{{ role_labels.get(u.role, u.role) }}</td>
        <td>
          {% if u.active %}<span class="tag
    tag--ok">启用</span>
```

```
            {% else %}<span class="tag tag--muted">停用</span>{% 5501
    endif %}                                                    5502
        </td>                                                   5503
        {% if current_user.role == "admin" %}                   5504
        <td class="nowrap">                                     5505
          <span class="row-actions">                            5506
            <a class="link-btn" href="/users/{{ u.id            5507
    }}/edit">编辑</a>                                           5508
            <form method="post" action="/users/{{ u.id          5509
    }}/delete"                                                  5510
                  onsubmit="return confirm('确定删除用户「{{    5511
    u.username }}」吗？\n该操作无法恢复。');">                  5512
              {{ form.hidden_tag() }}                           5513
              <button type="submit" class="link-btn             5514
    link-btn--danger"                                           5515
                      {% if u.id == current_user.id             5516
    %}disabled                                                  5517
                      title="不能删除当前登录的账号"{% endif    5518
    %}>删除</button>                                            5519
            </form>                                             5520
          </span>                                               5521
        </td>                                                   5522
        {% endif %}                                             5523
      </tr>                                                     5524
      {% endfor %}                                              5525
    </tbody>                                                    5526
  </table>                                                      5527
</div>                                                          5528
{% endblock %}                                                  5529
# ==== app/templates/auth/add_user.html ====                    5530
{% extends "base.html" %}                                       5531
{% block title %}新增用户{% endblock %}                         5532
{% block content %}                                             5533
<div class="card" style="max-width:560px;">                     5534
  <h2>新增用户</h2>                                             5535
  <form method="post">                                          5536
    {{ form.hidden_tag() }}                                     5537
    <div class="field">                                         5538
      <label class="field__label" for="username">用户名</label> 5539
      <input id="username" type="text" name="username"          5540
    required                                                    5541
             autocomplete="off" style="width:280px;">           5542
    </div>                                                      5543
    <div class="field">                                         5544
      <label class="field__label" for="password">密码</label>   5545
      <input id="password" type="password" name="password"      5546
    required                                                    5547
             autocomplete="new-password"                        5548
    style="width:280px;">                                       5549
    </div>                                                      5550
```

```
      <div class="field">
        <label class="field__label" for="role">角色</label>
        <select id="role" name="role">
          <option value="viewer">查看者（仅可查阅）</option>
          <option value="evaluator">评价员（可建项目、录数据、执行评价）</option
      >
          <option value="admin">管理员（可维护用户与基础数据）</option>
        </select>
      </div>
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/users">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/auth/edit_user.html ====
{% extends "base.html" %}
{% block title %}编辑用户 · {{ user.username }}{% endblock %}
{% block content %}
<div class="card" style="max-width:560px;">
  <h2>编辑用户 · {{ user.username }}</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="role">角色</label>
      <select id="role" name="role">
        {% for code, label in role_labels.items() %}
        <option value="{{ code }}" {% if user.role == code
    %}selected{% endif %}>{{ label }}</option>
        {% endfor %}
      </select>
    </div>
    <div class="field">
      <label class="checkbox" for="active">
        <input id="active" type="checkbox" name="active"
    value="1"
               {% if user.active %}checked{% endif %}>
        <span>启用该账号</span>
      </label>
      <p class="muted" style="margin:6px 0 0;">
        停用后该账号无法登录，但其历史操作记录会保留。
        系统至少需保留一名启用状态的管理员。
      </p>
    </div>
    <div class="field">
      <label class="field__label" for="password">重置密码（留空则不修改
    ）</label>
      <input id="password" type="password" name="password"
             autocomplete="new-password"
```

```
      style="width:280px;"                                                5601
                  placeholder="留空表示保持原密码">                          5602
      </div>                                                              5603
      <p class="actions" style="margin-top:18px;">                        5604
        <button type="submit" class="btn">保存</button>                   5605
        <a class="btn btn--ghost" href="/users">返回</a>                   5606
      </p>                                                                5607
    </form>                                                               5608
</div>                                                                    5609
{% endblock %}                                                            5610
# ==== app/templates/auth/logs.html ====                                  5611
{% extends "base.html" %}                                                 5612
{% block title %}操作日志{% endblock %}                                    5613
{% block content %}                                                       5614
<div class="card">                                                        5615
  <h2>操作日志</h2>                                                       5616
  <p class="muted">                                                       5617
    系统自动记录所有写操作（新建、编辑、删除、导入、执行评价等），          5618
    便于追溯数据变更。失败的操作也会记录并标注 HTTP 状态码。              5619
  </p>                                                                    5620
  <p class="actions" style="margin-bottom:12px;">                         5621
    <a class="btn btn--sm btn--ghost"                                     5622
    href="/users">返回用户管理</a>                                        5623
  </p>                                                                    5624
  {% if not records %}                                                    5625
  <p class="muted">暂无操作记录。</p>                                     5626
  {% else %}                                                              5627
  <table>                                                                 5628
    <caption class="muted">共 {{ total }} 条记录，当前第 {{ page }}      5629
    / {{ pages }} 页</caption>                                            5630
    <thead>                                                               5631
      <tr>                                                                5632
        <th scope="col">时间</th>                                         5633
        <th scope="col">操作人</th>                                       5634
        <th scope="col">操作</th>                                         5635
        <th scope="col">对象</th>                                         5636
        <th scope="col">说明</th>                                         5637
      </tr>                                                               5638
    </thead>                                                              5639
    <tbody>                                                               5640
      {% for r in records %}                                              5641
      <tr>                                                                5642
        <td class="mono nowrap">{{ r.created_at.strftime("%Y              5643
    -%m-%d %H:%M:%S") if r.created_at else "—" }}</td>                    5644
        <td>{{ users_map.get(r.user_id, "—") }}</td>                      5645
        <td style="text-align:left;">{{ r.action }}</td>                  5646
        <td>                                                              5647
          {% if r.target_type %}{{ r.target_type }}{% if                  5648
    r.target_id %}#{{ r.target_id }}{% endif %}                           5649
          {% else %}—{% endif %}                                          5650
```

```
            </td>                                                                5651
            <td style="text-align:left;">{{ r.detail or "—"                      5652
      }}</td>                                                                    5653
        </tr>                                                                    5654
        {% endfor %}                                                             5655
      </tbody>                                                                   5656
    </table>                                                                     5657
    {% if pages > 1 %}                                                           5658
    <nav class="actions" aria-label="日志分页">                                  5659
      {% if page > 1 %}                                                          5660
      <a class="btn btn--sm btn--ghost" href="/logs?page={{                      5661
      page - 1 }}">上一页</a>                                                    5662
      {% endif %}                                                                5663
      <span class="muted">第 {{ page }} / {{ pages }} 页</span>                  5664
      {% if page < pages %}                                                      5665
      <a class="btn btn--sm btn--ghost" href="/logs?page={{                      5666
      page + 1 }}">下一页</a>                                                    5667
      {% endif %}                                                                5668
    </nav>                                                                       5669
    {% endif %}                                                                  5670
    {% endif %}                                                                  5671
</div>                                                                           5672
{% endblock %}                                                                   5673
# ==== app/templates/project/index.html ====                                     5674
{% extends "base.html" %}                                                        5675
{% block title %}项目与矿山档案{% endblock %}                                    5676
{% block content %}                                                              5677
<div class="card">                                                               5678
    <h2>项目与矿山档案</h2>                                                     5679
    <p class="muted">                                                            5680
      项目是归档单位，用于组织同一矿区的多个图斑；                              5681
      <strong>限制性条件判定、微地质单元划分、指标数据与适宜性评价都是针对图斑进行的</strong>。  5682
      因此新建项目后，请继续在该项目下建立图斑，后续模块才可操作。             5683
    </p>                                                                         5684
    <p class="actions" style="margin-bottom:12px;">                              5685
      <a class="btn btn--sm" href="/project/add">新建项目</a>                    5686
    </p>                                                                         5687
    {% if not projects %}                                                        5688
    <p class="muted">                                                            5689
      暂无项目。点击上方「新建项目」开始建立评价项目，                          5690
      或运行 <span class="mono">python seed_demo.py</span>                       5691
      载入论文案例。                                                             5692
    </p>                                                                         5693
    {% else %}                                                                   5694
    <table>                                                                      5695
      <caption class="muted">共 {{ projects|length }}                            5696
      个项目</caption>                                                           5697
      <thead>                                                                    5698
        <tr>                                                                     5699
          <th scope="col">项目名称</th>                                          5700
```

```
            <th scope="col">编号</th>
            <th scope="col">状态</th>
            <th scope="col">图斑数</th>
            <th scope="col">操作</th>
          </tr>
        </thead>
        <tbody>
          {% for p in projects %}
          <tr>
            <td style="text-align:left;">{{ p.name }}</td>
            <td class="mono">{{ p.code or "—" }}</td>
            <td><span class="tag tag--muted">{{ p.status
        }}</span></td>
            <td class="num">
              {% if p.patches.count() %}
                {{ p.patches.count() }}
              {% else %}
                <span class="tag tag--warn">0</span>
              {% endif %}
            </td>
            <td class="nowrap">
              <span class="row-actions">
                <a class="link-btn" href="/project/{{ p.id
        }}/patches">{% if p.patches.count() %}矿山图斑{% else %}＋
        新建图斑{% endif %}</a>
                <a class="link-btn" href="/project/{{ p.id
        }}/edit">编辑</a>
                <form method="post" action="/project/{{ p.id
        }}/delete"
                      onsubmit="return confirm('确定删除项目「{{
        p.name }}」吗？\n该项目下的图斑、微地质单元与评价数据将一并删除，且无法恢复。');">
                  {{ form.hidden_tag() }}
                  <button type="submit" class="link-btn
        link-btn--danger">删除</button>
                </form>
              </span>
            </td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
      {% endif %}
    </div>
    {% endblock %}
    # ==== app/templates/project/edit.html ====
    {% extends "base.html" %}
    {% block title %}{{ "编辑项目" if project else "新建项目" }}{%
        endblock %}
    {% block content %}
    <div class="card" style="max-width:640px;">
```

```
  <h2>{{ "编辑项目" if project else "新建项目" }}</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="name">项目名称</label>
      <input id="name" type="text" name="name"
    style="width:100%;" required
             value="{{ project.name if project else '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="code">项目编号</label>
      <input id="code" type="text" name="code"
    style="width:240px;"
             value="{{ project.code or '' if project else
    '' }}">
    </div>
    {% if project %}
    <div class="field">
      <label class="field__label" for="status">状态</label>
      <select id="status" name="status">
        {% for s in ["进行中", "已完成", "已归档"] %}
        <option value="{{ s }}" {% if project.status == s
    %}selected{% endif %}>{{ s }}</option>
        {% endfor %}
      </select>
    </div>
    {% endif %}
    <div class="field">
      <label class="field__label" for="description">项目说明</la
    bel>
      <textarea id="description" name="description">{{
    project.description or '' if project else ''
    }}</textarea>
    </div>
    <p class="actions" style="margin-top:18px;">
      <button type="submit" class="btn">保存</button>
      <a class="btn btn--ghost" href="/project/">返回</a>
    </p>
  </form>
</div>
{% endblock %}
# ==== app/templates/project/patches.html ====
{% extends "base.html" %}
{% block title %}矿山图斑 · {{ project.name }}{% endblock %}
{% block content %}
<div class="card">
  <h2>矿山图斑 · {{ project.name }}</h2>
  <p class="actions" style="margin-bottom:12px;">
    <a class="btn btn--sm" href="/project/{{ project.id
    }}/patch/add">新建图斑</a>
```

```
    <a class="btn btn--sm btn--ghost"
    href="/project/">返回项目列表</a>
  </p>
  {% if not project.patches.count() %}
  <p class="muted">该项目下暂无图斑。点击上方「新建图斑」建立第一个矿山图斑。</p>
  {% else %}
  <table>
    <caption class="muted">共 {{ project.patches.count() }}
    个图斑</caption>
    <thead>
      <tr>
        <th scope="col">图斑名称</th>
        <th scope="col">行政区</th>
        <th scope="col">面积（公顷）</th>
        <th scope="col">开采矿种</th>
        <th scope="col">微地质单元</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in project.patches %}
      <tr>
        <td style="text-align:left;">{{ p.name }}</td>
        <td>{{ p.admin_region or "—" }}</td>
        <td class="num">{{ p.area_ha if p.area_ha is not
    none else "—" }}</td>
        <td>{{ p.mineral_type or "—" }}</td>
        <td class="num">{{ p.units.count() }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/unit/{{ p.id
    }}">微地质单元</a>
            <a class="link-btn" href="/constraint/{{ p.id
    }}">限制性判定</a>
            <a class="link-btn" href="/project/patch/{{
    p.id }}/edit">编辑</a>
            <form method="post" action="/project/patch/{{
    p.id }}/delete"
                  onsubmit="return confirm('确定删除图斑「{{
    p.name }}」吗？\n该图斑的微地质单元、指标数据与评价结果将一并删除，且无法恢复。');">
              {{ form.hidden_tag() }}
              <button type="submit" class="link-btn
    link-btn--danger">删除</button>
            </form>
          </span>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
```

```
  {% endif %}                                                          5851
</div>                                                                 5852
{% endblock %}                                                         5853
# ==== app/templates/project/patch_edit.html ====                      5854
{% extends "base.html" %}                                              5855
{% block title %}{{ "编辑图斑" if patch else "新建图斑" }}{%           5856
   endblock %}                                                         5857
{% block content %}                                                    5858
<div class="card" style="max-width:720px;">                            5859
  <h2>{{ "编辑图斑" if patch else "新建图斑" }} · {{ project.name     5860
    }}</h2>                                                            5861
  <form method="post">                                                 5862
    {{ form.hidden_tag() }}                                            5863
    <div class="field">                                                5864
      <label class="field__label" for="patch-name">图斑名称</lab5865
    el>                                                                5866
      <input id="patch-name" type="text" name="name"                   5867
    style="width:100%;" required                                       5868
             value="{{ patch.name if patch else '' }}">                5869
    </div>                                                             5870
    <div class="field">                                                5871
      <label class="field__label" for="admin-region">所在行政区</ 5872
    label>                                                             5873
      <input id="admin-region" type="text"                             5874
    name="admin_region" style="width:100%;"                            5875
             value="{{ patch.admin_region or '' if patch               5876
    else '' }}">                                                       5877
    </div>                                                             5878
    <div class="field">                                                5879
      <label class="field__label" for="area-ha">图斑面积（公顷）</la5880
    bel>                                                               5881
      <input id="area-ha" type="number" step="any"                     5882
    name="area_ha"                                                     5883
             value="{{ patch.area_ha if patch and                      5884
    patch.area_ha is not none else '' }}">                             5885
    </div>                                                             5886
    <div class="field">                                                5887
      <label class="field__label" for="mineral-type">开采矿种</l5888
    abel>                                                              5889
      <input id="mineral-type" type="text"                             5890
    name="mineral_type" style="width:240px;"                           5891
             value="{{ patch.mineral_type or '' if patch               5892
    else '' }}">                                                       5893
    </div>                                                             5894
    <div class="field">                                                5895
      <label class="field__label" for="mining-method">开采方式</ 5896
    label>                                                             5897
      <input id="mining-method" type="text"                            5898
    name="mining_method" style="width:240px;"                          5899
             value="{{ patch.mining_method or '' if patch              5900
```

```
else '' }}">
</div>
<div class="field">
  <label class="field__label" for="stop-years">停产年限</lab
el>
  <input id="stop-years" type="number" step="any"
name="stop_years"
         value="{{ patch.stop_years if patch and
patch.stop_years is not none else '' }}">
</div>
<div class="field">
  <label class="checkbox" for="responsible-missing">
    <input id="responsible-missing" type="checkbox"
name="responsible_entity_missing" value="1"
           {% if not patch or patch.responsible_entity_m
issing %}checked{% endif %}>
    <span>责任主体缺失</span>
  </label>
</div>
<h3>参照生态系统（指标标准化基准）</h3>
<p class="muted">
  论文 4.2.2 节取值：植被覆盖度 75%、Shannon-Wiener 多样性指数 0.7。
  该基准与图斑绑定管理，可按不同参照生态系统分别设定。
</p>
<div class="field">
  <label class="field__label" for="coverage-benchmark">植
被覆盖度基准（%）</label>
  <input id="coverage-benchmark" type="number"
step="any" name="coverage_benchmark"
         value="{{ reference.coverage_benchmark if
reference else 75 }}">
</div>
<div class="field">
  <label class="field__label" for="shannon-benchmark">Sh
annon-Wiener 多样性指数基准</label>
  <input id="shannon-benchmark" type="number"
step="any" name="shannon_benchmark"
         value="{{ reference.shannon_benchmark if
reference else 0.7 }}">
</div>
<div class="field">
  <label class="field__label" for="dominant-species">建群种
</label>
  <input id="dominant-species" type="text"
name="dominant_species" style="width:100%;"
         value="{{ reference.dominant_species or '' if
reference else '' }}">
</div>
<p class="actions" style="margin-top:18px;">
  <button type="submit" class="btn">保存</button>
```

```
        <a class="btn btn--ghost" href="/project/{{                5951
      project.id }}/patches">返回</a>                              5952
      </p>                                                         5953
    </form>                                                        5954
</div>                                                             5955
{% if patch %}                                                     5956
<div class="card" style="max-width:720px;">                        5957
    <h2>遥感影像挂接</h2>                                          5958
    <p class="muted">                                              5959
      挂接遥感底图或无人机正射影像，用于辅助微地质单元划分与植被覆盖度判读。  5960
      支持 PNG、JPG、GIF、WebP、BMP、TIFF，单个文件不超过 32 MB。  5961
    </p>                                                           5962
    {% if patch.image_path %}                                      5963
    <figure style="margin:0 0 16px;">                              5964
      <img src="{{ url_for('static', filename=patch.image_path     5965
      ) }}"                                                        5966
           alt="图斑「{{ patch.name }}」已挂接的遥感影像"          5967
           style="max-width:100%;border:1px solid                  5968
      var(--line);border-radius:6px;">                             5969
      <figcaption class="muted" style="margin-top:6px;">           5970
        {{ patch.image_note or "遥感影像" }}                       5971
        {% if patch.image_uploaded_at %}                           5972
        · 上传于 {{ patch.image_uploaded_at.strftime("%Y-%m-%d     5973
      %H:%M") }}                                                   5974
        {% endif %}                                                5975
      </figcaption>                                                5976
    </figure>                                                      5977
    <form method="post" action="/project/patch/{{ patch.id         5978
      }}/image/delete"                                             5979
          onsubmit="return confirm('确定移除该遥感影像吗？');">    5980
      {{ form.hidden_tag() }}                                      5981
      <button type="submit" class="btn btn--sm                     5982
      btn--danger">移除影像</button>                               5983
    </form>                                                        5984
    <hr style="margin:16px 0;border:none;border-top:1px solid      5985
      var(--line);">                                               5986
    {% endif %}                                                    5987
    <form method="post" action="/project/patch/{{ patch.id         5988
      }}/image"                                                    5989
          enctype="multipart/form-data">                           5990
      {{ form.hidden_tag() }}                                      5991
      <div class="field">                                          5992
        <label class="field__label" for="image-file">              5993
          {{ "替换影像文件" if patch.image_path else "选择影像文件" }}  5994
        </label>                                                   5995
        <input id="image-file" type="file" name="image"            5996
               accept=".png,.jpg,.jpeg,.gif,.webp,.bmp,.tif,.t     5997
      iff" required>                                               5998
      </div>                                                       5999
      <div class="field">                                          6000
```

```
        <label class="field__label" for="image-note">影像说明（如来源、时相）</label>
        <input id="image-note" type="text" name="image_note" style="width:100%;"
               placeholder="如：2025 年 6 月无人机正射影像"
               value="{{ patch.image_note or '' }}">
      </div>
      <p class="actions">
        <button type="submit" class="btn">{{ "替换影像" if patch.image_path else "上传并挂接" }}</button>
      </p>
    </form>
</div>
{% endif %}
{% endblock %}
# ==== app/templates/unit/list.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}微地质单元{% endblock %}
{% block content %}
<div class="card">
  <h2>微地质单元</h2>
  <p class="muted">
    按论文 4.1 节将矿山图斑划分为岩质边坡、碎石边坡、平台三类微地质单元。
    面积占比由面积自动计算，即论文式（9） 中决定各单元权重的系数。
  </p>
  {{ patch_guide(projects_without_patch) }}
  {% if not rows %}
  <p class="muted">
    暂无矿山图斑。请先到「项目与矿山档案」建立图斑。
  </p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">所属项目</th>
        <th scope="col">矿山图斑</th>
        <th scope="col">已划分单元</th>
        <th scope="col">占比合计</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rows %}
      <tr>
        <td>{{ r.patch.project.name if r.patch.project else "—" }}</td>
        <td style="text-align:left;">{{ r.patch.name }}</td>
        <td class="num">{{ r.count }} 个</td>
        <td class="num">
```

```
            {% if r.count %}                                                    6051
              {{ "%.2f%%"|format(r.ratio_sum * 100) }}                          6052
              {% if r.ratio_sum < 0.999 or r.ratio_sum >                        6053
    1.001 %}                                                                    6054
              <span class="warn">（不等于 100%）</span>                          6055
              {% endif %}                                                       6056
            {% else %}—{% endif %}                                              6057
          </td>                                                                 6058
          <td class="nowrap">                                                   6059
            <span class="row-actions">                                          6060
              <a class="link-btn" href="/unit/{{ r.patch.id                     6061
    }}">划分与维护</a>                                                          6062
              <a class="link-btn" href="/indicator/">录入指标数据</a            6063
    >                                                                           6064
            </span>                                                             6065
          </td>                                                                 6066
        </tr>                                                                   6067
        {% endfor %}                                                            6068
      </tbody>                                                                  6069
    </table>                                                                    6070
    {% endif %}                                                                 6071
</div>                                                                          6072
{% endblock %}                                                                  6073
#  ==== app/templates/unit/index.html ====                                      6074
{% extends "base.html" %}                                                       6075
{% block title %}微地质单元 · {{ patch.name }}{% endblock %}                    6076
{% block content %}                                                             6077
<div class="card">                                                              6078
  <h2>微地质单元 · {{ patch.name }}</h2>                                        6079
  <p class="muted">                                                             6080
    按论文 4.1 节将矿山图斑划分为岩质边坡、碎石边坡、平台三类微地质单元。      6081
    面积占比由面积自动计算，即论文式（9） 中的权重系数。                       6082
  </p>                                                                          6083
  {% if units %}                                                                6084
  <table>                                                                       6085
    <caption class="muted">                                                     6086
      共 {{ units|length }} 个单元；                                            6087
      面积合计 <span class="num">{{ "%.4f"|format(total_area)                   6088
    }}</span>；                                                                 6089
      占比合计 <span class="num">{{ "%.2f%%"|format(ratio_sum *                 6090
    100) }}</span>                                                              6091
      {% if ratio_sum and (ratio_sum < 0.999 or ratio_sum >                     6092
    1.001) %}                                                                   6093
      <span class="warn">（占比合计不等于 100%，请检查面积录入）</span>         6094
      {% endif %}                                                               6095
    </caption>                                                                  6096
    <thead>                                                                     6097
      <tr>                                                                      6098
        <th scope="col">微地质单元</th>                                         6099
        <th scope="col">面积</th>                                               6100
```

```
          <th scope="col">面积占比</th>                                    6101
          <th scope="col">说明</th>                                        6102
          <th scope="col">操作</th>                                        6103
        </tr>                                                              6104
      </thead>                                                             6105
      <tbody>                                                              6106
        {% for u in units %}                                               6107
        <tr>                                                               6108
          <td>{{ u.unit_type }}</td>                                       6109
          <td class="num">{{ u.area if u.area is not none                  6110
      else "—" }}</td>                                                     6111
          <td class="num">                                                 6112
            {% if u.ratio is not none %}{{                                 6113
      "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}              6114
          </td>                                                            6115
          <td style="text-align:left;">{{ u.description or                 6116
      "—" }}</td>                                                          6117
          <td class="nowrap">                                              6118
            <span class="row-actions">                                     6119
              <a class="link-btn" href="/unit/edit/{{ u.id                 6120
      }}">编辑</a>                                                         6121
              <form method="post" action="/unit/delete/{{                  6122
      u.id }}"                                                             6123
                    onsubmit="return confirm('确定删除微地质单元「{{       6124
      u.unit_type }}」吗？\n该单元已录入的指标数据将一并删除，且无法恢复。');">  6125
                {{ form.hidden_tag() }}                                    6126
                <button type="submit" class="link-btn                      6127
      link-btn--danger">删除</button>                                      6128
              </form>                                                      6129
            </span>                                                        6130
          </td>                                                            6131
        </tr>                                                              6132
        {% endfor %}                                                       6133
      </tbody>                                                             6134
    </table>                                                               6135
    {% else %}                                                             6136
    <p class="muted">尚未划分微地质单元。请在下方添加。</p>                  6137
    {% endif %}                                                            6138
    <h3>添加微地质单元</h3>                                                6139
    <form method="post" action="/unit/{{ patch.id }}/add">                 6140
      {{ form.hidden_tag() }}                                              6141
      <div class="actions">                                                6142
        <label class="field__label" for="add-unit-type"                    6143
      style="margin:0;">类型</label>                                       6144
        <select id="add-unit-type" name="unit_type">                       6145
          {% for t in unit_types %}<option value="{{ t }}">{{              6146
      t }}</option>{% endfor %}                                            6147
        </select>                                                          6148
        <label class="field__label" for="add-unit-area"                    6149
      style="margin:0;">面积</label>                                       6150
```

```
        <input id="add-unit-area" type="number" step="any"
      name="area">
        <label class="field__label" for="add-unit-desc"
      style="margin:0;">说明</label>
        <input id="add-unit-desc" type="text"
      name="description" style="width:220px;">
        <button type="submit" class="btn btn--sm">添加</button>
      </div>
    </form>
    <p class="actions" style="margin-top:20px;">
      <a class="btn btn--sm btn--ghost" href="/project/{{
      patch.project_id }}/patches">返回图斑列表</a>
      <a class="btn btn--sm btn--ghost"
      href="/indicator/">去录入指标数据</a>
    </p>
</div>
{% endblock %}
# ==== app/templates/unit/edit.html ====
{% extends "base.html" %}
{% block title %}编辑微地质单元{% endblock %}
{% block content %}
<div class="card">
    <h2>编辑微地质单元 · {{ patch.name }}</h2>
    <p class="muted">修改面积后，系统会自动重算本图斑各单元的面积占比。</p>
    <form method="post">
      {{ form.hidden_tag() }}
      <div class="field">
        <label class="field__label" for="unit-type">微地质单元类型</l
      abel>
        <select id="unit-type" name="unit_type">
          {% for t in unit_types %}
          <option value="{{ t }}" {% if unit.unit_type == t
      %}selected{% endif %}>{{ t }}</option>
          {% endfor %}
        </select>
      </div>
      <div class="field">
        <label class="field__label" for="unit-area">面积</label>
        <input id="unit-area" type="number" step="any"
      name="area"
               value="{{ unit.area if unit.area is not none
      else '' }}">
      </div>
      <div class="field">
        <label class="field__label" for="unit-desc">说明</label>
        <input id="unit-desc" type="text" name="description"
      style="width:420px;"
               value="{{ unit.description or '' }}">
      </div>
      <p class="actions" style="margin-top:16px;">
```

```
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/unit/{{
    unit.patch_id }}">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/standard/index.html ====
{% extends "base.html" %}
{% block title %}政策与标准知识库{% endblock %}
{% block content %}
<div class="card">
  <h2>政策与标准知识库</h2>
  <p class="muted">
    收录论文表 1 梳理的地方标准与政策文件，并将其判定条件结构化为可执行规则，
    支持同一指标在不同省份的阈值横向比对。
  </p>
  <p class="actions" style="margin-bottom:14px;">
    <a class="btn btn--sm" href="/standard/add">添加标准</a>
    <a class="btn btn--sm btn--ghost"
    href="/standard/compare">省际口径比对</a>
  </p>
  <h3>地方标准</h3>
  {% if local %}
  <table>
    <thead>
      <tr>
        <th scope="col">省份</th>
        <th scope="col">名称</th>
        <th scope="col">编号</th>
        <th scope="col">规则数</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for s in local %}
      <tr>
        <td>{{ s.province }}</td>
        <td style="text-align:left;">{{ s.name }}</td>
        <td class="mono">{{ s.doc_number or "—" }}</td>
        <td class="num">{{ s.rules|length }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/standard/{{ s.id
    }}/rules">规则</a>
            <a class="link-btn" href="/standard/{{ s.id
    }}/edit">编辑</a>
            <form method="post" action="/standard/{{ s.id
    }}/delete"
                  onsubmit="return confirm('确定删除「{{
```

```
    s.province }} {{ s.name }}」吗？\n该标准下的 {{ s.rules|length }} 条规则将一并删除，且无法恢复。');">
                {{ form.hidden_tag() }}
                <button type="submit" class="link-btn link-btn--danger">删除</button>
            </form>
          </span>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">暂无地方标准。运行 <span class="mono">python seed_demo.py</span> 可载入论文表 1 的 11 项标准与政策文件。</p>
  {% endif %}
  <h3>政策文件</h3>
  {% if policy %}
  <table>
    <thead>
      <tr>
        <th scope="col">省份</th>
        <th scope="col">名称</th>
        <th scope="col">文号</th>
        <th scope="col">规则数</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for s in policy %}
      <tr>
        <td>{{ s.province }}</td>
        <td style="text-align:left;">{{ s.name }}</td>
        <td class="mono">{{ s.doc_number or "—" }}</td>
        <td class="num">{{ s.rules|length }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/standard/{{ s.id }}/rules">规则</a>
            <a class="link-btn" href="/standard/{{ s.id }}/edit">编辑</a>
            <form method="post" action="/standard/{{ s.id }}/delete"
                  onsubmit="return confirm('确定删除「{{ s.province }} {{ s.name }}」吗？\n该文件下的 {{ s.rules|length }} 条规则将一并删除，且无法恢复。');">
                {{ form.hidden_tag() }}
                <button type="submit" class="link-btn link-btn--danger">删除</button>
            </form>
```

```
            </span>                                                    6301
          </td>                                                        6302
        </tr>                                                          6303
        {% endfor %}                                                   6304
      </tbody>                                                         6305
    </table>                                                           6306
    {% else %}                                                         6307
    <p class="muted">暂无政策文件。</p>                                  6308
    {% endif %}                                                        6309
</div>                                                                 6310
{% endblock %}                                                         6311
# ==== app/templates/standard/edit.html ====                           6312
{% extends "base.html" %}                                              6313
{% block title %}{{ "编辑标准" if standard else "添加标准" }}{%         6314
    endblock %}                                                        6315
{% block content %}                                                    6316
<div class="card" style="max-width:800px;">                            6317
    <h2>{{ "编辑标准" if standard else "添加标准" }}</h2>                6318
    <form method="post">                                               6319
      {{ form.hidden_tag() }}                                          6320
      <div class="field">                                              6321
        <label class="field__label" for="province">省份</label>        6322
        <input id="province" type="text" name="province"               6323
      style="width:160px;" required                                    6324
               value="{{ standard.province if standard else            6325
      '' }}">                                                          6326
      </div>                                                           6327
      <div class="field">                                              6328
        <label class="field__label" for="std-name">名称</label>        6329
        <input id="std-name" type="text" name="name"                   6330
      style="width:100%;" required                                     6331
               value="{{ standard.name if standard else ''             6332
      }}">                                                             6333
      </div>                                                           6334
      <div class="field">                                              6335
        <label class="field__label" for="doc-number">标准编号 /        6336
      文号</label>                                                     6337
        <input id="doc-number" type="text" name="doc_number"           6338
      style="width:320px;"                                             6339
               value="{{ standard.doc_number or '' if                  6340
      standard else '' }}">                                            6341
      </div>                                                           6342
      <div class="field">                                              6343
        <label class="field__label" for="category">类别</label>        6344
        <select id="category" name="category">                         6345
          {% for c in ["地方标准", "政策文件"] %}                      6346
          <option value="{{ c }}" {% if standard and                   6347
      standard.category == c %}selected{% endif %}>{{ c                6348
      }}</option>                                                      6349
          {% endfor %}                                                 6350
```

```
        </select>
      </div>
      <div class="field">
        <label class="field__label" for="publish-date">发布时间</l
      abel>
        <input id="publish-date" type="text"
      name="publish_date" style="width:200px;"
               placeholder="如 2023-03-30"
               value="{{ standard.publish_date or '' if
      standard else '' }}">
      </div>
      <div class="field">
        <label class="field__label" for="implement-date">实施时间<
      /label>
        <input id="implement-date" type="text"
      name="implement_date" style="width:200px;"
               placeholder="如 2023-04-30（未明确可留空）"
               value="{{ standard.implement_date or '' if
      standard else '' }}">
      </div>
      <div class="field">
        <label class="field__label" for="summary">特点 /
      判定标准</label>
        <textarea id="summary" name="summary"
      style="min-height:96px;">{{ standard.summary or '' if
      standard else '' }}</textarea>
      </div>
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        {% if standard %}
        <form method="post" action="/standard/{{ standard.id
      }}/delete" style="display:inline;"
              onsubmit="return confirm('确定删除「{{
      standard.province }} {{ standard.name
      }}」吗？\n该标准下的规则将一并删除，且无法恢复。');">
          {{ form.hidden_tag() }}
          <button type="submit" class="btn
      btn--danger">删除该标准</button>
        </form>
        {% endif %}
        <a class="btn btn--ghost" href="/standard/">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/standard/rules.html ====
{% extends "base.html" %}
{% block title %}结构化规则 · {{ standard.province }}{% endblock
    %}
{% block content %}
```

```
<div class="card">
  <h2>结构化规则 · {{ standard.province }} {{ standard.name }}</h2>
  <p class="muted">
    将标准条文拆解为「判定项 + 比较符 + 阈值 + 条件类型」的可执行规则，
    使同一指标在各省的阈值差异可自动比对。
  </p>
  {% if rules %}
  <table>
    <caption class="muted">共 {{ rules|length }} 条规则</caption>
    <thead>
      <tr>
        <th scope="col">判定项</th>
        <th scope="col">条件</th>
        <th scope="col">条件类型</th>
        <th scope="col">适用分区</th>
        <th scope="col">备注</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rules %}
      <tr>
        <td style="text-align:left;">{{ r.item }}</td>
        <td class="mono nowrap">{{ r.operator }} {{ r.threshold or "" }}</td>
        <td>{{ r.condition_type }}</td>
        <td>{{ r.region or "—" }}</td>
        <td style="text-align:left;">{{ r.note or "—" }}</td>
        <td>
          <form method="post" action="/standard/rule/{{ r.id }}/delete"
                onsubmit="return confirm('确定删除规则「{{ r.item }} {{ r.operator }} {{ r.threshold or '' }}」吗？');">
            {{ form.hidden_tag() }}
            <button type="submit" class="btn btn--sm btn--danger">删除</button>
          </form>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">该标准暂无规则。请在下方添加第一条规则。</p>
  {% endif %}
  <h3>添加规则</h3>
  <form method="post">
```

```
      {{ form.hidden_tag() }}
      <div class="actions">
        <div>
          <label class="field__label" for="r-item">判定项</label>
          <input id="r-item" type="text" name="item"
      placeholder="如：植被覆盖度" required>
        </div>
        <div>
          <label class="field__label" for="r-op">比较符</label>
          <select id="r-op" name="operator">
            {% for o in [">=", ">", "<=", "<", "=="]
      %}<option value="{{ o }}">{{ o }}</option>{% endfor %}
          </select>
        </div>
        <div>
          <label class="field__label" for="r-thr">阈值</label>
          <input id="r-thr" type="text" name="threshold"
      placeholder="如：50" style="width:110px;">
        </div>
        <div>
          <label class="field__label" for="r-cond">条件类型</label
      >
          <select id="r-cond" name="condition_type">
            <option value="必要条件">必要条件</option>
            <option value="充分条件">充分条件</option>
          </select>
        </div>
        <div>
          <label class="field__label" for="r-region">适用分区</lab
      el>
          <input id="r-region" type="text" name="region"
      placeholder="如宁夏分区" style="width:130px;">
        </div>
        <div>
          <label class="field__label" for="r-note">备注</label>
          <input id="r-note" type="text" name="note"
      style="width:220px;">
        </div>
        <div>
          <button type="submit" class="btn">添加规则</button>
        </div>
      </div>
    </form>
    <p class="actions" style="margin-top:20px;">
      <a class="btn btn--sm btn--ghost"
      href="/standard/">返回标准列表</a>
      <a class="btn btn--sm btn--ghost"
      href="/standard/compare">省际口径比对</a>
    </p>
</div>
```

```
{% endblock %}
# ==== app/templates/standard/compare.html ====
{% extends "base.html" %}
{% block title %}省际口径比对{% endblock %}
{% block content %}
<div class="card">
  <h2>省际口径比对</h2>
  <p class="muted">
    同一判定项在不同省份的阈值横向对比，避免人工翻阅多份标准文件。
  </p>
  {% if not groups %}
  <p class="muted">
    暂无规则数据。请先在「标准知识库」中为标准录入规则，
    或运行 <span class="mono">python seed_demo.py</span> 载入论文表 1 数据。
  </p>
  {% else %}
  {% for item, rs in groups.items() %}
  <h3>{{ item }}</h3>
  <table>
    <thead>
      <tr>
        <th scope="col">省份</th>
        <th scope="col">标准 / 文件</th>
        <th scope="col">条件</th>
        <th scope="col">条件类型</th>
        <th scope="col">备注</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rs %}
      <tr>
        <td>{{ r.standard.province }}</td>
        <td style="text-align:left;">{{ r.standard.name
    }}</td>
        <td class="mono nowrap">{{ r.operator }} {{
    r.threshold or "" }}</td>
        <td>{{ r.condition_type }}</td>
        <td style="text-align:left;">{{ r.note or "—"
    }}</td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endfor %}
  {% endif %}
  <p class="actions" style="margin-top:16px;">
    <a class="btn btn--ghost" href="/standard/">返回标准列表</a>
  </p>
</div>
```

```
{% endblock %}
# ==== app/templates/constraint/index.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}限制性指标判定{% endblock %}
{% block content %}
<div class="card">
  <h2>限制性指标判定</h2>
  <p class="muted">
    按论文表 2 与 4.1 节流程，对每项限制性条件逐项判定。
    任一项不通过即判定为「人工辅助或生态重建」，不再进入定量评价。
  </p>
  {{ patch_guide(projects_without_patch) }}
  {% if not patches %}
  <p class="muted">暂无矿山图斑。请先在「项目与矿山档案」中建立图斑。</p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">所属项目</th>
        <th scope="col">矿山图斑</th>
        <th scope="col">判定结果</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in patches %}
      {% set r = latest.get(p.id) %}
      <tr>
        <td>{{ p.project.name if p.project else "—" }}</td>
        <td style="text-align:left;">{{ p.name }}</td>
        <td style="text-align:left;">
          {% if r %}
            {% if r.passed %}
              <span class="tag tag--ok">通过</span>
            {% else %}
              <span class="tag tag--warn">不通过</span>
            {% endif %}
            <span class="muted">{{ r.reason }}</span>
          {% else %}
            <span class="tag tag--muted">未判定</span>
          {% endif %}
        </td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/constraint/{{ p.id
    }}">执行判定</a>
          </span>
        </td>
      </tr>
```

```
        {% endfor %}                                                    6601
      </tbody>                                                          6602
    </table>                                                            6603
    {% endif %}                                                         6604
</div>                                                                  6605
{% endblock %}                                                          6606
# ==== app/templates/constraint/check.html ====                         6607
{% extends "base.html" %}                                               6608
{% block title %}限制性条件判定 · {{ patch.name }}{% endblock %}        6609
{% block content %}                                                     6610
<div class="card" style="max-width:760px;">                             6611
  <h2>限制性条件判定 · {{ patch.name }}</h2>                            6612
  <p class="muted">                                                     6613
    五项条件全部通过方可进入定量评价；任一项不通过即判定为「人工辅助或生态重建」，  6614
    不再进行后续计算。                                                  6615
  </p>                                                                  6616
  <form method="post">                                                  6617
    {{ form.hidden_tag() }}                                             6618
    {% for field, name, val in field_states %}                          6619
    <fieldset style="border:1px solid                                   6620
    var(--line);border-radius:6px;padding:10px                          6621
    14px;margin:0 0 10px;">                                             6622
      <legend style="padding:0 6px;font-weight:600;">{{                 6623
    name }}</legend>                                                    6624
      <div class="actions">                                             6625
        <label class="checkbox">                                        6626
          <input type="radio" name="{{ field }}" value="1"              6627
                 {% if val is none or val %}checked{% endif             6628
    %}>                                                                 6629
          <span>通过</span>                                             6630
        </label>                                                        6631
        <label class="checkbox">                                        6632
          <input type="radio" name="{{ field }}" value="0"              6633
                 {% if val is not none and not val                      6634
    %}checked{% endif %}>                                               6635
          <span>不通过</span>                                           6636
        </label>                                                        6637
      </div>                                                            6638
    </fieldset>                                                         6639
    {% endfor %}                                                        6640
    <p class="actions" style="margin-top:16px;">                        6641
      <button type="submit" class="btn">保存判定</button>               6642
      <a class="btn btn--ghost" href="/constraint/">返回</a>            6643
    </p>                                                                6644
  </form>                                                               6645
  {% if record %}                                                       6646
  <div style="margin-top:18px;padding:14px;border-radius:6px            6647
    ;border-left:4px solid;                                             6648
                {% if record.passed %}background:var(--ok-soft          6649
    );border-left-color:var(--ok);                                      6650
```

```
                {% else %}background:var(--danger-soft);border
    -left-color:var(--danger);{% endif %}">
    <p style="margin:0 0 4px;font-weight:600;">
      判定结论：
      {% if record.passed %}
      <span class="ok">通过，可进入定量评价</span>
      {% else %}
      <span class="warn">不通过 —— 建议生态重建或人工辅助</span>
      {% endif %}
    </p>
    <p class="muted" style="margin:0;">{{ record.reason
    }}</p>
  </div>
  {% endif %}
</div>
{% endblock %}
# ==== app/templates/ahp/index.html ====
{% extends "base.html" %}
{% block title %}AHP 权重计算{% endblock %}
{% block content %}
<div class="card" style="max-width:720px;">
  <h2>AHP 权重计算 · 选择指标层级</h2>
  <p class="muted">
    选择微地质单元类型与指标层级后，系统按论文表 3 确定判断矩阵的规模。
    岩质边坡不设土壤指标、平台不设地形指标，系统会相应调整矩阵所包含的指标。
  </p>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="{{ form.unit_type.id
    }}">微地质单元类型</label>
      {{ form.unit_type(id=form.unit_type.id) }}
    </div>
    <div class="field">
      <label class="field__label" for="{{ form.level.id
    }}">指标层级</label>
      {{ form.level(id=form.level.id) }}
    </div>
    <div class="field">
      <label class="field__label" for="{{ form.l1.id }}">
        一级指标（层级选「二级指标」时需选择）
      </label>
      {{ form.l1(id=form.l1.id) }}
    </div>
    <p class="actions" style="margin-top:16px;">
      {{ form.submit(class_="btn") }}
      <a class="btn btn--ghost" href="/ahp/indicators">指标定义维
    护</a>
      <a class="btn btn--ghost" href="/ahp/experts">多专家矩阵</a
    >
```

```
      </p>
    </form>
</div>
<div class="card">
    <h2>各微地质单元当前生效权重</h2>
    <p class="muted">
      适宜性评价按此处列出的权重计算。默认采用论文表 3 的取值；
      若在本系统中用层次分析法算出了新权重并保存，则改按新权重计算。
    </p>
    <table>
      <thead>
        <tr>
          <th scope="col">微地质单元</th>
          <th scope="col">当前权重来源</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for r in weight_rows %}
        <tr>
          <td style="text-align:left;">{{ r.unit_type }}</td>
          <td style="text-align:left;">
            {% if r.custom %}
            <span class="tag tag--warn">{{ r.label }}</span>
            {% else %}
            <span class="tag tag--muted">{{ r.label }}</span>
            {% endif %}
          </td>
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/ahp/matrix?unit_type=
{{ r.unit_type }}&level=1">一级指标权重</a>
              <a class="link-btn" href="/ahp/matrix?unit_type=
{{ r.unit_type }}&level=1&calc=1">查看一致性</a>
              {% if r.custom %}
              <form method="post" action="/ahp/weights/reset"
                    onsubmit="return confirm('确定把「{{
r.unit_type }}」恢复为论文表 3 的默认权重吗？');">
                <input type="hidden" name="csrf_token"
value="{{ csrf_token() }}">
                <input type="hidden" name="unit_type"
value="{{ r.unit_type }}">
                <button type="submit" class="link-btn
link-btn--danger">恢复默认</button>
              </form>
              {% endif %}
            </span>
          </td>
        </tr>
        {% endfor %}
```

```
      </tbody>
    </table>
</div>
{% endblock %}
# ==== app/templates/ahp/matrix.html ====
{% extends "base.html" %}
{% block title %}判断矩阵录入{% endblock %}
{% block content %}
<div class="card">
    <h2>
      判断矩阵录入
      · {{ unit_type }}
      · {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{%
      endif %}
    </h2>
    <p class="muted">
      采用 1–9 标度：1 同等重要、3 稍微重要、5 明显重要、7 强烈重要、9 极端重要，
      2、4、6、8 为相邻判断的中间值。只需录入上三角，对角线为 1，下三角自动取倒数。
      矩阵已按论文表 3 的权重预填。
    </p>
    <form method="post">
      {{ form.hidden_tag() }}
      <table style="width:auto;">
        <thead>
          <tr>
            <th></th>
            {% for name in names %}<th scope="col">{{ name
      }}</th>{% endfor %}
          </tr>
        </thead>
        <tbody>
          {% for i in range(names|length) %}
          <tr>
            <th scope="row">{{ names[i] }}</th>
            {% for j in range(names|length) %}
              {% if i == j %}
              <td class="muted num">1</td>
              {% elif i < j %}
              <td>
                <label class="muted" style="position:absolute;
      left:-9999px;"
                       for="m-{{ i }}-{{ j }}">{{ names[i] }}
      相对 {{ names[j] }} 的重要性</label>
                <input id="m-{{ i }}-{{ j }}" type="number"
      step="0.01" min="0.01"
                       name="m_{{ i }}_{{ j }}" value="{{
      default[i][j] }}">
              </td>
              {% else %}
              <td class="muted num">{{ default[i][j] }}</td>
```

```
                {% endif %}                                                    6801
              {% endfor %}                                                     6802
            </tr>                                                              6803
            {% endfor %}                                                       6804
          </tbody>                                                             6805
        </table>                                                               6806
        <p class="actions">                                                    6807
          <button type="submit" class="btn" name="action"                      6808
        value="calc">计算权重</button>                                         6809
          {% if result and result.consistent %}                                6810
          <button type="submit" class="btn" name="action"                      6811
        value="save">保存并生效</button>                                       6812
          {% endif %}                                                          6813
          <a class="btn btn--ghost" href="/ahp/">重新选择</a>                  6814
        </p>                                                                   6815
      </form>                                                                  6816
      <p class="muted" style="margin-top:10px;">                               6817
        当前「{{ unit_type }}」生效权重：<strong>{{ source_label               6818
        }}</strong>。                                                          6819
        点击「保存并生效」后，本单元类型的适宜性评价将采用此处算出的权重；   6820
        如需回到论文取值，可在 <a href="/ahp/">权重首页</a> 恢复默认。        6821
      </p>                                                                     6822
      {% if result %}                                                          6823
      <hr style="margin:22px 0;border:none;border-top:1px solid                6824
        var(--line);">                                                         6825
      <h3>计算结果</h3>                                                        6826
      <table style="width:auto;">                                              6827
        <thead>                                                                6828
          <tr>                                                                 6829
            <th scope="col">指标</th>                                          6830
            {% for name in names %}<th scope="col">{{ name                     6831
        }}</th>{% endfor %}                                                    6832
          </tr>                                                                6833
        </thead>                                                               6834
        <tbody>                                                                6835
          <tr>                                                                 6836
            <th scope="row">权重</th>                                          6837
            {% for w in result.weights %}<td class="mono">{{ w                 6838
        }}</td>{% endfor %}                                                    6839
          </tr>                                                                6840
        </tbody>                                                               6841
      </table>                                                                 6842
      <div class="stat-row">                                                   6843
        <div>                                                                  6844
          <div class="stat__label">最大特征值 λmax</div>                       6845
          <div class="stat__value mono">{{ result.lambda_max                   6846
        }}</div>                                                               6847
        </div>                                                                 6848
        <div>                                                                  6849
          <div class="stat__label">一致性指标 CI</div>                         6850
```

```
        <div class="stat__value mono">{{                                    6851
      "%.4f"|format(result.ci) }}</div>                                     6852
      </div>                                                                6853
      <div>                                                                 6854
        <div class="stat__label">一致性比例 CR</div>                        6855
        <div class="stat__value mono">{{                                    6856
      "%.4f"|format(result.cr) }}</div>                                     6857
      </div>                                                                6858
      <div>                                                                 6859
        <div class="stat__label">一致性检验（要求 CR &lt; 0.1）</div>       6860
        <div class="stat__value">                                           6861
          {% if result.consistent %}<span class="ok">通过</span>            6862
          {% else %}<span class="warn">不通过</span>{% endif %}             6863
        </div>                                                              6864
      </div>                                                                6865
    </div>                                                                  6866
    {% if not result.consistent and result.most_inconsistent               6867
      %}                                                                    6868
    <p style="margin-top:14px;padding:12px;border-radius:6px;b              6869
      ackground:var(--danger-soft);                                         6870
                border-left:4px solid var(--danger);">                      6871
      <strong class="warn">建议优先调整：</strong>                          6872
      第 {{ result.most_inconsistent.i + 1 }} 行、第 {{                     6873
      result.most_inconsistent.j + 1 }} 列的比较                            6874
      （当前 <span class="mono num">{{ result.most_inconsistent.v           6875
      alue|round(3) }}</span>，                                             6876
      理想约 <span class="mono num">{{ result.most_inconsistent.e           6877
      xpected|round(3) }}</span>）                                          6878
    </p>                                                                    6879
    {% endif %}                                                             6880
  {% endif %}                                                               6881
</div>                                                                      6882
{% endblock %}                                                              6883
# ==== app/templates/ahp/indicators.html ====                               6884
{% extends "base.html" %}                                                   6885
{% block title %}指标定义维护{% endblock %}                                 6886
{% block content %}                                                         6887
<div class="card">                                                          6888
  <h2>指标体系维护</h2>                                                    6889
  <p class="muted">                                                         6890
    按论文表 2 与表 3 建立的指标树。此处可核对指标名称、单位、标准化类型、 6891
    理想取值与取值依据，并按需调整启用该指标的微地质单元。                 6892
    计算时的权威取值仍以 <span class="mono">app/utils/constants.py</sp     6893
    an> 为准，                                                              6894
    本页用于展示与核对。                                                    6895
  </p>                                                                      6896
  <p class="actions" style="margin-bottom:14px;">                           6897
    <a class="btn btn--sm btn--ghost"                                       6898
    href="/ahp/">返回权重计算</a>                                           6899
    <a class="btn btn--sm btn--ghost"                                       6900
```

```
    href="/ahp/experts">多专家矩阵</a>
    <form method="post" action="/ahp/indicators/reset"
    style="display:inline;"
          onsubmit="return confirm('将按论文取值重置全部指标定义，自定义修改会丢失。
    确定继续？');">
      {{ form.hidden_tag() }}
      <button type="submit" class="btn btn--sm
    btn--danger">按论文取值重置</button>
    </form>
  </p>
  {% for level, title in [(1, "一级指标"), (2, "二级指标"), (3,
    "三级指标")] %}
  {% set level_items = all_ind|selectattr("level",
    "equalto", level)|list %}
  {% if level_items %}
  <h3>{{ title }}（{{ level_items|length }} 项）</h3>
  <table>
    <thead>
      <tr>
        <th scope="col">指标</th>
        <th scope="col">上级指标</th>
        <th scope="col">单位</th>
        <th scope="col">标准化类型</th>
        <th scope="col">理想取值</th>
        <th scope="col">启用单元</th>
        <th scope="col">限制性</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for i in level_items %}
      <tr>
        <td style="text-align:left;">
          <span class="mono muted">{{ i.code
    }}</span><br>{{ i.name }}
        </td>
        <td>{{ (all_ind|selectattr("code", "equalto",
    i.parent_code)|list|first).name if i.parent_code else
    "—" }}</td>
        <td>{{ i.unit or "—" }}</td>
        <td>{{ type_label(i.ideal_type) }}</td>
        <td class="mono">{{ i.ideal_value or "—" }}</td>
        <td style="text-align:left;">{{ i.enabled_units or
    "—" }}</td>
        <td>
          {% if i.is_constraint %}<span class="tag
    tag--warn">是</span>
          {% else %}<span class="tag tag--muted">否</span>{%
    endif %}
        </td>
```

```
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/ahp/indicators/{{ i.code }}/edit">编辑</a>
            </span>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    {% endif %}
    {% endfor %}
</div>
{% endblock %}
# ==== app/templates/ahp/indicator_edit.html ====
{% extends "base.html" %}
{% block title %}编辑指标 · {{ ind.name }}{% endblock %}
{% block content %}
<div class="card" style="max-width:680px;">
    <h2>编辑指标 · {{ ind.name }}</h2>
    <p class="muted">
      指标编码 <span class="mono">{{ ind.code }}</span>、层级 {{ ind.level }} 级，
      由论文指标体系确定，不可修改。
    </p>
    <form method="post">
      {{ form.hidden_tag() }}
      <div class="field">
        <label class="field__label" for="name">指标名称</label>
        <input id="name" type="text" name="name"
               style="width:100%;" required
               value="{{ ind.name }}">
      </div>
      <div class="field">
        <label class="field__label" for="unit">计量单位</label>
        <input id="unit" type="text" name="unit"
               style="width:180px;"
               value="{{ ind.unit or '' }}">
      </div>
      <div class="field">
        <label class="field__label" for="ideal-type">标准化类型</label>
        <select id="ideal-type" name="ideal_type">
          {% for code, label in [("benefit","越大越优"),("cost","越小越优"),("mid","适中型"),("discrete","离散赋值")] %}
          <option value="{{ code }}" {% if ind.ideal_type == code %}selected{% endif %}>{{ label }}</option>
          {% endfor %}
        </select>
      </div>
```

```
      <div class="field">                                                    7001
        <label class="field__label" for="ideal-value">理想取值</label>      7002
      <label>                                                                7003
        <input id="ideal-value" type="text"                                  7004
      name="ideal_value" style="width:100%;"                                 7005
               value="{{ ind.ideal_value or '' }}">                         7006
        <p class="muted" style="margin:4px 0 0;">                            7007
          如土石比 0.8、土层厚度 20、pH 5~9、植被覆盖度 75。                 7008
        </p>                                                                 7009
      </div>                                                                 7010
      <fieldset style="border:1px solid                                      7011
      var(--line);border-radius:6px;padding:10px                             7012
      14px;margin:0 0 12px;">                                                7013
        <legend style="padding:0 6px;font-weight:600;">启用该指标的微       7014
      地质单元</legend>                                                      7015
        {% for t in unit_types %}                                            7016
        <label class="checkbox" style="margin-right:18px;">                  7017
          <input type="checkbox" name="enabled_units"                        7018
      value="{{ t }}"                                                        7019
                 {% if t in ind.units_list() %}checked{%                     7020
      endif %}>                                                              7021
          <span>{{ t }}</span>                                               7022
        </label>                                                             7023
        {% endfor %}                                                         7024
        <p class="muted" style="margin:8px 0 0;">                            7025
          论文设定：岩质边坡不启用土壤指标，平台不启用地形指标。            7026
        </p>                                                                 7027
      </fieldset>                                                            7028
      <div class="field">                                                    7029
        <label class="checkbox" for="is-constraint">                         7030
          <input id="is-constraint" type="checkbox"                          7031
      name="is_constraint" value="1"                                         7032
                 {% if ind.is_constraint %}checked{% endif %}>               7033
          <span>同时作为限制性指标（参与一票否决判定）</span>                7034
        </label>                                                             7035
      </div>                                                                 7036
      <div class="field">                                                    7037
        <label class="field__label" for="description">取值依据说明</       7038
      label>                                                                 7039
        <textarea id="description" name="description"                        7040
      style="min-height:88px;">{{ ind.description or ''                      7041
      }}</textarea>                                                          7042
      </div>                                                                 7043
      <p class="actions" style="margin-top:16px;">                           7044
        <button type="submit" class="btn">保存</button>                      7045
        <a class="btn btn--ghost" href="/ahp/indicators">返回指标列          7046
      表</a>                                                                 7047
      </p>                                                                   7048
    </form>                                                                  7049
</div>                                                                       7050
```

```
{% endblock %}
# ==== app/templates/ahp/experts.html ====
{% extends "base.html" %}
{% block title %}多专家判断矩阵{% endblock %}
{% block content %}
<div class="card">
  <h2>多专家判断矩阵 · 录入与汇总</h2>
  <p class="muted">
    论文明确权重由「组织 10 位专家对各项指标权重进行打分」确定。
    此页支持多位专家分别录入判断矩阵，再用几何平均汇总，
    保持矩阵互反性，并重新做一致性检验。
  </p>
  <form method="get" class="actions" style="margin-bottom:14px;">
    <div>
      <label class="field__label" for="unit-type">微地质单元类型</label>
      <select id="unit-type" name="unit_type"
              onchange="this.form.submit()">
        {% for t in ["岩质边坡", "碎石边坡", "平台"] %}
          <option value="{{ t }}" {% if unit_type == t %}selected{% endif %}>{{ t }}</option>
        {% endfor %}
      </select>
    </div>
    <div>
      <label class="field__label" for="level">指标层级</label>
      <select id="level" name="level"
              onchange="this.form.submit()">
        <option value="1" {% if level == 1 %}selected{% endif %}>一级指标</option>
        <option value="2" {% if level == 2 %}selected{% endif %}>二级指标</option>
      </select>
    </div>
  </form>
  <p class="muted">
    当前范围：<strong>{{ unit_type }}</strong> ·
    {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif %}。
    若选二级指标，请先到
    <a href="/ahp/">权重计算</a> 选择一级指标后进入本页。
  </p>
  {% if rows %}
  <table>
    <caption class="muted">已录入 {{ rows|length }}
    位专家的矩阵</caption>
    <thead>
      <tr>
        <th scope="col">专家</th>
```

```
          <th scope="col">矩阵规模</th>
          <th scope="col">该矩阵 CR</th>
          <th scope="col">一致性</th>
          <th scope="col">录入时间</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for r in rows %}
        <tr>
          <td>{{ r.expert_name }}</td>
          <td class="num">{{ n }} × {{ n }}</td>
          <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is
      not none else "—" }}</td>
          <td>
            {% if r.cr is not none and r.cr < 0.1 %}<span
      class="tag tag--ok">通过</span>
            {% else %}<span class="tag tag--warn">不通过</span>{%
      endif %}
          </td>
          <td class="mono nowrap">{{ r.created_at.strftime("%Y
      -%m-%d %H:%M") if r.created_at else "—" }}</td>
          <td>
            <form method="post" action="/ahp/experts/{{ r.id
      }}/delete"
                  onsubmit="return confirm('确定删除专家「{{
      r.expert_name }}」的矩阵吗？');">
              {{ form.hidden_tag() }}
              <button type="submit" class="btn btn--sm
      btn--danger">删除</button>
            </form>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    <p class="actions" style="margin-bottom:10px;">
      <a class="btn" href="/ahp/experts/aggregate?unit_type={{
      unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{%
      endif %}">
        汇总计算（几何平均）
      </a>
    </p>
    {% else %}
    <p class="muted">尚未录入任何专家矩阵。请在下方录入第一位专家的判断矩阵。</p>
    {% endif %}

    <h3>录入一位专家的判断矩阵</h3>
    <form method="post">
      {{ form.hidden_tag() }}
      <input type="hidden" name="action" value="add">
```

```
<input type="hidden" name="unit_type" value="{{ unit_type }}">
<input type="hidden" name="level" value="{{ level }}">
{% if l1 %}<input type="hidden" name="l1" value="{{ l1 }}">{% endif %}
<div class="field">
  <label class="field__label" for="expert-name">专家姓名</label>
  <input id="expert-name" type="text" name="expert_name"
         style="width:220px;" placeholder="如：张工">
</div>
<table style="width:auto;">
  <thead>
    <tr>
      <th></th>
      {% for name in names %}<th scope="col">{{ name }}</th>{% endfor %}
    </tr>
  </thead>
  <tbody>
    {% for i in range(n) %}
    <tr>
      <th scope="row">{{ names[i] }}</th>
      {% for j in range(n) %}
        {% if i == j %}
        <td class="muted num">1</td>
        {% elif i < j %}
        <td>
          <label class="muted" style="position:absolute;left:-9999px;"
                 for="e-{{ i }}-{{ j }}">{{ names[i] }} 相对 {{ names[j] }} 的重要性</label>
          <input id="e-{{ i }}-{{ j }}" type="number" step="0.01" min="0.01"
                 name="m_{{ i }}_{{ j }}" value="{{ default[i][j] }}">
        </td>
        {% else %}
        <td class="muted num">{{ default[i][j] }}</td>
        {% endif %}
      {% endfor %}
    </tr>
    {% endfor %}
  </tbody>
</table>
<p class="actions" style="margin-top:12px;">
  <button type="submit" class="btn">录入该专家矩阵</button>
  <a class="btn btn--ghost" href="/ahp/indicators">指标维护</a>
  <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>
```

```
      </p>                                                                    7201
    </form>                                                                   7202
</div>                                                                        7203
{% endblock %}                                                                7204
#  ==== app/templates/ahp/experts_aggregate.html ====                         7205
{% extends "base.html" %}                                                     7206
{% block title %}多专家矩阵汇总{% endblock %}                                  7207
{% block content %}                                                           7208
<div class="card">                                                            7209
  <h2>多专家矩阵汇总 · {{ unit_type }} · {% if level == 1 %}一级指标{%       7210
    else %}二级指标（{{ l1 }}）{% endif %}</h2>                               7211
  <p class="muted">                                                           7212
    对各位专家的判断矩阵按对应元素取几何平均：                                7213
    a<sub>ij</sub> = (a<sub>ij</sub><sup>(1)</sup> × … ×                      7214
    a<sub>ij</sub><sup>(k)</sup>)<sup>1/k</sup>。                             7215
    该方式可保持矩阵的互反性，汇总后重新计算权重与一致性比例。                7216
  </p>                                                                        7217
  {% if not result %}                                                         7218
  <p class="muted">                                                           7219
    该范围内尚未录入专家矩阵。                                                7220
    <a href="/ahp/experts?unit_type={{ unit_type                              7221
    }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif                       7222
    %}">去录入</a>                                                            7223
  </p>                                                                        7224
  {% else %}                                                                  7225
  <div class="stat-row">                                                      7226
    <div>                                                                     7227
      <div class="stat__label">参与汇总的专家</div>                           7228
      <div class="stat__value">{{ result.count }} 位</div>                    7229
    </div>                                                                    7230
    <div>                                                                     7231
      <div class="stat__label">λmax</div>                                     7232
      <div class="stat__value mono">{{ result.lambda_max                      7233
    }}</div>                                                                  7234
    </div>                                                                    7235
    <div>                                                                     7236
      <div class="stat__label">CI</div>                                       7237
      <div class="stat__value mono">{{                                        7238
    "%.4f"|format(result.ci) }}</div>                                         7239
    </div>                                                                    7240
    <div>                                                                     7241
      <div class="stat__label">CR（要求 &lt; 0.1）</div>                      7242
      <div class="stat__value mono">{{                                        7243
    "%.4f"|format(result.cr) }}</div>                                         7244
    </div>                                                                    7245
    <div>                                                                     7246
      <div class="stat__label">一致性检验</div>                               7247
      <div class="stat__value">                                               7248
        {% if result.consistent %}<span class="ok">通过</span>                7249
        {% else %}<span class="warn">不通过</span>{% endif %}                 7250
```

```
      </div>                                                          7251
    </div>                                                            7252
  </div>                                                              7253
  <h3>汇总后权重</h3>                                                 7254
  <table style="width:auto;">                                         7255
    <thead>                                                           7256
      <tr>                                                            7257
        <th scope="col">指标</th>                                     7258
        {% for name in names %}<th scope="col">{{ name                7259
    }}</th>{% endfor %}                                               7260
      </tr>                                                           7261
    </thead>                                                          7262
    <tbody>                                                           7263
      <tr>                                                            7264
        <th scope="row">权重</th>                                     7265
        {% for w in result.weights %}<td class="mono">{{ w            7266
    }}</td>{% endfor %}                                               7267
      </tr>                                                           7268
    </tbody>                                                          7269
  </table>                                                            7270
  <h3>各专家矩阵的一致性</h3>                                         7271
  <table style="width:auto;">                                         7272
    <thead>                                                           7273
      <tr><th scope="col">专家</th><th scope="col">该矩阵             7274
    CR</th><th scope="col">判定</th></tr>                             7275
    </thead>                                                          7276
    <tbody>                                                           7277
      {% for r in rows %}                                             7278
      <tr>                                                            7279
        <td>{{ r.expert_name }}</td>                                  7280
        <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is            7281
    not none else "—" }}</td>                                         7282
        <td>                                                          7283
          {% if r.cr is not none and r.cr < 0.1 %}<span               7284
    class="ok">通过</span>                                            7285
          {% else %}<span class="warn">不通过</span>{% endif %}       7286
        </td>                                                         7287
      </tr>                                                           7288
      {% endfor %}                                                    7289
    </tbody>                                                          7290
  </table>                                                            7291
  {% if not result.consistent and result.most_inconsistent            7292
    %}                                                                7293
  <p style="padding:12px;border-radius:6px;background:var(--          7294
    danger-soft);                                                     7295
             border-left:4px solid var(--danger);">                   7296
    <strong class="warn">汇总矩阵一致性不通过，建议优先调整：</strong>  7297
    第 {{ result.most_inconsistent.i + 1 }} 行、第 {{                 7298
    result.most_inconsistent.j + 1 }} 列的比较                        7299
    （当前 <span class="mono">{{ result.most_inconsistent.value       7300
```

```
      |round(3) }}</span>,                                                    7301
      理想约 <span class="mono">{{ result.most_inconsistent.expec             7302
      ted|round(3) }}</span>)。                                              7303
      该偏差通常来自个别专家打分，可先核对一致性较差的专家矩阵。                7304
    </p>                                                                     7305
    {% endif %}                                                              7306
    {% endif %}                                                              7307
    <p class="actions" style="margin-top:16px;">                             7308
      <form method="post"                                                    7309
            action="/ahp/experts?unit_type={{ unit_type                      7310
      }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}"                7311
            style="display:inline;"                                          7312
            onsubmit="return confirm('保存后，「{{ unit_type                 7313
      }}」的适宜性评价将改用这组汇总权重计算。确定继续吗？');">               7314
        {{ form.hidden_tag() }}                                              7315
        <input type="hidden" name="action" value="save_agg">                 7316
        <button type="submit" class="btn" {% if not                          7317
      result.consistent %}disabled{% endif %}>                               7318
          保存为生效权重                                                     7319
        </button>                                                            7320
      </form>                                                                7321
      <a class="btn btn--ghost" href="/ahp/experts?unit_type={               7322
      { unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1                   7323
      }}{% endif %}">返回录入</a>                                            7324
      <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>                 7325
    </p>                                                                     7326
    {% if not result.consistent %}                                           7327
    <p class="muted" style="margin-top:8px;">                                7328
      汇总矩阵一致性未通过，暂不能保存为生效权重。                           7329
    </p>                                                                     7330
    {% endif %}                                                              7331
</div>                                                                       7332
{% endblock %}                                                               7333
# ==== app/templates/indicator/index.html ====                               7334
{% extends "base.html" %}                                                    7335
{% from "_macros.html" import patch_guide %}                                 7336
{% block title %}指标数据采集{% endblock %}                                  7337
{% block content %}                                                          7338
<div class="card">                                                           7339
    <h2>指标数据采集与管理</h2>                                              7340
    <p class="muted">                                                        7341
      按微地质单元组织录入。系统按单元类型自动决定应录入的指标：             7342
      岩质边坡不启用土壤指标、平台不启用地形指标。                           7343
    </p>                                                                     7344
    <p class="actions" style="margin-bottom:12px;">                          7345
      <a class="btn btn--sm" href="/indicator/import">批量导入（CSV          7346
      / Excel）</a>                                                          7347
    </p>                                                                     7348
    {{ patch_guide(projects_without_patch) }}                                7349
    {% if not patches %}                                                     7350
```

```
<p class="muted">
  暂无矿山图斑。请先在「项目与矿山档案」中建立图斑，
  或运行 <span class="mono">python seed_demo.py</span>
  载入论文案例。
</p>
{% else %}
{% for p in patches %}
<div class="patch-panel">
  <div class="patch-panel__head">
    <h3 class="patch-panel__name">{{ p.name }}</h3>
    <span class="patch-panel__meta">
      所属项目：{{ p.project.name if p.project else "—" }}
    </span>
    <span class="patch-panel__meta">
      微地质单元：{{ p.units.count() }} 个
    </span>
    {% if p.units.count() %}
    <span class="patch-panel__meta patch-panel__tail">
      <a class="link-btn" href="/indicator/import">批量导入本图斑
  数据</a>
    </span>
    {% endif %}
  </div>
  {% if not p.units.count() %}
  <div class="notice notice--warn">
    <p class="notice__title">该图斑尚未划分微地质单元，暂不能录入指标数据。</p>
    <p style="margin:0;">
      指标数据是按微地质单元（岩质边坡 / 碎石边坡 / 平台）分别组织的，
      请先完成划分，再回到本页录入。
    </p>
    <p style="margin:10px 0 0;">
      <a class="btn btn--sm" href="/unit/{{ p.id
  }}">去划分微地质单元</a>
    </p>
  </div>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">面积占比</th>
        <th scope="col">数据状态</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for u in p.units %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">
```

```
                {% if u.ratio is not none %}{{                          7401
    "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}             7402
            </td>                                                       7403
            <td>                                                        7404
                {% if u.indicator_data %}                               7405
                <span class="tag tag--ok">已录入</span>                  7406
                {% else %}                                              7407
                <span class="tag tag--muted">未录入</span>               7408
                {% endif %}                                             7409
            </td>                                                       7410
            <td class="nowrap">                                         7411
                <span class="row-actions">                              7412
                  <a class="link-btn" href="/indicator/{{ u.id          7413
    }}">录入 / 编辑</a>                                                 7414
                </span>                                                 7415
            </td>                                                       7416
          </tr>                                                         7417
          {% endfor %}                                                  7418
        </tbody>                                                        7419
      </table>                                                          7420
      {% endif %}                                                       7421
    </div>                                                              7422
    {% endfor %}                                                        7423
    {% endif %}                                                         7424
</div>                                                                  7425
{% endblock %}                                                          7426
# ==== app/templates/indicator/edit.html ====                           7427
{% extends "base.html" %}                                               7428
{% block title %}指标数据录入 · {{ unit.unit_type }}{% endblock %}      7429
{% block content %}                                                     7430
<div class="card" style="max-width:820px;">                             7431
  <h2>指标数据录入 · {{ unit.unit_type }}</h2>                          7432
  <p class="muted">                                                     7433
    输入框留空表示该指标「不适用」，与「实测值为 0」含义不同，计算时会被跳过。   7434
    岩质边坡不设土壤指标、平台不设地形指标，系统已按单元类型隐藏不相关字段。     7435
  </p>                                                                  7436
  <form method="post">                                                  7437
    {{ form.hidden_tag() }}                                             7438
    {% for group_name, fields, has_hm in groups %}                      7439
    <h3>{{ group_name }}</h3>                                           7440
    {% for fname, cname, unit_ in fields %}                             7441
    <div class="field">                                                 7442
      <label class="field__label" for="f-{{ fname }}">                  7443
        {{ cname }}{% if unit_ %} ({{ unit_ }}) {% endif %}             7444
      </label>                                                          7445
      {% if fname == "aspect" %}                                        7446
      <select id="f-{{ fname }}" name="aspect">                         7447
        <option value="">— 不适用 —</option>                            7448
        {% for a in aspect_types %}                                     7449
        <option value="{{ a }}" {% if current.get("aspect")             7450
```

```
      == a %}selected{% endif %}>{{ a }}</option>                    7451
        {% endfor %}                                                 7452
      </select>                                                      7453
      {% else %}                                                     7454
      <input id="f-{{ fname }}" type="number" step="any"             7455
    name="{{ fname }}"                                               7456
             value="{{ current.get(fname) if                         7457
    current.get(fname) is not none else '' }}">                      7458
      {% endif %}                                                    7459
    </div>                                                           7460
    {% endfor %}                                                     7461
    {% endfor %}                                                     7462
    {% for group_name, fields, has_hm in groups %}                   7463
      {% if has_hm %}                                                7464
      <h3>重金属含量（mg/kg）</h3>                                    7465
      <p class="muted">未检测的金属可留空，系统按论文注释视为未受污染处理。</p>   7466
      <table style="width:auto;">                                    7467
        <thead>                                                      7468
          <tr>                                                       7469
            {% for m in heavy_metals %}<th scope="col">{{ m          7470
    }}</th>{% endfor %}                                              7471
          </tr>                                                      7472
        </thead>                                                     7473
        <tbody>                                                      7474
          <tr>                                                       7475
            {% for m in heavy_metals %}                              7476
            <td>                                                     7477
              <label class="muted" style="position:absolute;         7478
    left:-9999px;"                                                   7479
                     for="hm-{{ m }}">{{ m }} 含量</label>           7480
              <input id="hm-{{ m }}" type="number"                   7481
    step="any" name="hm_{{ m }}"                                     7482
                     value="{{ current.get('heavy',                  7483
    {}).get(m) if current.get('heavy', {}).get(m) is not             7484
    none else '' }}">                                                7485
            </td>                                                    7486
            {% endfor %}                                             7487
          </tr>                                                      7488
        </tbody>                                                     7489
      </table>                                                       7490
      {% endif %}                                                    7491
    {% endfor %}                                                     7492
    <p class="actions" style="margin-top:18px;">                     7493
      <button type="submit" class="btn">保存</button>                7494
      <a class="btn btn--ghost" href="/indicator/">返回</a>          7495
    </p>                                                             7496
  </form>                                                            7497
</div>                                                               7498
{% endblock %}                                                       7499
# ==== app/templates/indicator/import.html ====                      7500
```

```
{% extends "base.html" %}
{% block title %}批量导入指标数据{% endblock %}
{% block content %}
<div class="card" style="max-width:860px;">
  <h2>批量导入指标数据</h2>
  <p class="muted">
    支持 CSV 与 Excel（.xlsx/.xls）。文件每行对应一个微地质单元，
    系统按「单元类型」匹配图斑内已划分的单元后写入数据。
    先下载模板按格式填写，可减少出错。
  </p>
  <p class="actions" style="margin-bottom:14px;">
    <a class="btn btn--sm btn--ghost"
    href="/indicator/import/template">下载导入模板（CSV）</a>
    <a class="btn btn--sm btn--ghost"
    href="/indicator/">返回指标数据</a>
  </p>
  <form method="post" enctype="multipart/form-data">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="patch-id">目标矿山图斑</lab
    el>
      <select id="patch-id" name="patch_id" required>
        <option value="">— 请选择 —</option>
        {% for p in patches %}
        <option value="{{ p.id }}">{{ p.name }}（{{
    p.units.count() }} 个单元）</option>
        {% endfor %}
      </select>
    </div>
    <div class="field">
      <label class="field__label" for="datafile">数据文件</label
    >
      <input id="datafile" type="file" name="datafile"
             accept=".csv,.xlsx,.xls" required>
    </div>
    <p class="actions" style="margin-top:16px;">
      <button type="submit" class="btn">开始导入</button>
    </p>
  </form>
  <h3>模板列说明</h3>
  <p class="muted">
    共 {{ headers|length }} 列。留空表示该指标「不适用」。
    坡向须填「阳坡 / 阴阳坡 / 阴坡」之一。未检测的重金属留空即可
    （按论文注释视为未受污染）。
  </p>
  <p class="mono muted" style="line-height:1.9;">{{
    headers|join(" , ") }}</p>
  <h3>导入规则</h3>
  <ul class="muted" style="padding-left:20px;line-height:1.9
    ;">
```

```
      <li>「单元类型」必须与图斑内已划分的单元一致，否则该行会被跳过并在结果中提示。</li>
      <li>同一单元类型若在文件中出现多行，以最后一行为准。</li>
      <li>已有数据的单元会被覆盖更新；没有数据的单元将新建记录。</li>
      <li>数值列请填纯数字，不要带单位（如填 <span class="mono">10</span> 而非
      <span class="mono">10cm</span>）。</li>
    </ul>
</div>
{% if report %}
<div class="card" style="max-width:860px;">
    <h2>导入结果</h2>
    <div class="stat-row">
      <div>
        <div class="stat__label">目标图斑</div>
        <div style="font-size:15px;">{{ report.patch }}</div>
      </div>
      <div>
        <div class="stat__label">成功导入</div>
        <div class="stat__value ok">{{ report.saved }}
      个单元</div>
      </div>
      <div>
        <div class="stat__label">问题行</div>
        <div class="stat__value {% if report.errors %}warn{%
      endif %}">{{ report.errors|length }}</div>
      </div>
    </div>
    {% if report.errors %}
    <h3>未导入的行</h3>
    <ul class="muted" style="padding-left:20px;line-height:1.9
      ;">
      {% for e in report.errors %}<li>{{ e }}</li>{% endfor %}
    </ul>
    {% else %}
    <p class="ok" style="margin-top:12px;">全部数据行均已成功导入。</p>
    {% endif %}
    <p class="actions" style="margin-top:16px;">
      <a class="btn btn--ghost" href="/indicator/">查看指标数据</a>
    </p>
</div>
{% endif %}
{% endblock %}
# ==== app/templates/evaluate/index.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}适宜性定量评价{% endblock %}
{% block content %}
<div class="card">
    <h2>适宜性定量评价</h2>
    <p class="muted">
      系统按论文式（8） 对各项指标做标准化，按式（9） 分单元赋分并按面积占比加权合成，
```

```
    最后按论文表 4 判定适宜性等级。未满足前置条件的图斑不能执行评价，
    下表会列出所缺步骤。
</p>
{{ patch_guide(projects_without_patch) }}
{% if not patches %}
<p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>
{% else %}
<table>
  <thead>
    <tr>
      <th scope="col">所属项目</th>
      <th scope="col">矿山图斑</th>
      <th scope="col">前置条件</th>
      <th scope="col">最近评价结果</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for p in patches %}
    {% set pr = progress.get(p.id) %}
    <tr>
      <td>{{ p.project.name if p.project else "—" }}</td>
      <td style="text-align:left;">{{ p.name }}</td>
      <td style="text-align:left;">
        {% if pr and pr.ready %}
        <span class="tag tag--ok">已具备</span>
        {% elif pr %}
        <span class="tag tag--warn">缺：{{
pr.blocking[0].label }}</span>
        <span class="muted">{{ pr.blocking[0].reason
}}</span>
        {% endif %}
      </td>
      <td>
        {% if latest.get(p.id) %}
        F = <span class="mono num">{{
latest[p.id].final_score }}</span>,
        <span class="tag tag--ok">{{ latest[p.id].grade
}}</span>
        {% else %}
        <span class="tag tag--muted">未评价</span>
        {% endif %}
      </td>
      <td class="nowrap">
        <span class="row-actions">
          <a class="link-btn" href="/evaluate/{{ p.id
}}">执行评价</a>
          {% if latest.get(p.id) %}
          <a class="link-btn" href="/evaluate/result/{{
latest[p.id].id }}">查看结果</a>
```

```
                {% endif %}
              </span>
            </td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
      {% endif %}
</div>
{% endblock %}
# ==== app/templates/evaluate/run.html ====
{% extends "base.html" %}
{% import "_macros.html" as ui %}
{% block title %}执行评价 · {{ patch.name }}{% endblock %}
{% block content %}
<div class="card" style="max-width:880px;">
  <h2>执行适宜性评价 · {{ patch.name }}</h2>
  <p class="muted">
    系统按论文式（8） 对各项指标做标准化，按式（9） 分单元赋分并按面积占比加权合成，
    最后按论文表 4 判定适宜性等级。评价结果会保存，可随时回看。
  </p>
  {{ ui.workflow_steps(progress) }}
  {{ ui.workflow_blocking(progress) }}
  {% if progress.ready %}
  <form method="post">
    {{ form.hidden_tag() }}
    <p class="actions">
      <button type="submit" class="btn">开始评价</button>
      <a class="btn btn--ghost" href="/evaluate/">返回</a>
    </p>
  </form>
  {% else %}
  <p class="actions">
    <button type="button" class="btn" disabled>开始评价</button>
    <a class="btn btn--ghost" href="/evaluate/">返回</a>
  </p>
  <p class="muted">前置步骤全部完成后，该按钮才可用。</p>
  {% endif %}
</div>
<div class="card" style="max-width:880px;">
  <h2>当前流程状态</h2>
  <p class="muted">按论文 4.1 节的顺序列出，限制性条件判定在前。</p>
  <table>
    <thead>
      <tr><th scope="col">步骤</th><th scope="col">状态</th><th
    scope="col">说明</th></tr>
    </thead>
    <tbody>
      <tr>
        <td>限制性条件判定</td>
```

```
          <td>                                                                    7701
            {% if progress.constraint.passed %}<span                              7702
      class="tag tag--ok">已通过</span>                                           7703
            {% elif progress.constraint.done %}<span                              7704
      class="tag tag--warn">未通过</span>                                         7705
            {% else %}<span class="tag tag--muted">待完成</span>{                 7706
      % endif %}                                                                  7707
          </td>                                                                   7708
          <td style="text-align:left;">{{                                        7709
      progress.constraint.detail }}</td>                                          7710
        </tr>                                                                     7711
        <tr>                                                                      7712
          <td>划分微地质单元</td>                                                 7713
          <td>                                                                    7714
            {% if progress.units.done %}<span class="tag                          7715
      tag--ok">已完成</span>                                                      7716
            {% else %}<span class="tag tag--muted">待完成</span>{                 7717
      % endif %}                                                                  7718
          </td>                                                                   7719
          <td style="text-align:left;">{{                                        7720
      progress.units.detail }}</td>                                               7721
        </tr>                                                                     7722
        <tr>                                                                      7723
          <td>录入指标数据</td>                                                   7724
          <td>                                                                    7725
            {% if progress.data.ready %}<span class="tag                          7726
      tag--ok">已完成</span>                                                      7727
            {% else %}<span class="tag tag--muted">待完成</span>{                 7728
      % endif %}                                                                  7729
          </td>                                                                   7730
          <td style="text-align:left;">{{                                        7731
      progress.data.detail }}</td>                                                7732
        </tr>                                                                     7733
      </tbody>                                                                    7734
    </table>                                                                      7735
</div>                                                                            7736
{% if weight_sources %}                                                           7737
<div class="card" style="max-width:880px;">                                       7738
    <h2>本次将采用的指标权重</h2>                                                 7739
    <p class="muted">                                                             7740
      权重决定各指标对评价结果的影响程度。默认采用论文表 3 的取值；              7741
      若已用层次分析法算出新权重并保存，则按新权重计算。                         7742
    </p>                                                                          7743
    <table style="max-width:720px;">                                              7744
      <thead>                                                                     7745
        <tr><th scope="col">微地质单元</th><th                                    7746
      scope="col">权重来源</th></tr>                                              7747
      </thead>                                                                    7748
      <tbody>                                                                     7749
        {% for ut, label in weight_sources.items() %}                             7750
```

```
        <tr>                                                                    7751
          <td style="text-align:left;">{{ ut }}</td>                           7752
          <td style="text-align:left;">                                         7753
            {% if "自定义" in label %}                                          7754
            <span class="tag tag--warn">{{ label }}</span>                      7755
            {% else %}                                                          7756
            <span class="tag tag--muted">{{ label }}</span>                     7757
            {% endif %}                                                         7758
          </td>                                                                 7759
        </tr>                                                                   7760
        {% endfor %}                                                            7761
      </tbody>                                                                  7762
    </table>                                                                    7763
</div>                                                                          7764
{% endif %}                                                                     7765
{% endblock %}                                                                  7766
# ==== app/templates/evaluate/result.html ====                                  7767
{% extends "base.html" %}                                                       7768
{% block title %}评价结果{% endblock %}                                          7769
{% block content %}                                                             7770
<div class="card">                                                              7771
  <h2>适宜性评价结果 · {{ patch.name if patch else "" }}</h2>                    7772
  <div class="stat-row">                                                        7773
    <div>                                                                       7774
      <div class="stat__label">最终评价因子 F</div>                             7775
      <div class="stat__value mono">{{ r.final_score                            7776
    }}</div>                                                                    7777
    </div>                                                                      7778
    <div>                                                                       7779
      <div class="stat__label">适宜性等级</div>                                 7780
      <div class="stat__value ok">{{ r.grade }}</div>                           7781
    </div>                                                                      7782
    <div style="flex:1;min-width:240px;">                                       7783
      <div class="stat__label">推荐措施</div>                                   7784
      <div style="font-size:15px;">{{ r.measures }}</div>                       7785
    </div>                                                                      7786
  </div>                                                                        7787
  {% if unit_scores %}                                                          7788
  <h3>各微地质单元得分</h3>                                                     7789
  <table style="max-width:860px;">                                              7790
    <thead>                                                                     7791
      <tr>                                                                      7792
        <th scope="col">微地质单元</th>                                         7793
        <th scope="col">面积占比</th>                                           7794
        <th scope="col">单元评价因子 F′</th>                                    7795
        <th scope="col">加权贡献</th>                                           7796
        <th scope="col">权重来源</th>                                           7797
      </tr>                                                                     7798
    </thead>                                                                    7799
    <tbody>                                                                     7800
```

```
        {% for u in unit_scores %}
        <tr>
          <td>{{ u.unit_type }}</td>
          <td class="num">{{ "%.2f%%"|format(u.ratio * 100)
      }}</td>
          <td class="mono">{{ u.unit_score }}</td>
          <td class="mono">{{ "%.4f"|format(u.ratio *
      u.unit_score) }}</td>
          <td class="muted">{{ u.weight_source or "论文表 3
      默认权重" }}</td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    {% endif %}
    <p class="actions" style="margin-top:18px;">
      <a class="btn btn--ghost" href="/report/{{ r.patch_id
      }}">查看得分明细</a>
      <a class="btn btn--ghost" href="/evaluate/">返回评价列表</a>
    </p>
</div>
{% endblock %}
# ==== app/templates/report/index.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}报告输出{% endblock %}
{% block content %}
<div class="card">
    <h2>报告与结果输出</h2>
    <p class="muted">
      可查看逐层得分明细与结果图表，并导出 PDF 或 Word 评价报告。
      报告是评价结论的呈现，因此仅在评价所依赖的前置步骤全部完成后才可生成。
    </p>
    {{ patch_guide(projects_without_patch) }}
    {% if not patches %}
    <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>
    {% else %}
    <table>
      <thead>
        <tr>
          <th scope="col">所属项目</th>
          <th scope="col">矿山图斑</th>
          <th scope="col">可生成报告</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for p in patches %}
        {% set pr = progress.get(p.id) %}
        <tr>
```

```
            <td>{{ p.project.name if p.project else "—" }}</td>
            <td style="text-align:left;">{{ p.name }}</td>
            <td style="text-align:left;">
              {% if pr and pr.ready %}
              <span class="tag tag--ok">可以</span>
              {% elif pr %}
              <span class="tag tag--warn">暂不可</span>
              <span class="muted">{{ pr.blocking[0].reason
      }}</span>
              {% endif %}
            </td>
            <td class="nowrap">
              {% if pr and pr.ready %}
              <span class="row-actions">
                <a class="link-btn" href="/report/{{ p.id
      }}">查看得分明细</a>
                <a class="link-btn" href="/report/{{ p.id
      }}/pdf">导出 PDF</a>
                <a class="link-btn" href="/report/{{ p.id
      }}/word">导出 Word</a>
              </span>
              {% else %}
              <span class="muted">请先完成前置步骤</span>
              {% endif %}
            </td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
      {% endif %}
</div>
{% endblock %}
# ==== app/templates/report/detail.html ====
{% extends "base.html" %}
{% block title %}得分明细 · {{ patch.name }}{% endblock %}
{% block content %}
<div class="card">
  <h2>得分明细 · {{ patch.name }}</h2>
  <div class="stat-row">
    <div>
      <div class="stat__label">最终评价因子 F</div>
      <div class="stat__value mono">{{
    "%.4f"|format(result.final_score) }}</div>
    </div>
    <div>
      <div class="stat__label">适宜性等级</div>
      <div class="stat__value ok">{{ result.grade }}</div>
    </div>
    <div style="flex:1;min-width:240px;">
      <div class="stat__label">推荐措施</div>
```

```
        <div style="font-size:15px;">{{ result.grade_info.measure }}</div>
      </div>
    </div>
    <p class="muted" style="margin-top:12px;">
      本页与导出报告均取自<strong>{% if record %}{{ record.created_at.strftime("%Y-%m-%d %H:%M:%S") }}{% endif %}</strong>记录的那次评价结果（评价记录编号 {{ record.id if record else "—" }}），不会因事后修改指标数据而变化。
    </p>
    <p class="actions" style="margin-top:16px;">
      <a class="btn" href="/report/{{ patch.id }}/pdf">导出 PDF 报告</a>
      <a class="btn btn--ghost" href="/report/{{ patch.id }}/word">导出 Word 报告</a>
      <a class="btn btn--ghost" href="/report/">返回</a>
    </p>
  </div>
  <div class="card">
    <h2>结果图表</h2>
    <h3>各微地质单元得分对比</h3>
    <div>{{ chart_units|safe }}</div>
    <h3>各一级指标平均得分</h3>
    <p class="muted">反映各一级指标对评价结果的实际影响，便于识别关键控制项。</p>
    <div>{{ chart_weight|safe }}</div>
  </div>
  {% for u in result.unit_results %}
  <div class="card">
    <h2>{{ u.unit_type }} · 占比 {{ "%.2f%%"|format(u.ratio * 100) }} · 单元得分 {{ "%.4f"|format(u.unit_score) }}</h2>
    {% if u.weight_source %}
    <p class="muted" style="margin-top:-6px;">权重来源：{{ u.weight_source }}</p>
    {% endif %}
    <table>
      <thead>
        <tr>
          <th scope="col">一级指标</th>
          <th scope="col">二级指标</th>
          <th scope="col">实测值</th>
          <th scope="col">标准化值</th>
          <th scope="col">权重</th>
          <th scope="col">贡献</th>
        </tr>
      </thead>
      <tbody>
        {% for d in u.details %}
        <tr>
```

```
          <td>{{ d.l1 }}</td>                                                    7951
          <td style="text-align:left;">{{ d.l2 }}</td>                          7952
          <td class="mono" style="text-align:left;">{{ "—" if                   7953
    d.raw is none else d.raw }}</td>                                             7954
          <td class="mono">{{ "—" if d.std_value is none else                    7955
    "%.4f"|format(d.std_value) }}</td>                                           7956
          <td class="mono">{{ "%.4f"|format(d.weight) }}</td>                    7957
          <td class="mono">{{ "—" if d.contribution is none                      7958
    else "%.4f"|format(d.contribution) }}</td>                                   7959
        </tr>                                                                    7960
        {% endfor %}                                                             7961
    </tbody>                                                                     7962
  </table>                                                                       7963
  <p class="muted" style="margin:0;">                                            7964
    一级指标得分：                                                               7965
    {% for k, v in u.l1_scores.items() %}{{ k }} = <span                         7966
    class="num">{{ "%.4f"|format(v) }}</span>{% if not                           7967
    loop.last %};{% endif %}{% endfor %}                                         7968
  </p>                                                                           7969
</div>                                                                           7970
{% endfor %}                                                                     7971
{% endblock %}                                                                   7972
# ==== app/templates/report/print.html ====                                      7973
<!DOCTYPE html>                                                                  7974
<html lang="zh-CN">                                                              7975
<head>                                                                           7976
<meta charset="UTF-8">                                                           7977
<title>自然恢复适宜性评价报告 · {{ patch.name }}</title>                         7978
<style>                                                                          7979
  @page { size: A4; margin: 18mm 16mm; }                                         7980
  * { box-sizing: border-box; }                                                  7981
  body {                                                                         7982
    margin: 0; color: #23262b; background: #fff;                                 7983
    font-family: "Microsoft YaHei", "SimSun", sans-serif;                        7984
    font-size: 10.5pt; line-height: 1.65;                                        7985
  }                                                                              7986
  h1 { font-size: 17pt; text-align: center; margin: 0 0                          7987
    6px; }                                                                       7988
  h2 { font-size: 13pt; color: #1f3a5f; margin: 20px 0 8px;                      7989
       border-bottom: 1px solid #dfe3e8; padding-bottom:                         7990
    4px; }                                                                       7991
  h3 { font-size: 11.5pt; color: #1f3a5f; margin: 16px 0                         7992
    6px; }                                                                       7993
  .sub { text-align: center; color: #6b7280; font-size:                          7994
    9.5pt; margin-bottom: 18px; }                                                7995
  table { width: 100%; border-collapse: collapse; margin:                        7996
    8px 0 14px;                                                                  7997
          font-variant-numeric: tabular-nums; }                                  7998
  th, td { border: 1px solid #c9ccd1; padding: 4px 8px;                          7999
    text-align: center; }                                                        8000
```

```
  th { background: #eef2f7; font-weight: 600; }
  .left { text-align: left; }
  .mono { font-family: Consolas, "Courier New", monospace; }
  .kv { width: auto; }
  .kv th { width: 30%; text-align: left; }
  .kv td { text-align: left; }
  ul { margin: 6px 0 12px; padding-left: 20px; }
  .note { color: #6b7280; font-size: 9.5pt; }
  .chart { margin: 10px 0 16px; }
  .sig { margin-top: 28px; font-size: 10pt; }
  .avoid-break { break-inside: avoid; }
  svg { max-width: 100%; }
</style>
</head>
<body>
  <h1>历史遗留废弃矿山自然恢复适宜性评价报告</h1>
  <p class="sub">
    {{ patch.name }}
    {% if patch.admin_region %} · {{ patch.admin_region
    }}{% endif %}
    <br>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1.0 ·
    生成时间 {{ generated_at }}
  </p>
  <h2>一、图斑基本信息</h2>
  <table class="kv">
    <tr><th>图斑名称</td><td>{{ patch.name }}</td></tr>
    <tr><th>所在行政区</td><td>{{ patch.admin_region or "—"
    }}</td></tr>
    <tr><th>图斑面积（公顷）</td><td>{{ patch.area_ha if
    patch.area_ha is not none else "—" }}</td></tr>
    <tr><th>开采矿种</td><td>{{ patch.mineral_type or "—"
    }}</td></tr>
    <tr><th>开采方式</td><td>{{ patch.mining_method or "—"
    }}</td></tr>
    <tr><th>责任主体缺失</td><td>{{ "是" if patch.responsible_entit
    y_missing else "否" }}</td></tr>
    {% if reference %}
    <tr><th>参照生态系统</th>
        <td>植被覆盖度基准 {{ reference.coverage_benchmark }}%，
            多样性指数基准 {{ reference.shannon_benchmark
    }}</td></tr>
    {% endif %}
  </table>
  <h2>二、评价结论</h2>
  <table class="kv">
    <tr><th>最终评价因子 F</th><td class="mono"><strong>{{
    "%.4f"|format(result.final_score) }}</strong></td></tr>
    <tr><th>适宜性等级</td><td><strong>{{ result.grade
    }}</strong></td></tr>
    <tr><th>判定依据</td><td>{{ result.grade_info.meaning
```

```
    }}</td></tr>                                                    8051
    <tr><th>推荐措施</th><td>{{ result.grade_info.measure           8052
    }}</td></tr>                                                    8053
</table>                                                            8054
<h2>三、各微地质单元得分</h2>                                        8055
<table>                                                             8056
    <thead>                                                         8057
      <tr><th>微地质单元</th><th>面积占比</th><th>单元评价因子        8058
    F'</th><th>加权贡献</th></tr>                                    8059
    </thead>                                                        8060
    <tbody>                                                         8061
      {% for u in result.unit_results %}                            8062
      <tr>                                                          8063
        <td>{{ u.unit_type }}</td>                                  8064
        <td class="mono">{{ "%.2f%%"|format(u.ratio * 100)          8065
    }}</td>                                                         8066
        <td class="mono">{{ "%.4f"|format(u.unit_score)             8067
    }}</td>                                                         8068
        <td class="mono">{{ "%.4f"|format(u.weighted)               8069
    }}</td>                                                         8070
      </tr>                                                         8071
      {% endfor %}                                                  8072
    </tbody>                                                        8073
</table>                                                            8074
<div class="chart">{{ chart_units|safe }}</div>                     8075
<h2>四、一级指标权重分布</h2>                                        8076
<div class="chart">{{ chart_weight|safe }}</div>                    8077
<h2>五、得分明细</h2>                                                8078
{% for u in result.unit_results %}                                  8079
<h3>{{ u.unit_type }}（占比 {{ "%.2f%%"|format(u.ratio *            8080
    100) }}，单元得分 {{ "%.4f"|format(u.unit_score) }}）</h3>       8081
<table>                                                             8082
    <thead>                                                         8083
      <tr>                                                          8084
        <th>一级指标</th><th>二级指标</th><th>实测值</th>            8085
        <th>标准化值</th><th>权重</th><th>贡献</th>                  8086
      </tr>                                                         8087
    </thead>                                                        8088
    <tbody>                                                         8089
      {% for d in u.details %}                                      8090
      <tr>                                                          8091
        <td>{{ d.l1 }}</td>                                         8092
        <td class="left">{{ d.l2 }}</td>                            8093
        <td class="mono left">{{ "—" if d.raw is none else          8094
    d.raw }}</td>                                                   8095
        <td class="mono">{{ "—" if d.std_value is none else         8096
    "%.4f"|format(d.std_value) }}</td>                              8097
        <td class="mono">{{ "%.4f"|format(d.weight) }}</td>         8098
        <td class="mono">{{ "—" if d.contribution is none           8099
    else "%.4f"|format(d.contribution) }}</td>                      8100
```

```
        </tr>                                                                    8101
        {% endfor %}                                                             8102
      </tbody>                                                                   8103
    </table>                                                                     8104
    <p class="note">                                                             8105
      一级指标得分：                                                             8106
      {% for k, v in u.l1_scores.items() %}{{ k }} = {{                          8107
      "%.4f"|format(v) }}{% if not loop.last %};{% endif %}{%                    8108
      endfor %}                                                                  8109
    </p>                                                                         8110
    {% endfor %}                                                                 8111
    <h2>六、评价依据说明</h2>                                                    8112
    <ul class="note">                                                            8113
      <li>指标标准化按论文式（8） 执行，按指标性质分为越大越优、越小越优、     8114
        适中型与离散赋值四类。</li>                                              8115
      <li>单元评价因子按论文式（9） 由二级指标权重加权求得；                    8116
        整矿评价因子由各微地质单元按面积占比加权合成。</li>                     8117
      <li>适宜性等级按论文表 4 划分：F&lt;0.3 不适宜、0.3≤F≤0.7 基本适宜、F≥0.7 8118
      适宜。</li>                                                                8119
      <li>指标权重取自论文表 3，按岩质边坡、碎石边坡、平台三类微地质单元分别确定。</li>  8120
    </ul>                                                                        8121
    <p class="sig">                                                              8122
      评价人（签字）：_____　　日期：_____                                      8123
    </p>                                                                         8124
</body>                                                                          8125
</html>                                                                          8126
#  ==== app/templates/search.html ====                                           8127
{% extends "base.html" %}                                                        8128
{% from "_macros.html" import icon %}                                            8129
{% block title %}搜索 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}     8130
{% block content %}                                                              8131
<div class="card">                                                               8132
  <h2>搜索结果</h2>                                                             8133
  {% if not q %}                                                                 8134
  <p class="muted">请输入关键词。可检索项目名称与编号、图斑名称与行政区/矿种、  8135
    标准名称与文号。</p>                                                        8136
  {% else %}                                                                     8137
  <p class="muted">                                                              8138
    关键词「<strong>{{ q }}</strong>」共找到                                     8139
    <span class="num">{{ total }}</span> 条结果。                                8140
  </p>                                                                           8141
  {% if not total %}                                                             8142
  <p class="muted">没有匹配的内容。换个词试试，或检查是否有错别字。</p>         8143
  {% endif %}                                                                    8144
  {% endif %}                                                                    8145
</div>                                                                           8146
{% if projects %}                                                                8147
<div class="card">                                                               8148
  <h2>评价项目（{{ projects|length }}）</h2>                                     8149
  <table>                                                                        8150
```

```
      <thead>                                                                  8151
        <tr>                                                                   8152
          <th scope="col">项目名称</th>                                        8153
          <th scope="col">编号</th>                                            8154
          <th scope="col">状态</th>                                            8155
          <th scope="col">图斑数</th>                                          8156
          <th scope="col">操作</th>                                            8157
        </tr>                                                                  8158
      </thead>                                                                 8159
      <tbody>                                                                  8160
        {% for p in projects %}                                                8161
        <tr>                                                                   8162
          <td style="text-align:left;">{{ p.name }}</td>                       8163
          <td class="mono">{{ p.code or "—" }}</td>                            8164
          <td><span class="tag tag--muted">{{ p.status                         8165
      }}</span></td>                                                           8166
          <td class="num">{{ p.patches.count() }}</td>                         8167
          <td class="nowrap">                                                  8168
            <span class="row-actions">                                         8169
              <a class="link-btn" href="/project/{{ p.id                       8170
      }}/patches">矿山图斑</a>                                                 8171
              <a class="link-btn" href="/project/{{ p.id                       8172
      }}/edit">编辑</a>                                                        8173
            </span>                                                            8174
          </td>                                                                8175
        </tr>                                                                  8176
        {% endfor %}                                                           8177
      </tbody>                                                                 8178
    </table>                                                                   8179
</div>                                                                         8180
{% endif %}                                                                    8181
{% if patches %}                                                               8182
<div class="card">                                                             8183
    <h2>矿山图斑（{{ patches|length }}）</h2>                                  8184
    <table>                                                                    8185
      <thead>                                                                  8186
        <tr>                                                                   8187
          <th scope="col">所属项目</th>                                        8188
          <th scope="col">图斑名称</th>                                        8189
          <th scope="col">行政区</th>                                          8190
          <th scope="col">矿种</th>                                            8191
          <th scope="col">操作</th>                                            8192
        </tr>                                                                  8193
      </thead>                                                                 8194
      <tbody>                                                                  8195
        {% for m in patches %}                                                 8196
        <tr>                                                                   8197
          <td>{{ m.project.name if m.project else "—" }}</td>                  8198
          <td style="text-align:left;">{{ m.name }}</td>                       8199
          <td>{{ m.admin_region or "—" }}</td>                                 8200
```

```
            <td>{{ m.mineral_type or "—" }}</td>                    8201
            <td class="nowrap">                                      8202
              <span class="row-actions">                             8203
                <a class="link-btn" href="/constraint/{{ m.id        8204
      }}">限制性判定</a>                                             8205
                <a class="link-btn" href="/unit/{{ m.id              8206
      }}">微地质单元</a>                                             8207
                <a class="link-btn" href="/evaluate/{{ m.id          8208
      }}">执行评价</a>                                               8209
              </span>                                                8210
            </td>                                                    8211
          </tr>                                                      8212
          {% endfor %}                                               8213
        </tbody>                                                     8214
    </table>                                                         8215
</div>                                                               8216
{% endif %}                                                          8217
{% if standards %}                                                   8218
<div class="card">                                                   8219
    <h2>标准与政策文件（{{ standards|length }}）</h2>                  8220
    <table>                                                          8221
      <thead>                                                        8222
        <tr>                                                         8223
          <th scope="col">类别</th>                                  8224
          <th scope="col">省份</th>                                  8225
          <th scope="col">名称</th>                                  8226
          <th scope="col">文号</th>                                  8227
          <th scope="col">操作</th>                                  8228
        </tr>                                                        8229
      </thead>                                                       8230
      <tbody>                                                        8231
        {% for s in standards %}                                     8232
        <tr>                                                         8233
          <td>{{ s.category }}</td>                                  8234
          <td>{{ s.province }}</td>                                  8235
          <td style="text-align:left;">{{ s.name }}</td>             8236
          <td class="mono">{{ s.doc_number or "—" }}</td>            8237
          <td class="nowrap">                                        8238
            <span class="row-actions">                               8239
              <a class="link-btn" href="/standard/{{ s.id            8240
      }}/rules">规则</a>                                             8241
            </span>                                                  8242
          </td>                                                      8243
        </tr>                                                        8244
        {% endfor %}                                                 8245
      </tbody>                                                       8246
    </table>                                                         8247
</div>                                                               8248
{% endif %}                                                          8249
{% endblock %}                                                       8250
```

```
# ==== seed_demo.py ====
# -*- coding: utf-8 -*-
"""内置玉田县案例：把论文第 5 节案例数据写入数据库。
用于功能验证与操作演示。数据逐字取自论文表 5 与第 5 节。
运行：python seed_demo.py                    幂等，只补写缺失的演示数据，
                                                不会删除用户自建的项目与图斑
      python seed_demo.py --reset            连同标准库与指标定义一并重建
"""
import json
import sys
sys.stdout.reconfigure(encoding="utf-8")
from app import create_app
from app.models import (db, User, Project, MinePatch,
    ReferenceEcosystem,
                        MicroUnit, IndicatorData, Standard,
    StandardRule,
                        ConstraintCheck, EvaluationResult,
    ScoreDetail)
app = create_app()
# 内置演示案例的项目编号，用于区分「演示数据」与「用户自建数据」
DEMO_CODE = "DEMO-001"
# 论文表 1：11 项地方标准与政策文件
STANDARDS = [
    dict(category="地方标准", province="辽宁",
    name="矿山废弃地植被自然恢复技术规程",
        doc_number="DB21/T 3702—2023",
    publish_date="2023-03-30",
        implement_date="2023-04-30",
        summary="首个正式发布的矿山自然恢复地方标准；提出必要条件和充分条件",
        rules=[("植被覆盖度", ">=", "50", "必要条件", None),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None),
               ("与参照系差距", "<", "20", "充分条件", "%")]),
    dict(category="地方标准", province="湖北",
    name="废弃露天矿山生态环境自然恢复标准",
        doc_number="DB42/T 2421—2025",
    publish_date="2025-08-20",
        implement_date="2025-10-20",
        summary="结合矿山生态需求进行自然恢复适宜性分级判断",
        rules=[("自然恢复适宜性分级", "==", "是", "必要条件", None)]),
    dict(category="地方标准", province="内蒙古",
    name="历史遗留废弃矿山自然恢复判定规范",
        doc_number="DB15/T 4280—2026",
    publish_date="2026-01-30",
        implement_date="2026-03-01",
        summary="调查内容包含参照生态系统调查",
        rules=[("地质灾害发育程度", "<=", "弱发育", "必要条件", None),
               ("地形地貌与周边协调", "==", "是", "必要条件", None),
               ("有利于植被恢复的表层土壤", "==", "是", "必要条件", None),
               ("已形成稳定植被覆盖", "==", "是", "必要条件", None)]),
```

```
    dict(category="地方标准", province="江西",                                  8301
    name="废弃矿山自然恢复技术规范",                                            8302
        doc_number="DB36/T 2221—2026",                                         8303
    publish_date="2026-02-02",                                                 8304
        implement_date="2026-08-01",                                           8305
        summary="对参照生态系统进行了要求；提出自然恢复适宜性指数",              8306
        rules=[("参照生态系统调查", "==", "是", "必要条件", None),              8307
               ("自然恢复适宜性指数", ">=", "阈值", "充分条件", None)]),        8308
    dict(category="地方标准", province="宁夏", name="矿山自然恢复认定规程",     8309
        doc_number=None, publish_date="2026-06-18",                            8310
    implement_date=None,                                                       8311
        summary="分区差异化：三个一级生态修复分区及 22 个二级分区",            8312
        rules=[("无地质灾害风险", "==", "是", "必要条件", None),               8313
               ("植被覆盖度趋势", "==", "稳定或正增长", "必要条件", None),     8314
               ("与参照系差距", "<", "20", "充分条件", "%")]),                 8315
    dict(category="政策文件", province="安徽",                                 8316
        name="安徽省自然资源厅关于规范废弃露天矿山采取自然恢复方式进行治理的通知",  8317
        doc_number="皖自然资修函[2020]34号",                                   8318
    publish_date="2020-03-03", implement_date=None,                            8319
        summary="自然恢复条件：责任主体为政府、无地质灾害、不在三线三边、水土条件良好",8320
        rules=[("责任主体", "==", "政府", "必要条件", None),                   8321
               ("无地质灾害", "==", "是", "必要条件", None),                   8322
               ("无水土污染", "==", "是", "必要条件", None)]),                 8323
    dict(category="政策文件", province="山东",                                 8324
        name="山东省自然资源厅关于进一步规范历史遗留废弃露天开采矿山采取自然恢复方式进行治理的通知",  8325
        doc_number="鲁自然资字[2022]104号",                                    8326
    publish_date="2022-08-02", implement_date=None,                            8327
        summary="条件与安徽省相同，另增加自然恢复认定标准",                    8328
        rules=[("无地质灾害", "==", "是", "必要条件", None),                   8329
               ("无水土污染", "==", "是", "必要条件", None)]),                 8330
    dict(category="政策文件", province="湖南",                                 8331
        name="关于印发《湖南省历史遗留矿山自然恢复指导意见（试行）》的通知",  8332
        doc_number="湘自资办发[2023]58号",                                     8333
    publish_date="2023-07-20", implement_date=None,                            8334
        summary="自然恢复认定标准：无地质灾害或无威胁对象、无水土污染、设备拆除、符合规划、植被覆盖度8335
    >75%、连续3年正向增长",                                                    8336
        rules=[("植被覆盖度", ">", "75", "必要条件", "%"),                     8337
               ("植被覆盖率趋势", "==", "连续3年正向增长", "必要条件", None),8338
               ("无地质灾害", "==", "是", "必要条件", None),                   8339
               ("符合国土空间规划", "==", "是", "必要条件", None)]),           8340
    dict(category="政策文件", province="新疆",                                 8341
        name="自治区历史遗留废弃矿山自然恢复方式销号图斑工作指南（试行）",     8342
        doc_number="新自然资源办函〔2024〕219号",                              8343
    publish_date="2024-12-20", implement_date=None,                            8344
        summary="荒漠区沙地/裸土地/裸岩石砾地满足无地质灾害、地形协调、无水土污染即可判定",8345
        rules=[("植被覆盖度", ">=", "周边30%", "必要条件", None),              8346
               ("无地质灾害", "==", "是", "必要条件", None),                   8347
               ("地形平缓地貌稳定", "==", "是", "必要条件", None)]),           8348
    dict(category="政策文件", province="广西",                                 8349
        name="广西壮族自治区历史遗留矿山自然恢复技术指南",                     8350
```

```
            doc_number="桂自然资办〔2024〕250号",
        publish_date="2024-12-30", implement_date=None,
            summary="除高陡边坡和裸露岩体外，水土水热条件良好、植被自然恢复趋势明显",
            rules=[("无地质灾害", "==", "是", "必要条件", None),
                   ("植被自然恢复趋势", "==", "明显", "必要条件", None),
                   ("生产设备拆除完毕", "==", "是", "必要条件", None)]),
        dict(category="政策文件", province="西藏",
            name="西藏自治区历史遗留废弃矿山生态自然恢复认定指导意见（试行）",
            doc_number="藏自然资发〔2025〕19号",
        publish_date="2025-09-24", implement_date=None,
            summary="含修复成本高、生态修复效益不显著等适用条件",
            rules=[("责任主体", "==", "政府", "必要条件", None),
                   ("地形地貌完整与周边协调", "==", "是", "必要条件", None),
                   ("无地质灾害", "==", "是", "必要条件", None)]),
]
CASE = {
    "name": "玉田县典型历史遗留废弃矿山案例",
    "patch_name": "玉田县孤树镇白云岩采石场",
    "area_ha": 5.6,
    "mineral_type": "白云岩",
    "units": [
        {
            "unit_type": "岩质边坡", "ratio": 0.1386,
            "data": {"coverage": 68.88, "shannon": 0.00,
                     "slope": 63.33, "aspect": "阴阳坡"},
        },
        {
            "unit_type": "碎石边坡", "ratio": 0.6295,
            "data": {"soil_ratio": 0.34, "thickness": 10,
    "ph": 9.24,
                     "organic_matter": 4.89,
                     "heavy_metals": {"Hg": 0.05, "As":
    5.17, "Ni": 12.60,
                                      "Cu": 11.70, "Cd":
    0.04, "Pb": 6.50},
                     "coverage": 87.92, "shannon": 0.60,
                     "slope": 28.75, "aspect": "阳坡"},
        },
        {
            "unit_type": "平台", "ratio": 0.2319,
            "data": {"soil_ratio": 0.56, "thickness": 30,
    "ph": 8.72,
                     "organic_matter": 10.92,
                     "heavy_metals": {"Hg": 0.02, "As":
    3.13, "Ni": 12.33,
                                      "Cu": 11.14, "Cd":
    0.09, "Pb": 9.98},
                     "coverage": 58.42, "shannon": 0.72},
        },
    ],
```

```
}
def clean_case(reset=False):
    """清理数据。
    默认只清理内置演示案例（code = DEMO-001）自身的数据，**不触碰
    用户自己建立的项目与图斑**。加 --reset 时才额外清空标准库与指标定义，
    以便把演示环境恢复到初始状态。
    注意：不要在此处执行 Project.query.delete()，否则用户新建的项目
    会在每次启动时被删除。
    """
    from app.models import EvaluationResult, ScoreDetail,
    ConstraintCheck
    demo_projects = Project.query.filter_by(code=DEMO_CODE).
    all()
    demo_ids = [p.id for p in demo_projects]
    patch_ids = []
    if demo_ids:
        patch_ids = [p.id for p in MinePatch.query.filter(
            MinePatch.project_id.in_(demo_ids)).all()]
    if patch_ids:
        unit_ids = [u.id for u in MicroUnit.query.filter(
            MicroUnit.patch_id.in_(patch_ids)).all()]
        if unit_ids:
            IndicatorData.query.filter(
                IndicatorData.unit_id.in_(unit_ids)).delete(
                synchronize_session=False)
        result_ids = [r.id for r in EvaluationResult.query.f
    ilter(
            EvaluationResult.patch_id.in_(patch_ids)).all()]
        if result_ids:
            ScoreDetail.query.filter(
                ScoreDetail.result_id.in_(result_ids)).delet
    e(
                synchronize_session=False)
        EvaluationResult.query.filter(
            EvaluationResult.patch_id.in_(patch_ids)).delete
    (
            synchronize_session=False)
        ConstraintCheck.query.filter(
            ConstraintCheck.patch_id.in_(patch_ids)).delete(
            synchronize_session=False)
        ReferenceEcosystem.query.filter(
            ReferenceEcosystem.patch_id.in_(patch_ids)).dele
    te(
            synchronize_session=False)
        MicroUnit.query.filter(MicroUnit.patch_id.in_(patch_
    ids)).delete(
            synchronize_session=False)
        MinePatch.query.filter(MinePatch.id.in_(patch_ids)).
    delete(
            synchronize_session=False)
```

```
    if demo_ids:                                                    8451
        Project.query.filter(Project.id.in_(demo_ids)).delet        8452
    e(                                                              8453
            synchronize_session=False)                              8454
    if reset:                                                       8455
        StandardRule.query.delete()                                 8456
        Standard.query.delete()                                     8457
    #  兜底：清理指向已不存在评价记录的孤儿得分明细，                8458
    #  避免长期运行后 score_details 表堆积无效数据。                 8459
    _purge_orphan_details()                                         8460
    db.session.commit()                                             8461
    return len(demo_ids)                                            8462
def demo_patch():                                                   8463
    """取内置演示案例的图斑。                                       8464
    库中可能存在用户自建项目或验证算例（编号 VERIFY-），            8465
    故不能简单用 MinePatch.query.first()，否则取到的对象不确定。    8466
    没有演示案例时才退回第一个图斑。                                8467
    """                                                             8468
    proj = Project.query.filter_by(code=DEMO_CODE).first()          8469
    if proj is not None:                                            8470
        m = MinePatch.query.filter_by(project_id=proj.id).fi        8471
    rst()                                                           8472
        if m is not None:                                           8473
            return m                                                8474
    return MinePatch.query.first()                                  8475
def _purge_orphan_details():                                        8476
    """清理指向已不存在评价记录的孤儿得分明细。                     8477
    长期运行后 score_details 表可能堆积无效明细，启动时顺带清一次。 8478
    """                                                             8479
    live_ids = [r.id for r in EvaluationResult.query.all()]         8480
    if live_ids:                                                    8481
        ScoreDetail.query.filter(~ScoreDetail.result_id.in_(        8482
    live_ids)).delete(                                              8483
            synchronize_session=False)                              8484
    else:                                                           8485
        ScoreDetail.query.delete()                                  8486
    db.session.commit()                                             8487
def seed(reset=False):                                              8488
    with app.app_context():                                         8489
        db.create_all()          #  全新环境下先建表，保证本脚本可独立运行  8490
        _purge_orphan_details()                                     8491
        admin = User.query.filter_by(username="admin").first        8492
    ()                                                              8493
        if not admin:                                               8494
            from werkzeug.security import                           8495
    generate_password_hash                                          8496
            admin = User(username="admin",                          8497
                         password_hash=generate_password_has        8498
    h("admin123"),                                                  8499
                         role="admin")                              8500
```

```
            db.session.add(admin)                                                    8501
            db.session.commit()                                                      8502
        #  幂等：演示案例已存在就不重复写入，更不清空用户自己的项目                                  8503
        existing = Project.query.filter_by(code=DEMO_CODE).f                         8504
irst()                                                                               8505
        if existing and not reset:                                                   8506
            print("演示案例已存在，跳过写入（用户自建项目不受影响）")                                  8507
            _report()                                                                8508
            _seed_reference_data(reset=False)                                        8509
            return                                                                   8510
        clean_case(reset=reset)                                                      8511
        project = Project(name=CASE["name"], code=DEMO_CODE,                         8512
                          description="论文第 5 节典型案例",                                 8513
created_by=admin.id)                                                                 8514
        db.session.add(project)                                                      8515
        db.session.flush()                                                           8516
        patch = MinePatch(project_id=project.id,                                     8517
name=CASE["patch_name"],                                                             8518
                          admin_region="河北省唐山市玉田县",                                 8519
                          area_ha=CASE["area_ha"],                                   8520
                          mineral_type=CASE["mineral_type"],                         8521
                          mining_method="露天开采",                                    8522
stop_years=None,                                                                     8523
                          responsible_entity_missing=True)                           8524
        db.session.add(patch)                                                        8525
        db.session.flush()                                                           8526
        ref = ReferenceEcosystem(patch_id=patch.id,                                  8527
                                 coverage_benchmark=75.0,                            8528
shannon_benchmark=0.7)                                                               8529
        db.session.add(ref)                                                          8530
        for u in CASE["units"]:                                                      8531
            unit = MicroUnit(patch_id=patch.id,                                      8532
unit_type=u["unit_type"],                                                            8533
                             ratio=u["ratio"], area=None)                            8534
            db.session.add(unit)                                                     8535
            db.session.flush()                                                       8536
            d = u["data"]                                                            8537
            ind = IndicatorData(                                                     8538
                unit_id=unit.id,                                                     8539
                soil_ratio=d.get("soil_ratio"),                                      8540
                thickness=d.get("thickness"),                                        8541
                ph=d.get("ph"),                                                      8542
                organic_matter=d.get("organic_matter"),                              8543
                heavy_metals=json.dumps(d["heavy_metals"],                           8544
ensure_ascii=False)                                                                  8545
                                if d.get("heavy_metals") else                        8546
None,                                                                                8547
                coverage=d.get("coverage"),                                          8548
                shannon=d.get("shannon"),                                            8549
                slope=d.get("slope"),                                                8550
```

```
                    aspect=d.get("aspect"),                                  8551
                )                                                            8552
                db.session.add(ind)                                          8553
            #  限制性条件判定记录：演示案例按论文第 5 节描述属「无地质灾害隐患点」、          8554
            #  具备自然恢复条件，故五项全部通过。缺此记录会使后续评价被流程门禁拦住。          8555
            db.session.add(ConstraintCheck(                                  8556
                patch_id=patch.id,                                           8557
                policy_pass=True, location_pass=True,                        8558
    pollution_pass=True,                                                     8559
                geo_pass=True, farmland_pass=True,                           8560
                passed=True, reason="全部通过"))                              8561
            db.session.commit()                                              8562
            print("已写入案例：", CASE["name"])                               8563
            print("  图斑：", CASE["patch_name"])                            8564
            print("  微地质单元：岩质边坡 13.86% / 碎石边坡 62.95% / 平台      8565
    23.19%")                                                                 8566
            print("  管理员账号：admin / admin123")                          8567
            _seed_reference_data(reset=reset)                                8568
            _report()                                                        8569
def _seed_reference_data(reset=False):                                       8570
    """写入标准库与指标定义。                                                 8571
    参考资料只在库为空（或显式 --reset）时写入，避免覆盖用户自行维护          8572
    的标准条目与指标定义。                                                    8573
    """                                                                      8574
    if reset or Standard.query.count() == 0:                                 8575
        seed_standards()                                                     8576
    from app.services.indicator_service import                               8577
    seed_indicators                                                          8578
    n = seed_indicators()                                                    8579
    print("指标定义：%d 项" % n)                                             8580
def _report():                                                               8581
    total_p = Project.query.count()                                          8582
    total_patch = MinePatch.query.count()                                    8583
    print()                                                                  8584
    print("当前库内：项目 %d 个、图斑 %d 个" % (total_p, total_patch))        8585
    if total_p > total_patch:                                                8586
        print("  提示：项目是归档单位，评价针对的是图斑。")                   8587
        print("        新项目需先建立图斑，才会出现在后续各模块中。")         8588
def seed_standards():                                                        8589
    """写入论文表 1 的 11 项标准与政策文件及其结构化规则。"""                 8590
    StandardRule.query.delete()                                              8591
    Standard.query.delete()                                                  8592
    db.session.commit()                                                      8593
    for s in STANDARDS:                                                      8594
        std = Standard(category=s["category"],                               8595
    province=s["province"],                                                  8596
                       name=s["name"],                                       8597
    doc_number=s["doc_number"],                                              8598
                       publish_date=s["publish_date"],                       8599
                       implement_date=s["implement_date"],                   8600
```

```
                                summary=s["summary"])                    8601
            db.session.add(std)                                          8602
            db.session.flush()                                           8603
            for item, op, thr, ctype, note in s["rules"]:                8604
                db.session.add(StandardRule(standard_id=std.id,          8605
        item=item,                                                       8606
                                                operator=op,             8607
        threshold=thr,                                                   8608
                                                                         8609
        condition_type=ctype, note=note))                                8610
        db.session.commit()                                              8611
        print("已写入标准/政策文件：%d 项，规则 %d 条"                    8612
              % (len(STANDARDS), sum(len(s["rules"]) for s in            8613
        STANDARDS)))                                                     8614
if __name__ == "__main__":                                               8615
        #  --reset：把演示环境恢复到初始状态（清空标准库与指标定义后重建）。  8616
        #  无论是否 --reset，都不会删除用户自建的项目与图斑。              8617
        seed(reset="--reset" in sys.argv[1:])                            8618
#  ==== 测试/test_ahp.py ====                                            8619
#  -*- coding: utf-8 -*-                                                 8620
"""地基自检：验证 AHP 引擎与应用可启动。"""                               8621
import sys                                                               8622
sys.stdout.reconfigure(encoding="utf-8")                                 8623
import numpy as np                                                       8624
import os                                                                8625
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，              8626
#  使 `from app import ...` 与 app/app.db 的定位不受影响。               8627
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil             8628
        e__)))                                                           8629
if ROOT not in sys.path:                                                 8630
        sys.path.insert(0, ROOT)                                         8631
from app.services import ahp_service as ahp                              8632
LINE = "=" * 66                                                          8633
#  ------------------------------------------------------------------    8634
        ------------------                                               8635
#  1） 完全一致矩阵 → 权重应精确还原，CR=0                               8636
#  ------------------------------------------------------------------    8637
        ------------------                                               8638
print(LINE)                                                              8639
print("测试 1  完全一致判断矩阵的权重还原与一致性")                       8640
print(LINE)                                                              8641
#  由权重构造完全一致矩阵：a_ij = w_i / w_j                              8642
w_true = np.array([0.44, 0.31, 0.25])                                    8643
A = np.outer(w_true, 1.0 / w_true)                                       8644
r = ahp.weights_and_consistency(A)                                       8645
print("  判断矩阵 A：")                                                 8646
for row in A:                                                            8647
        print("      " + "  ".join(f"{v:8.4f}" for v in row))            8648
print()                                                                  8649
print("  计算权重 :", "  ".join(f"{v:.4f}" for v in                     8650
```

```
        r["weights"]))                                                    8651
print("   真实权重 :", "  ".join(f"{v:.4f}" for v in w_true))            8652
print("   λmax = %.6f    CI = %.3e    CR = %.3e    一致: %s"              8653
      % (r["lambda_max"], r["ci"], r["cr"],                               8654
        r["consistent"]))                                                 8655
assert np.allclose(r["weights"], w_true, atol=1e-6),                      8656
      "权重未精确还原"                                                    8657
assert r["consistent"], "完全一致矩阵应通过检验"                          8658
# ------------------------------------------------------------------------8659
      ------------------                                                  8660
# 2) 上三角输入 → 完整矩阵 → 一致性定位                                   8661
# ------------------------------------------------------------------------8662
      ------------------                                                  8663
print()                                                                   8664
print(LINE)                                                               8665
print("测试 2  上三角录入与最不一致项定位")                               8666
print(LINE)                                                               8667
# 一个不太一致的 3 阶矩阵：上三角 [2, 5, 3]（a12=2, a13=5, a23=3）        8668
M = ahp.matrix_from_upper([2, 5, 3])                                      8669
r2 = ahp.weights_and_consistency(M)                                       8670
bad = ahp.find_most_inconsistent(M)                                       8671
print("   由 [a12=2, a13=5, a23=3] 构造矩阵：")                          8672
for row in M:                                                             8673
      print("      " + "  ".join(f"{v:8.4f}" for v in row))               8674
print("   权重:", "  ".join(f"{v:.4f}" for v in r2["weights"]),           8675
      "  CR = %.3f  一致: %s" % (r2["cr"], r2["consistent"]))             8676
print("   最不一致比较项: 第(%d,%d) 项，当前 %.3f，理想 %.3f"             8677
      % (bad["i"] + 1, bad["j"] + 1, bad["value"],                        8678
      bad["expected"]))                                                   8679
# ------------------------------------------------------------------------8680
      ------------------                                                  8681
# 3) 多专家聚合（几何平均）                                              8682
# ------------------------------------------------------------------------8683
      ------------------                                                  8684
print()                                                                   8685
print(LINE)                                                               8686
print("测试 3  多专家判断矩阵聚合（几何平均）")                           8687
print(LINE)                                                               8688
m1 = ahp.matrix_from_upper([2, 4, 3])                                     8689
m2 = ahp.matrix_from_upper([3, 5, 2])                                     8690
agg = ahp.aggregate_matrices([m1, m2])                                    8691
print("   专家1 上三角 [2,4,3]，专家2 上三角 [3,5,2]")                  8692
print("   聚合后上三角: a12=%.4f a13=%.4f a23=%.4f"                      8693
      % (agg[0, 1], agg[0, 2], agg[1, 2]))                                8694
assert abs(agg[0, 1] - np.sqrt(2 * 3)) < 1e-9, "几何平均错误"             8695
# ------------------------------------------------------------------------8696
      ------------------                                                  8697
# 4) 论文表 3 权重可还原性                                               8698
# ------------------------------------------------------------------------8699
      ------------------                                                  8700
```

```
print()
print(LINE)
print("测试 4  论文表 3 碎石边坡一级权重 (0.44, 0.31, 0.25)")
print(LINE)
j = ahp.judge_from_weights([0.44, 0.31, 0.25])
print("  λmax = %.6f    CR = %.3e    一致: %s" %
      (j["lambda_max"], j["cr"], j["consistent"]))
assert j["consistent"], "论文权重应能作为完全一致矩阵主特征向量"
# ----------------------------------------------------------------
      ----------------
# 5) 应用可启动 + 建表
# ----------------------------------------------------------------
      ----------------
print()
print(LINE)
print("测试 5  Flask 应用启动与建表")
print(LINE)
from app import create_app
from app.models import db
app = create_app()
with app.app_context():
      db.create_all()
      tables = sorted(db.metadata.tables.keys())
print("  已创建 %d 张表:" % len(tables))
for t in tables:
      print("    -", t)
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
# ==== 测试/test_evaluate.py ====
# -*- coding: utf-8 -*-
"""评价引擎自检:用论文玉田县案例跑通完整计算链。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
# 本脚本位于 测试/ 子目录,先把仓库根目录加入模块搜索路径,
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
      e__)))
if ROOT not in sys.path:
      sys.path.insert(0, ROOT)
from app.services import evaluation_service as ev
from app.services import standardization as std
LINE = "=" * 70
# ----------------------------------------------------------------
      ----------------
# 论文第 5 节案例数据(表 5,逐字抄录)
# ----------------------------------------------------------------
      ----------------
```

```
#  三类单元面积占比：岩质边坡  １３．８６%、碎石边坡  ６２．９５%、平台  ２３．１９%
#  注：岩质边坡无土壤指标、平台无地形指标，对应"不适用"。
case = [
    {
        "unit_type": "岩质边坡",
        "ratio": 0.1386,
        "data": {
            "coverage": 68.88, "shannon": 0.00,
            "slope": 63.33, "aspect": "阴阳坡",
        },
    },
    {
        "unit_type": "碎石边坡",
        "ratio": 0.6295,
        "data": {
            "soil_ratio": 0.34, "thickness": 10, "ph":
    9.24, "organic_matter": 4.89,
            "heavy_metals": {"Hg": 0.05, "As": 5.17, "Ni":
    12.60,
                             "Cu": 11.70, "Cd": 0.04, "Pb":
    6.50},
            "coverage": 87.92, "shannon": 0.60,
            "slope": 28.75, "aspect": "阳坡",
        },
    },
    {
        "unit_type": "平台",
        "ratio": 0.2319,
        "data": {
            "soil_ratio": 0.56, "thickness": 30, "ph":
    8.72, "organic_matter": 10.92,
            "heavy_metals": {"Hg": 0.02, "As": 3.13, "Ni":
    12.33,
                             "Cu": 11.14, "Cd": 0.09, "Pb":
    9.98},
            "coverage": 58.42, "shannon": 0.72,
        },
    },
]
#  ----------------------------------------------------------------------
    ------------------
#  逐单元输出标准化与得分
#  ----------------------------------------------------------------------
    ------------------
print(LINE)
print("玉田县案例 · 评价引擎完整计算")
print(LINE)
for u in case:
    r = ev.score_unit(u["unit_type"], u["data"])
    print()
```

```
        print("   【%s】   （面积占比 %.2f%%）" % (u["unit_type"],
        u["ratio"] * 100))
        print("   " + "-" * 60)
        for l1, score in r["l1_scores"].items():
            print("      %s 得分 = %.4f" % (l1, score))
        print("      单元评价因子 F' = %.4f" % r["unit_score"])
print()
print(LINE)
print("面积加权合成（式(9)）")
print(LINE)
res = ev.score_mine(case)
total = res["final_score"]
print()
for ur in res["unit_results"]:
        print("   %s  %.2f%% × %.4f = %.4f"
              % (ur["unit_type"], ur["ratio"] * 100,
        ur["unit_score"], ur["weighted"]))
print("   " + "-" * 60)
print("   最终评价因子 F = %.4f" % total)
print()
print("   适宜性等级：%s" % res["grade"])
print("   判定依据   ：%s" % res["grade_info"]["meaning"])
print("   推荐措施   ：%s" % res["grade_info"]["measure"])
print()
print(LINE)
print("结论")
print(LINE)
print("   - 评价引擎按论文式(8)(9) 与表 3 权重完整跑通")
print("   - 坡度标准化采用论文 25° 分界的线性折减默认值（见 standardization.py）")
print("   - 最终 F ≈ %.2f，判定为「%s」，与论文「最终适宜自然恢复」的结论一致"
      % (total, res["grade"]))
print("   - 论文表 5 所载最终得分为 0.76；差异集中在坡度、坡向的标准化口径，")
print("     该口径在 standardization.std_slope / aspect_score
     中集中定义、可配置")
print(LINE)
#  ==== 测试/test_web_ahp.py ====
#  -*- coding: utf-8 -*-
"""AHP 网页界面自检：走通选择→录入→计算 全流程。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file
    __)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
#  测试配置：关闭 CSRF 以简化测试客户端（真实运行仍开启 CSRF）
```

```
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///:memory:"
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 建立表并创建测试用户，登录后再访问业务页面
with app.app_context():
    from app.models import db, User
    from werkzeug.security import generate_password_hash
    db.create_all()
    if not User.query.filter_by(username="admin").first():
        db.session.add(User(username="admin",
                            password_hash=generate_password_
    hash("admin123"),
                            role="admin"))
        db.session.commit()
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  首页与 AHP 入口")
print(LINE)
r = client.get("/")
print("  首页状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
r = client.get("/ahp/")
print("  AHP 入口状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
assert "微地质单元类型" in r.get_data(as_text=True)
print()
print(LINE)
print("测试 2  选择碎石边坡一级指标 → 跳转矩阵页")
print(LINE)
r = client.post("/ahp/", data={
    "unit_type": "碎石边坡", "level": "1", "submit": "生成判断矩阵",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转后页面含「判断矩阵录入」：", "判断矩阵录入" in body)
print("  含三个一级指标（土壤/植被/地形）：", all(n in body for n in
    ("土壤指标", "植被指标", "地形指标")))
assert "判断矩阵录入" in body
assert all(n in body for n in ("土壤指标", "植被指标", "地形指标"))
print()
print(LINE)
print("测试 3  提交论文表 3 一致矩阵 → 还原权重")
print(LINE)
# 碎石边坡一级权重 (0.44, 0.31, 0.25) 对应的完全一致矩阵上三角：
# a12 = 0.44/0.31 = 1.4194, a13 = 0.44/0.25 = 1.76, a23 =
```

```
    0.31/0.25 = 1.24
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  结果页含「计算结果」:", "计算结果" in body)
print("  含权重 0.44:", "0.44" in body, " 含 0.31:", "0.31" in
    body, " 含 0.25:", "0.25" in body)
print("  含一致性检验通过:", "通过" in body)
assert "计算结果" in body
assert "0.44" in body and "0.31" in body and "0.25" in body
print()
print(LINE)
print("测试 4  不一致矩阵应提示不通过并定位")
print(LINE)
# 明显不一致：a12=9, a13=1, a23=9
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "9", "m_0_2": "1", "m_1_2": "9",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  含「不通过」:", "不通过" in body)
print("  含「建议优先调整」:", "建议优先调整" in body)
assert "不通过" in body
assert "建议优先调整" in body
print()
print(LINE)
print("网页界面全部测试通过 √")
print(LINE)
# ==== 测试/test_web_modules.py ====
# -*- coding: utf-8 -*-
"""模块自检：登录、项目档案、微地质单元、标准库、限制性判定。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
    e__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
```

```
client = app.test_client()
print(LINE)
print("测试 1  未登录应被重定向到登录页")
print(LINE)
r = client.get("/project/")
print("  /project/ 状态码:", r.status_code, " 重定向到:",
      r.headers.get("Location"))
assert r.status_code == 302 and "login" in
      r.headers.get("Location", "")
print()
print(LINE)
print("测试 2  登录")
print(LINE)
r = client.post("/login", data={"username": "admin",
      "password": "admin123"},
                  follow_redirects=True)
body = r.get_data(as_text=True)
#  工作台:页首横幅 + 评价流程分区 + 数据概览
print("  登录后含「评价流程」:", "评价流程" in body)
print("  含用户名 admin:", "admin" in body)
print("  含页首横幅:", 'class="hero"' in body)
print("  含数据概览:", 'class="stat-card"' in body)
assert "评价流程" in body and 'class="hero"' in body
print()
print(LINE)
print("测试 3  项目与矿山档案")
print(LINE)
r = client.get("/project/")
body = r.get_data(as_text=True)
print("  项目列表状态码:", r.status_code)
print("  含案例项目:", "玉田县典型历史遗留废弃矿山案例" in body)
assert "玉田县典型历史遗留废弃矿山案例" in body
#  新建项目
client.post("/project/add", data={"name": "测试项目A", "code":
      "T-001"})
r = client.get("/project/")
print("  新建项目后含「测试项目A」:", "测试项目A" in r.get_data(as_text=True
      ))
assert "测试项目A" in r.get_data(as_text=True)
#  清理测试数据,避免污染演示库
with app.app_context():
      from app.models import db, Project
      for p in Project.query.filter_by(name="测试项目A").all():
            db.session.delete(p)
      db.session.commit()
print("  已清理测试项目")
print()
print(LINE)
print("测试 4  微地质单元(面积占比自动计算)")
print(LINE)
```

```
from app.models import MinePatch
with app.app_context():
    patch = MinePatch.query.filter_by(name="玉田县孤树镇白云岩采石场").first()
    pid = patch.id
r = client.get("/unit/%d" % pid)
body = r.get_data(as_text=True)
print("  单元列表含三类单元：", all(t in body for t in ("岩质边坡", "碎石边坡", "平台")))
print("  含占比合计：", "占比合计" in body)
assert "岩质边坡" in body and "占比合计" in body
print()
print(LINE)
print("测试 5  标准知识库与省际口径比对")
print(LINE)
r = client.get("/standard/")
body = r.get_data(as_text=True)
print("  含辽宁/湖南/新疆标准：", all(p in body for p in ("辽宁", "湖南", "新疆")))
assert "辽宁" in body
r = client.get("/standard/compare")
body = r.get_data(as_text=True)
print("  口径比对页含「植被覆盖度」分组：", "植被覆盖度" in body)
print("  含多省份并列：", "湖南" in body and "新疆" in body)
assert "植被覆盖度" in body
print()
print(LINE)
print("测试 6  限制性指标判定（一票否决）")
print(LINE)
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  含五项条件：", all(x in body for x in ("政策约束", "区位约束", "土壤污染", "地质灾害", "耕地属性")))
assert all(x in body for x in ("政策约束", "区位约束", "土壤污染", "地质灾害", "耕地属性"))
# 全部通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  全通过后显示「通过，可进入定量评价」：", "可进入定量评价" in body)
assert "可进入定量评价" in body
# 一项不通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "0",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
```

```
body = r.get_data(as_text=True)
print("  一项不通过后显示「不通过」:", "不通过" in body)
print("  含失败项「区位约束」:", "区位约束" in body)
assert "不通过" in body
# 恢复为全部通过，避免把演示数据留在「不通过」状态影响后续步骤
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
print("  已恢复为全部通过:", "可进入定量评价" in r.get_data(as_text=True))
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
# ==== 测试/test_web_evaluate.py ====
# -*- coding: utf-8 -*-
"""端到端自检：录数据 → 执行评价 → 查看结果。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
# 与真实配置共用同一个 app.db，但关闭 CSRF 便于测试客户端
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  指标数据列表")
print(LINE)
r = client.get("/indicator/")
body = r.get_data(as_text=True)
print("  状态码:", r.status_code)
print("  含案例图斑:", "玉田县孤树镇白云岩采石场" in body)
print("  含三个单元:", all(n in body for n in ("岩质边坡", "碎石边坡",
```

```
        "平台"))) 9101
assert r.status_code == 200 9102
assert "岩质边坡" in body and "碎石边坡" in body and "平台" in body 9103
print() 9104
print(LINE) 9105
print("测试 2  指标数据录入页（按单元类型显示字段）") 9106
print(LINE) 9107
from app.models import MicroUnit, MinePatch 9108
with app.app_context(): 9109
    unit_rock = MicroUnit.query.filter_by(unit_type="岩质边坡"). 9110
    first() 9111
    unit_gravel = MicroUnit.query.filter_by(unit_type="碎石边坡" 9112
    ).first() 9113
r = client.get("/indicator/%d" % unit_rock.id) 9114
b_rock = r.get_data(as_text=True) 9115
print("  岩质边坡页：无「土石比」字段：", "土石比" not in b_rock, "（应 True）") 9116
print("  岩质边坡页：含「植被覆盖度」：", "植被覆盖度" in b_rock) 9117
assert "土石比" not in b_rock and "植被覆盖度" in b_rock 9118
r = client.get("/indicator/%d" % unit_gravel.id) 9119
b_gravel = r.get_data(as_text=True) 9120
print("  碎石边坡页：含「土石比」字段：", "土石比" in b_gravel) 9121
print("  碎石边坡页：含 8 种重金属：", all(m in b_gravel for m in 9122
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn"))) 9123
assert "土石比" in b_gravel and all(m in b_gravel for m in 9124
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn")) 9125
print() 9126
print(LINE) 9127
print("测试 3  执行评价") 9128
print(LINE) 9129
with app.app_context(): 9130
    import seed_demo 9131
    patch = seed_demo.demo_patch() 9132
r = client.post("/evaluate/%d" % patch.id, data={}, 9133
    follow_redirects=True) 9134
body = r.get_data(as_text=True) 9135
print("  跳转至结果页：", "适宜性评价结果" in body) 9136
print("  含最终评价因子 F：", "最终评价因子 F" in body) 9137
print("  含等级「适宜」：", "适宜" in body) 9138
print("  含推荐措施「自然恢复」：", "自然恢复" in body) 9139
assert "最终评价因子 F" in body and "自然恢复" in body 9140
print() 9141
print(LINE) 9142
print("测试 4  结果持久化") 9143
print(LINE) 9144
from app.models import EvaluationResult 9145
with app.app_context(): 9146
    res = EvaluationResult.query.order_by(EvaluationResult.i 9147
    d.desc()).first() 9148
    print("  最终得分 F =", res.final_score) 9149
    print("  等级 =", res.grade) 9150
```

```
    assert res.grade == "适宜"
    assert 0.7 <= res.final_score < 1.0
print()
print(LINE)
print("端到端全部测试通过 √")
print(LINE)
#  ==== 测试/test_web_report.py ====
#  -*- coding: utf-8 -*-
"""报告输出自检：得分明细页 + Word 导出。"""
import os
import sys
import io
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file
    __)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
#  登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
from app.models import MinePatch, EvaluationResult
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
    pid = patch.id
#  报告取自已保存的评价记录，因此先按网页路径真实执行一次评价
from app.views.evaluate import load_patch_input,
    save_evaluation
with app.app_context():
    units = load_patch_input(pid)
    rec = save_evaluation(pid, units)
    rid = rec.id
    print("前置：已执行评价并保存记录 id=%d，F=%.4f（%s）"
          % (rid, rec.final_score, rec.grade))
    from app.models import ScoreDetail
    n_detail = ScoreDetail.query.filter_by(result_id=rid).co
    unt()
```

```
        print("          逐层得分明细已落库：%d 条" % n_detail)                    9201
print()                                                                          9202
print(LINE)                                                                      9203
print("测试 1  报告列表")                                                        9204
print(LINE)                                                                      9205
r = client.get("/report/")                                                       9206
print("  状态码：", r.status_code)                                              9207
body = r.get_data(as_text=True)                                                  9208
print("  含「所属项目」列：", "所属项目" in body)                                9209
assert r.status_code == 200                                                      9210
print()                                                                          9211
print(LINE)                                                                      9212
print("测试 2  得分明细页")                                                      9213
print(LINE)                                                                      9214
r = client.get("/report/%d" % pid)                                               9215
body = r.get_data(as_text=True)                                                  9216
print("  状态码：", r.status_code)                                              9217
print("  含「得分明细」：", "得分明细" in body)                                  9218
print("  含「最终评价因子 F」：", "最终评价因子 F" in body)                      9219
print("  含二级指标明细（植被覆盖度/坡度/坡向）：", all(n in body for n in      9220
      ("植被覆盖度", "坡度", "坡向")))                                           9221
print("  含权重/标准化值列：", "标准化值" in body and "贡献" in body)            9222
assert r.status_code == 200                                                      9223
assert "植被覆盖度" in body and "标准化值" in body                               9224
print()                                                                          9225
print(LINE)                                                                      9226
print("测试 3  Word 报告导出")                                                   9227
print(LINE)                                                                      9228
r = client.get("/report/%d/word" % pid)                                          9229
print("  状态码：", r.status_code)                                              9230
print("  Content-Type:", r.headers.get("Content-Type",                           9231
      "")[:60])                                                                  9232
data = r.data                                                                    9233
print("  文件大小：", len(data), "字节")                                         9234
print("  是否为有效 docx（PK 头）：", data[:2] == b"PK")                         9235
assert r.status_code == 200 and data[:2] == b"PK"                                9236
# 用 python-docx 打开验证内容                                                   9237
import docx                                                                      9238
d = docx.Document(io.BytesIO(data))                                              9239
text = "\n".join(p.text for p in d.paragraphs)                                   9240
print("  含报告标题：", "自然恢复适宜性评价报告" in text)                        9241
print("  含评价结论章节：", "评价结论" in text)                                  9242
print("  含得分明细章节：", "得分明细" in text)                                  9243
print("  段落数：", len(d.paragraphs), " 表格数：", len(d.tables))               9244
assert "自然恢复适宜性评价报告" in text and len(d.tables) >= 3                   9245
print()                                                                          9246
print(LINE)                                                                      9247
print("报告输出全部测试通过 √")                                                 9248
print(LINE)                                                                      9249
# ==== 测试/test_ui_fixes.py ====                                               9250
```

```
# -*- coding: utf-8 -*-
"""界面修复自检：「记住我」与各模块删除功能。"""
import os
import re
import sys
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
sys.stdout.reconfigure(encoding="utf-8")
LINE = "=" * 70
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    REMEMBER_COOKIE_DURATION = __import__("datetime").timedelta(days=30)
app = create_app(TestConfig)
client = app.test_client()
# ------------------------------------------------------------------
print(LINE)
print("一、「记住我」")
print(LINE)
html = client.get("/login").get_data(as_text=True)
# 1) 复选框必须有可点击关联的 label（这是原先失效的根因）
wrapped = re.search(r'<label[^>]*class="checkbox"[^>]*>\s*<input[^>]*name="remember"',
                    html) is not None
print("  %s 「记住我」文字与复选框包在同一 label 内" % ("[符合]" if wrapped
      else "[不符合]"))
assert wrapped, "复选框未与文字关联，点击文字不会勾选"
has_for = 'for="remember"' in html
print("  %s label 带 for 属性指向复选框" % ("[符合]" if has_for else
      "[不符合]"))
# 2) 勾选后应下发带过期时间的 remember_token
c = app.test_client()
r = c.post("/login", data={"username": "admin", "password":
      "admin123",
                           "remember": "y"},
      follow_redirects=False)
cookies = r.headers.getlist("Set-Cookie")
rt = [h for h in cookies if h.startswith("remember_token=")]
print("  %s 勾选后下发 remember_token" % ("[符合]" if rt else
```

```
        "[不符合]"))                                                    9301
assert rt, "未下发 remember_token"                                      9302
exp = re.search(r"Expires=([^;]+)", rt[0])                              9303
print("   %s 该 cookie 带过期时间：%s"                                  9304
      % ("[符合]" if exp else "[不符合]", exp.group(1) if exp           9305
        else "无（会话 cookie）"))                                      9306
assert exp, "remember_token 无过期时间，关浏览器即失效"                 9307
# 3) 不勾选时不应下发 remember_token                                    9308
c2 = app.test_client()                                                  9309
r2 = c2.post("/login", data={"username": "admin",                       9310
        "password": "admin123"},                                        9311
                follow_redirects=False)                                 9312
rt2 = [h for h in r2.headers.getlist("Set-Cookie") if                   9313
        h.startswith("remember_token=")]                                9314
print("   %s 未勾选时不下发 remember_token" % ("[符合]" if not rt2      9315
        else "[不符合]"))                                               9316
assert not rt2                                                          9317
# 4) 携带 remember_token 的新会话应自动登录                             9318
c3 = app.test_client()                                                  9319
c3.set_cookie("remember_token",                                         9320
              rt[0].split(";")[0].split("=", 1)[1],                     9321
        domain="localhost")                                             9322
r3 = c3.get("/", follow_redirects=False)                                9323
print("   %s 仅凭 remember_token 访问首页可自动登录（状态码 %d）"       9324
      % ("[符合]" if r3.status_code == 200 else "[不符合]",             9325
        r3.status_code))                                                9326
assert r3.status_code == 200                                            9327
# ------------------------------------------------------------------    9328
        ----------------                                                9329
print()                                                                 9330
print(LINE)                                                             9331
print("二、删除功能")                                                   9332
print(LINE)                                                             9333
client.post("/login", data={"username": "admin",                        9334
        "password": "admin123"})                                        9335
# 项目：列表页须有删除按钮                                              9336
html = client.get("/project/").get_data(as_text=True)                   9337
has_del = "/delete" in html and "btn--danger" in html                   9338
print("   %s 项目列表含删除按钮" % ("[符合]" if has_del else "[不符合]"))9339
assert has_del, "项目列表缺少删除按钮"                                  9340
print("   %s 删除前有二次确认" % ("[符合]" if "confirm(" in html else   9341
        "[不符合]"))                                                    9342
assert "confirm(" in html                                               9343
# 图斑列表                                                              9344
with app.app_context():                                                 9345
        from app.models import MinePatch, Project                       9346
        import seed_demo                                                9347
        patch = seed_demo.demo_patch()                                  9348
        patch_id = patch.id                                             9349
html = client.get("/project/%d/patches" %                               9350
```

```
        patch_id).get_data(as_text=True)                                    9351
print("  %s 图斑列表含删除按钮" % ("[符合]" if "/project/patch/" in        9352
        html and "btn--danger" in html else "[不符合]"))                    9353
assert "btn--danger" in html                                                9354
#  微地质单元                                                               9355
html = client.get("/unit/%d" % patch_id).get_data(as_text=Tr                9356
        ue)                                                                 9357
print("  %s 微单元列表含删除按钮" % ("[符合]" if "/unit/delete/" in        9358
        html else "[不符合]"))                                              9359
assert "/unit/delete/" in html                                              9360
print("  %s 微单元编辑为独立页面（表单不再嵌在表格行内）"                  9361
      % ("[符合]" if "/unit/edit/" in html else "[不符合]"))                9362
#  标准                                                                     9363
html = client.get("/standard/").get_data(as_text=True)                      9364
print("  %s 标准列表含删除按钮" % ("[符合]" if "btn--danger" in html       9365
        else "[不符合]"))                                                   9366
assert "btn--danger" in html                                                9367
#  实际执行一次删除：新建项目 → 删除 → 确认消失                            9368
client.post("/project/add", data={"name": "待删除测试项目",                 9369
        "code": "DEL-1"})                                                   9370
with app.app_context():                                                     9371
        from app.models import db, Project                                  9372
        tmp = Project.query.filter_by(name="待删除测试项目").first()        9373
        tmp_id = tmp.id                                                     9374
html = client.get("/project/").get_data(as_text=True)                       9375
print("  %s 新建项目成功" % ("[符合]" if "待删除测试项目" in html else      9376
        "[不符合]"))                                                        9377
r = client.post("/project/%d/delete" % tmp_id,                              9378
        follow_redirects=True)                                              9379
html = r.get_data(as_text=True)                                             9380
gone = "待删除测试项目" not in html                                         9381
print("  %s 删除项目后列表中不再出现" % ("[符合]" if gone else "[不符合]"))  9382
assert gone                                                                 9383
print("  %s 有删除成功提示" % ("[符合]" if "已删除" in html else            9384
        "[不符合]"))                                                        9385
print()                                                                     9386
print(LINE)                                                                 9387
print("界面修复自检全部通过 √")                                            9388
print(LINE)                                                                 9389
#  ==== 测试/test_new_features.py ====                                      9390
#  -*- coding: utf-8 -*-                                                    9391
"""补齐模块自检：操作日志与用户维护、遥感影像、指标与多专家矩阵、          9392
批量导入、图表与 PDF 导出。                                                 9393
对应计划书功能清单与系统实现的差异核对结论。                               9394
"""                                                                         9395
import io                                                                   9396
import json                                                                 9397
import os                                                                   9398
import re                                                                   9399
import sys                                                                  9400
```

```
sys.stdout.reconfigure(encoding="utf-8")
import fitz
#  本脚本位于  测试/  子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, User, Project, MinePatch,
    MicroUnit, IndicatorData,
                        Indicator, ExpertMatrix,
    OperationLog)
from app.utils.schema import ensure_schema
from app.utils import browser, charts
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    UPLOAD_FOLDER = os.path.join(BASE, "app", "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
app = create_app(TestConfig)
with app.app_context():
    db.create_all()
    ensure_schema(db)
    ExpertMatrix.query.delete()
    db.session.commit()
    #  固定取内置演示案例的图斑，避免受库中用户自建项目  /  验证算例的影响
    import seed_demo
    patch_id = seed_demo.demo_patch().id
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
def ok(flag, label, extra=""):
    print("  %s %-44s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    assert flag, label
# ------------------------------------------------------------------
    ----------------
print(LINE)
print("操作日志与用户维护")
print(LINE)
```

```
before = None
with app.app_context():
    before = OperationLog.query.count()
c.post("/project/add", data={"name": "日志核对项目", "code":
    "LOG-T"})
with app.app_context():
    after = OperationLog.query.count()
    last = OperationLog.query.order_by(OperationLog.id.desc(
    )).first()
ok(after == before + 1, "写操作被写入操作日志", "%d -> %d" % (before,
    after))
ok(last.action == "新建项目", "日志记录了可读动作名", last.action)
ok("project" in (last.target_type or ""), "日志记录了对象类型",
    last.target_type)
r = c.get("/logs")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "操作日志" in body, "操作日志查看页可访问")
ok("新建项目" in body, "日志页显示操作记录")
# 用户编辑 / 删除
with app.app_context():
    u = User(username="tmp_user", role="viewer",
             password_hash=User.query.first().password_hash)
    db.session.add(u); db.session.commit()
    uid = u.id
r = c.get("/users/%d/edit" % uid)
ok(r.status_code == 200, "用户编辑页可访问")
c.post("/users/%d/edit" % uid, data={"role": "evaluator",
    "active": "1"})
with app.app_context():
    ok(db.session.get(User, uid).role == "evaluator",
    "角色已修改并保存")
# 最后一个管理员保护
with app.app_context():
    admin_id = User.query.filter_by(role="admin").first().id
c.post("/users/%d/edit" % admin_id, data={"role": "viewer",
    "active": "1"})
with app.app_context():
    ok(db.session.get(User, admin_id).role == "admin",
    "拒绝把最后一个管理员降级")
c.post("/users/%d/delete" % uid, follow_redirects=True)
with app.app_context():
    ok(db.session.get(User, uid) is None, "用户删除生效")
# 清理日志核对用的项目
with app.app_context():
    for p in Project.query.filter_by(name="日志核对项目").all():
        db.session.delete(p)
    db.session.commit()
# ------------------------------------------------------------
    ----------------
print()
```

```
print(LINE)
print("遥感影像挂接")
print(LINE)
doc = fitz.open()
pg = doc.new_page(width=320, height=200)
pg.insert_text((20, 60), "RS image", fontname="china-s",
    fontsize=16)
png = doc[0].get_pixmap(dpi=100).tobytes("png")
doc.close()
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(png), "影像.png"),
                 "image_note": "2025年6月无人机正射影像"},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("已挂接" in body and "<img" in body, "影像上传并显示")
with app.app_context():
    rel = db.session.get(MinePatch, patch_id).image_path
ok(bool(rel), "数据库记录影像路径", rel)
abs_path = os.path.join(TestConfig.UPLOAD_FOLDER,
    rel.replace("uploads/", "", 1))
ok(os.path.isfile(abs_path), "影像文件落盘")
ok(c.get("/static/" + rel).status_code == 200, "影像可经 URL
    访问")
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(b"x"), "bad.txt")},
           content_type="multipart/form-data",
    follow_redirects=True)
ok("不支持" in r.get_data(as_text=True), "拒绝非图片格式")
c.post("/project/patch/%d/image/delete" % patch_id,
    follow_redirects=True)
with app.app_context():
    ok(db.session.get(MinePatch, patch_id).image_path is
    None, "影像移除后字段清空")
ok(not os.path.isfile(abs_path), "影像文件已删除")
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("指标定义维护与多专家矩阵")
print(LINE)
r = c.get("/ahp/indicators")
body = r.get_data(as_text=True)
with app.app_context():
    n_ind = Indicator.query.count()
ok(r.status_code == 200 and n_ind > 0, "指标定义已入库并可查看", "%d
    项" % n_ind)
ok(all(k in body for k in ("一级指标", "二级指标", "三级指标")),
   "指标树按层级展示")
ok("岩质边坡" in body and "平台" in body, "显示各指标启用的微地质单元")
```

```
c.post("/ahp/indicators/coverage/edit",
          data={"name": "植被覆盖度", "unit": "%", "ideal_type":
      "benefit",
                "ideal_value": "75", "enabled_units": ["岩质边坡"],
                "description": "自检修改"})
with app.app_context():
      ok(Indicator.query.filter_by(code="coverage").first().de
      scription == "自检修改",
          "指标定义可编辑")
c.post("/ahp/indicators/reset")
with app.app_context():
      ok("自检修改" not in (Indicator.query.filter_by(code="covera
      ge").first().description or ""),
          "可按论文取值重置")
base = {"m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24"}
for name in ("张工", "李工", "王工"):
      d = dict(base); d.update({"action": "add", "unit_type":
      "碎石边坡",
                                "level": "1", "expert_name":
      name})
      c.post("/ahp/experts", data=d)
with app.app_context():
      ok(ExpertMatrix.query.filter_by(unit_type="碎石边坡",
      level=1).count() == 3,
          "录入 3 位专家的判断矩阵")
r = c.get("/ahp/experts/aggregate?unit_type=碎石边坡&level=1")
body = r.get_data(as_text=True)
ok("汇总后权重" in body and "各专家矩阵的一致性" in body,
    "多专家矩阵几何平均汇总可用")
ws = re.search(r"<th scope=\"row\">权重</th>\s*(.*?)</tr>",
      body, re.S)
vals = re.findall(r">([0-9.]+)</td>", ws.group(1)) if ws
      else []
ok(len(vals) == 3 and abs(sum(map(float, vals)) - 1.0) <
      0.01,
    "汇总权重归一", "、".join(vals))
ok(abs(float(vals[0]) - 0.44) < 0.02, "汇总权重接近论文表 3 取值",
      vals[0] if vals else "")
# ----------------------------------------------------------------
      ------------------
print()
print(LINE)
print("批量导入")
print(LINE)
r = c.get("/indicator/import/template")
tpl = r.data.decode("utf-8-sig")
ok(r.status_code == 200 and "单元类型" in tpl, "导入模板可下载")
ok("土石比" in tpl and "植被覆盖度" in tpl, "模板包含指标列")
csv_text = (
      "单元类型,土石比,土层厚度,pH,有机质,Hg,As,Ni,Cu,Cd,Pb,植被覆盖度,Shannon,坡度
```

```
        ,坡向\n"                                                                    9601
        "碎石边坡,0.34,10,9.24,4.89,0.05,5.17,12.60,11.70,0.04,6.50,               9602
        87.92,0.60,28.75,阳坡\n"                                                    9603
        "平台,0.56,30,8.72,10.92,0.02,3.13,12.33,11.14,0.09,9.98,5                  9604
        8.42,0.72,,,\n"                                                             9605
        "不存在单元,1,1,7,1,,,,,,,,,,\n"                                           9606
)                                                                                   9607
r = c.post("/indicator/import",                                                     9608
                data={"patch_id": str(patch_id),                                    9609
                      "datafile": (io.BytesIO(csv_text.encode("ut                   9610
        f-8")), "d.csv")},                                                          9611
                content_type="multipart/form-data",                                 9612
        follow_redirects=True)                                                      9613
body = r.get_data(as_text=True)                                                     9614
ok("导入结果" in body, "导入结果页展示")                                              9615
ok("不存在单元" in body, "错误行被报告")                                              9616
with app.app_context():                                                             9617
        u = MicroUnit.query.filter_by(patch_id=patch_id,                            9618
        unit_type="碎石边坡").first()                                                9619
        ind = IndicatorData.query.filter_by(unit_id=u.id).first(                    9620
        )                                                                           9621
        ok(abs(ind.soil_ratio - 0.34) < 1e-6 and                                    9622
        abs(ind.coverage - 87.92) < 1e-6,                                           9623
            "导入数据已写入数据库")                                                  9624
r = c.post("/indicator/import",                                                     9625
                data={"patch_id": str(patch_id),                                    9626
                      "datafile": (io.BytesIO(b"x"), "d.pdf")},                     9627
                content_type="multipart/form-data",                                 9628
        follow_redirects=True)                                                      9629
ok("仅支持" in r.get_data(as_text=True), "拒绝不支持的导入格式")                      9630
# ------------------------------------------------------------------------------    9631
        ------------------                                                          9632
print()                                                                             9633
print(LINE)                                                                         9634
print("图表与 PDF 导出")                                                            9635
print(LINE)                                                                         9636
svg = charts.hbars([("岩质边坡", 0.54, "0.5401"), ("平台", 0.84,                    9637
        "0.8376")])                                                                 9638
ok("<svg" in svg and "0.8376" in svg, "条形图生成且带数值标签")                      9639
svg2 = charts.donut([("植被指标", 0.6, "0.6000"), ("土壤指标", 0.4,                  9640
        "0.4000")])                                                                 9641
ok("<svg" in svg2 and "占比" in svg2, "环形图生成且带图例")                          9642
# 报告是评价结论的呈现，需先真实执行一次评价才能打开                                  9643
with app.app_context():                                                             9644
        from app.views.evaluate import load_patch_input,                            9645
        save_evaluation                                                             9646
        save_evaluation(patch_id, load_patch_input(patch_id))                       9647
r = c.get("/report/%d" % patch_id)                                                  9648
body = r.get_data(as_text=True)                                                     9649
ok(body.count("<svg") >= 2, "得分明细页含图表", "%d 个 SVG" %                        9650
```

```
    body.count("<svg"))
exe = browser.browser_path()
if exe is None:
    print("  [跳过] 未找到浏览器，PDF 导出无法验证")
else:
    r = c.get("/report/%d/pdf" % patch_id)
    data = r.data
    ok(data[:4] == b"%PDF", "PDF 报告导出成功", "%d 字节" %
    len(data))
    d = fitz.open(stream=data, filetype="pdf")
    txt = "".join(d[i].get_text() for i in
    range(d.page_count))
    ok("自然恢复适宜性评价报告" in txt, "PDF 含报告标题")
    ok("评价结论" in txt and "得分明细" in txt, "PDF 含结论与明细章节")
    ok(d.page_count >= 2, "PDF 页数合理", "%d 页" % d.page_count)
    d.close()
print()
print(LINE)
print("补齐模块自检全部通过 √")
print(LINE)
#  ==== 测试/test_workflow.py ====
#  -*- coding: utf-8 -*-
"""流程门禁自检：前置步骤未完成时，后续模块必须拦得住。
论文 4.1 节确定的评价流程存在先后关系，缺任何一环都不应产出评价结论。
本测试逐个构造「缺前置」的场景，验证系统不会放行。
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file
    __)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult)
from app.services import workflow as wf
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
```

```
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
def new_patch(tag):
    with app.app_context():
        p = Project(name="门禁-" + tag, code="WF-" + tag)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="门禁图斑-" +
    tag)
        db.session.add(patch); db.session.flush()
        pid = patch.id
        db.session.commit()
    return pid
def add_units(pid):
    with app.app_context():
        db.session.add(MicroUnit(patch_id=pid,
    unit_type="碎石边坡",
                                 area=1.0, ratio=1.0))
        db.session.commit()
def add_constraint(pid, passed=True):
    with app.app_context():
        db.session.add(ConstraintCheck(
            patch_id=pid, policy_pass=True,
    location_pass=passed,
            pollution_pass=True, geo_pass=True,
    farmland_pass=True,
            passed=passed, reason="全部通过" if passed else
    "未通过：区位约束"))
        db.session.commit()
def add_data(pid):
    with app.app_context():
        u = MicroUnit.query.filter_by(patch_id=pid).first()
        db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.34, thickness=10,
                                     ph=9.24,
    organic_matter=4.89, coverage=87.92,
                                     shannon=0.60,
    slope=28.75, aspect="阳坡"))
        db.session.commit()
def cleanup(pid, tag):
    with app.app_context():
        from app.models import ScoreDetail
        #  先删得分明细，再删评价记录，避免留下孤儿明细
        rids = [r.id for r in EvaluationResult.query.filter_
```

```
    by(patch_id=pid)]
        if rids:
            ScoreDetail.query.filter(ScoreDetail.result_id.i
    n_(rids)).delete(
                synchronize_session=False)
        EvaluationResult.query.filter_by(patch_id=pid).delet
    e()
        ConstraintCheck.query.filter_by(patch_id=pid).delete
    ()
        uids = [u.id for u in MicroUnit.query.filter_by(patc
    h_id=pid)]
        if uids:
            IndicatorData.query.filter(IndicatorData.unit_id
    .in_(uids)).delete(
                synchronize_session=False)
        MicroUnit.query.filter_by(patch_id=pid).delete()
        m = db.session.get(MinePatch, pid)
        if m:
            db.session.delete(m)
        pr = Project.query.filter_by(name="门禁-" +
    tag).first()
        if pr:
            db.session.delete(pr)
        db.session.commit()
def results_of(pid):
    with app.app_context():
        return EvaluationResult.query.filter_by(patch_id=pid
    ).count()
def post_eval(pid):
    return c.post("/evaluate/%d" % pid, data={},
    follow_redirects=False)
def get_report(pid):
    return c.get("/report/%d" % pid, follow_redirects=False)
# ----------------------------------------------------------------
    ----------------
print(LINE)
print("一、前置条件缺失时，评价必须被拦住")
print(LINE)
print()
print("场景 1：连微地质单元都没划分")
pid = new_patch("1")
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未划分单元 → 不产生评价结果",
   "HTTP %d，结果数 %d" % (r.status_code, n))
with app.app_context():
    pr = wf.patch_progress(pid)
steps = [b["step"] for b in pr["blocking"]]
ok(not pr["ready"] and "units" in steps,
   "拦原因包含「划分微地质单元」", "、".join(steps))
```

```
ok(steps == ["constraint", "units"],
    "拦截顺序符合论文 4.1 节：限制性判定在前，划分单元在后",
    "、".join(steps))
cleanup(pid, "1")
print()
print("场景 2：有单元，但未做限制性条件判定")
pid = new_patch("2"); add_units(pid); add_data(pid)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未做限制性判定 → 不产生评价结果",
    "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "constraint" for b in pr["blocking"]),
    "拦原因指向「执行限制性条件判定」")
cleanup(pid, "2")
print()
print("场景 3：限制性条件判定【未通过】")
pid = new_patch("3"); add_units(pid); add_data(pid);
    add_constraint(pid, passed=False)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "判定未通过 → 不进入定量评价",
    "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(not pr["constraint"]["passed"], "流程状态显示判定未通过")
blocked = [b for b in pr["blocking"] if b["step"] ==
    "constraint"]
ok(bool(blocked) and "人工辅助" in blocked[0]["reason"],
    "拦原因说明了按论文应判「人工辅助或生态重建」")
cleanup(pid, "3")
print()
print("场景 4：判定通过，但指标数据全空")
pid = new_patch("4"); add_units(pid); add_constraint(pid,
    passed=True)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "数据未录入 → 不产生评价结果",
    "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "data" for b in pr["blocking"]),
    "拦原因指向「录入指标数据」")
ok(pr["data"]["missing"] == ["碎石边坡"],
    "指出具体缺哪个单元的数据", "、".join(pr["data"]["missing"]))
cleanup(pid, "4")
print()
print("场景 5：单元面积占比合计不为 100%")
pid = new_patch("5"); add_constraint(pid, passed=True)
```

```
with app.app_context():
    u = MicroUnit(patch_id=pid, unit_type="碎石边坡", area=1.0,
    ratio=0.5)
    db.session.add(u); db.session.commit()
    db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.3, thickness=10,
                                 ph=8.0,
    organic_matter=5.0, coverage=80.0,
                                 shannon=0.6, slope=20.0,
    aspect="阳坡"))
    db.session.commit()
r = post_eval(pid)
ok(results_of(pid) == 0, "占比不足 100% → 不产生评价结果")
cleanup(pid, "5")
print()
print(LINE)
print("二、前置条件齐备时才放行")
print(LINE)
pid = new_patch("OK"); add_units(pid); add_constraint(pid,
    passed=True); add_data(pid)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(pr["ready"], "单元 + 判定通过 + 数据齐备 → 判定为可执行")
r = post_eval(pid)
n = results_of(pid)
ok(n == 1, "此时执行评价成功产生结果", "结果数 %d" % n)
print("  重定向到:", r.headers.get("Location"))
print()
print(LINE)
print("三、报告不得先于评价产生")
print(LINE)
pid2 = new_patch("R")    # 什么都不做
r = get_report(pid2)
ok(r.status_code == 302 and "/report/" ==
    (r.headers.get("Location") or ""),
   "前置缺失时 /report/<id> 被退回列表", "HTTP %d" % r.status_code)
for suffix in ("/pdf", "/word"):
    rr = c.get("/report/%d%s" % (pid2, suffix),
    follow_redirects=False)
    ok(rr.status_code == 302, "导出 %s 同样被拦住" % suffix, "HTTP
    %d" % rr.status_code)
cleanup(pid2, "R")
cleanup(pid, "OK")
print()
print(LINE)
print("四、前置齐备但从未执行评价时，报告仍不得产生")
print(LINE)
#  报告是评价结论的呈现：即使前置步骤都做完了，只要没真正执行过定量评价，
#  也不应凭空推算出结论。
pid3 = new_patch("N")
```

```
add_units(pid3); add_constraint(pid3, passed=True);
    add_data(pid3)
with app.app_context():
    pr = wf.report_progress(pid3)
ok(not pr["ready"], "无评价记录 → 报告流程判定为未就绪")
ok(bool(pr["blocking"]) and pr["blocking"][0]["step"] ==
    "evaluate",
   "拦原因指向「执行适宜性评价」")
r = get_report(pid3)
ok(r.status_code == 302 and results_of(pid3) == 0,
   "直接访问 /report/<id> 被退回，且不产生任何评价结果",
   "HTTP %d" % r.status_code)
# 真正执行评价之后，报告才放行，且取值与评价记录一致
r = post_eval(pid3)
ok(results_of(pid3) == 1, "先执行评价 → 产生评价记录")
with app.app_context():
    pr = wf.report_progress(pid3)
    rec = pr["result"]
ok(pr["ready"] and rec is not None, "有评价记录后，报告流程就绪")
rr = get_report(pid3)
body = rr.get_data(as_text=True)
ok(rr.status_code == 200, "此时可打开得分明细页", "HTTP %d" %
    rr.status_code)
ok(("%.4f" % rec.final_score) in body,
   "报告数值取自评价记录，而非重新计算",
   "F=%.4f" % rec.final_score)
cleanup(pid3, "N")
print()
print(LINE)
print("五、内置演示数据的清理不得波及用户自建数据")
print(LINE)
# 回归保护：seed_demo.clean_case() 曾执行 Project.query.delete(),
# 导致用户新建的项目在每次启动时被连带删除。
import seed_demo as sd
with app.app_context():
    keep_p = Project(name="用户自建项目-守护", code="KEEP-001")
    db.session.add(keep_p); db.session.flush()
    keep_patch = MinePatch(project_id=keep_p.id,
    name="用户自建图斑-守护")
    db.session.add(keep_patch); db.session.flush()
    keep_pid = keep_p.id
    keep_patch_id = keep_patch.id
    db.session.commit()
with app.app_context():
    sd.clean_case(reset=False)
with app.app_context():
    still_p = db.session.get(Project, keep_pid)
    still_patch = db.session.get(MinePatch, keep_patch_id)
ok(still_p is not None, "清理演示数据后，用户自建项目仍存在")
ok(still_patch is not None, "清理演示数据后，用户自建图斑仍存在")
```

```
# 演示数据清掉后应能重新写入，保证演示环境可恢复
sd.seed()
with app.app_context():
    demo_back = Project.query.filter_by(code=sd.DEMO_CODE).f
    irst()
ok(demo_back is not None, "演示案例可重新写入（幂等恢复）")
with app.app_context():
    m = db.session.get(MinePatch, keep_patch_id)
    if m:
        db.session.delete(m)
    p = db.session.get(Project, keep_pid)
    if p:
        db.session.delete(p)
    db.session.commit()
print()
print(LINE)
print("六、同一图斑不得重复划分同类型微地质单元")
print(LINE)
# 论文 4.1 节把图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
# 允许重复会让面积占比、按类型组织的指标数据与批量导入对不上号。
pid4 = new_patch("D")
c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "1.0"},
       follow_redirects=True)
with app.app_context():
    n1 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n1 == 1, "首次添加「平台」成功", "单元数 %d" % n1)
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "2.0"},
           follow_redirects=True)
with app.app_context():
    n2 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n2 == 1, "重复添加同类型被拒绝", "单元数 %d" % n2)
ok("同一类型只能划分一个" in r.get_data(as_text=True), "给出重复提示")
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "火山口",
    "area": "1.0"},
           follow_redirects=True)
with app.app_context():
    n3 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n3 == 1, "非法单元类型被拒绝", "单元数 %d" % n3)
c.post("/unit/%d/add" % pid4, data={"unit_type": "碎石边坡",
    "area": "3.0"},
       follow_redirects=True)
with app.app_context():
    uid = MicroUnit.query.filter_by(patch_id=pid4,
                                    unit_type="碎石边坡").first(
    ).id
r = c.post("/unit/edit/%d" % uid,
           data={"unit_type": "平台", "area": "3.0"},
    follow_redirects=True)
```

```
with app.app_context():
    types = sorted(u.unit_type for u in
                   MicroUnit.query.filter_by(patch_id=pid4)
    all())
ok(types == ["平台", "碎石边坡"], "改类型时同样拦截重复", "、".join(types))
cleanup(pid4, "D")
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("流程门禁自检全部通过 √")
print(LINE)
#  ==== 测试/test_weights.py ====
#  -*- coding: utf-8 -*-
"""权重生效链路自检：AHP 算出的权重必须真正参与评价计算。
此前 weight_schemes / weight_items 两张表从未被写入，评价引擎也只用
代码里的论文表 3 默认权重，相当于权重模块算完即弃。本测试守住这条链路：
    默认 → 论文表 3
    保存 AHP 权重 → 评价改用新权重，结果随之变化
    恢复默认 → 结果回到论文取值
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    WeightScheme,
                        WeightItem)
from app.services import workflow as wf
from app.services import weight_service as ws
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
```

```
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
TAG = "W"
UNIT_TYPE = "碎石边坡"
def build_case():
    """建一个只含单一微地质单元的图斑，便于观察权重变化对得分的影响。"""
    with app.app_context():
        p = Project(name="权重测试-%s" % TAG, code="WEIGHT-%s"
    % TAG)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="权重测试图斑-%s"
    % TAG)
        db.session.add(patch); db.session.flush()
        db.session.add(MicroUnit(patch_id=patch.id,
    unit_type=UNIT_TYPE,
                                 area=1.0, ratio=1.0))
        db.session.flush()
        u = MicroUnit.query.filter_by(patch_id=patch.id).fir
    st()
        db.session.add(IndicatorData(
            unit_id=u.id, soil_ratio=0.34, thickness=10,
    ph=9.24,
            organic_matter=4.89, coverage=87.92,
    shannon=0.60,
            slope=28.75, aspect="阳坡"))
        db.session.add(ConstraintCheck(
            patch_id=patch.id, policy_pass=True,
    location_pass=True,
            pollution_pass=True, geo_pass=True,
    farmland_pass=True,
            passed=True, reason="全部通过"))
        db.session.commit()
        return patch.id
def cleanup(pid):
    with app.app_context():
        from app.models import ScoreDetail
        rids = [r.id for r in EvaluationResult.query.filter_
    by(patch_id=pid)]
        if rids:
            ScoreDetail.query.filter(ScoreDetail.result_id.i
    n_(rids)).delete(
                synchronize_session=False)
```

```
        EvaluationResult.query.filter_by(patch_id=pid).delet
    e()
        ConstraintCheck.query.filter_by(patch_id=pid).delete
    ()
        uids = [u.id for u in MicroUnit.query.filter_by(pat
    h_id=pid)]
        if uids:
            IndicatorData.query.filter(IndicatorData.unit_id
    .in_(uids)).delete(
                synchronize_session=False)
        MicroUnit.query.filter_by(patch_id=pid).delete()
        m = db.session.get(MinePatch, pid)
        if m:
            db.session.delete(m)
        p = Project.query.filter_by(code="WEIGHT-%s" %
    TAG).first()
        if p:
            db.session.delete(p)
        db.session.commit()
def run_eval(pid):
    c.post("/evaluate/%d" % pid, data={},
    follow_redirects=False)
    with app.app_context():
        return EvaluationResult.query.filter_by(patch_id=pid
    ) \
            .order_by(EvaluationResult.id.desc()).first().fi
    nal_score
print(LINE)
print("一、默认应使用论文表 3 的权重")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
    w, label, scheme = ws.effective_weights(UNIT_TYPE)
ok(scheme is None and "论文表" in label, "未配置时权重来源为论文表 3",
    label)
ok(abs(w["l1"]["土壤指标"] - 0.44) < 1e-9, "一级权重取自论文表 3",
   "土壤指标 %.2f" % w["l1"]["土壤指标"])
pid = build_case()
base_score = run_eval(pid)
print("  默认权重下的评价值 F = %.4f" % base_score)
print()
print(LINE)
print("二、保存 AHP 算出的权重后，评价应改用新权重")
print(LINE)
with app.app_context():
    # 模拟 AHP 计算结果：把土壤指标权重提高、植被与地形降低（仍和为 1）
    ws.save_weights(UNIT_TYPE, [("soil", 0.70), ("veg",
    0.20),
                                ("terrain", 0.10)])
    n_items = WeightItem.query.count()
```

```
    ok(n_items >= 3, "weight_items 表已写入权重条目", "%d 条" % n_items)
with app.app_context():
    w2, label2, scheme2 = ws.effective_weights(UNIT_TYPE)
ok(scheme2 is not None and "自定义" in label2, "权重来源变为自定义方案",
   label2)
ok(abs(w2["l1"]["土壤指标"] - 0.70) < 1e-9, "新权重已生效",
   "土壤指标 %.2f" % w2["l1"]["土壤指标"])
ok(abs(w2["l1"]["植被指标"] - 0.20) < 1e-9, "同批保存的其他项也已生效")
ok(abs(w2["l2"]["土壤指标"]["土石比"] - 0.25) < 1e-9,
   "未覆盖的二级权重仍保留论文取值",
   "土石比 %.2f" % w2["l2"]["土壤指标"]["土石比"])
new_score = run_eval(pid)
print("  新权重下的评价值 F = %.4f" % new_score)
ok(abs(new_score - base_score) > 1e-4,
   "权重变化确实改变了评价结果",
   "%.4f → %.4f" % (base_score, new_score))
print()
print(LINE)
print("三、结果应记录所用权重来源")
print(LINE)
import json
with app.app_context():
    rec = EvaluationResult.query.filter_by(patch_id=pid) \
          .order_by(EvaluationResult.id.desc()).first()
    stored = json.loads(rec.unit_scores)
ok(any(u.get("weight_source") for u in stored),
   "评价记录中标注了权重来源",
   stored[0].get("weight_source") or "")
print()
print(LINE)
print("四、恢复默认后应回到论文取值")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
    w3, label3, scheme3 = ws.effective_weights(UNIT_TYPE)
ok(scheme3 is None and "论文表" in label3, "已恢复为论文表 3 权重",
   label3)
back_score = run_eval(pid)
print("  恢复后评价值 F = %.4f（默认时 %.4f）" % (back_score,
   base_score))
ok(abs(back_score - base_score) < 1e-9, "评价结果与默认时一致")
print()
print(LINE)
print("五、权重首页概览")
print(LINE)
with app.app_context():
    rows = ws.summary()
ok(len(rows) == 3, "三种微地质单元均有权重来源说明", "%d 行" % len(rows))
r = c.get("/ahp/")
body = r.get_data(as_text=True)
```

```
ok(r.status_code == 200 and "当前生效权重" in body, "权重首页正常展示")
print()
print(LINE)
print("六、一致性不通过的矩阵不得保存生效")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
n = 3
# 明显矛盾的矩阵：a12=9, a13=1/9, a23=9 —— CR 必然超限
mats = [(0, 1, "9"), (0, 2, "0.1111"), (1, 2, "9")]
data = {"csrf_token": "", "action": "save"}
for i, j, v in mats:
    data["m_%d_%d" % (i, j)] = v
r = c.post("/ahp/matrix?unit_type=%s&level=1" % UNIT_TYPE,
    data=data,
           follow_redirects=True)
with app.app_context():
    _, label4, scheme4 = ws.effective_weights(UNIT_TYPE)
ok(scheme4 is None, "CR 超限的矩阵未被保存生效", label4)
ok("一致性" in r.get_data(as_text=True), "页面给出了未通过一致性的提示")
cleanup(pid)
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("权重生效链路自检全部通过 √")
print(LINE)
# ==== app/services/__init__.py ====
# -*- coding: utf-8 -*-
# ==== app/utils/__init__.py ====
# -*- coding: utf-8 -*-
# ==== app/views/__init__.py ====
# -*- coding: utf-8 -*-
```